A Regular First-Order Calculus for the Wasserstein–Fisher–Rao Geometry

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Abstract

We introduce a notion of intrinsic differentiability for functionals on the space of finite nonnegative measures equipped with the Wasserstein–Fisher–Rao (WFR) distance, also known as the Hellinger–Kantorovich (HK) distance. The resulting static first-order calculus accounts simultaneously for spatial transport and mass reweighting. We characterize regular barycentric logarithms through optimal semicouplings and measurable selections at the transport threshold, and prove their tangency at arbitrary finite source measures. The defining expansion extends to non-optimal semicouplings, with remainder of smaller order than the square root of their cost. This yields differentiation formulas for simultaneous transport and reweighting and uniqueness of the gradients. We also give a sufficient criterion in terms of regular flat derivatives. We characterize differentiability of one half the squared WFR distance by a common optimal Monge transport–growth representation with no target mass created from the cone vertex. Equivalently, the regular barycentric logarithm is unique and its norm equals the distance; the gradient is minus this logarithm. Finally, we formulate the corresponding calculus on probability measures. On $\mathcal{P}_2(\mathbb{R}^d)$, WFR differentiability implies intrinsic Wasserstein differentiability, with the transport component of the WFR gradient coinciding with the Wasserstein gradient. Whenever a flat derivative exists, one quarter of the reaction component agrees almost everywhere with its centered representative. As a byproduct we obtain a Wasserstein differentiability result for the squared WFR distance.

Contents

1 Introduction **2**
- 1.1 Main contributions . . . 3
  - 1.1.1 Regular barycentric logarithms and tangency . . . 4
  - 1.1.2 Strong first-order expansions . . . 4
  - 1.1.3 Differentiability of the squared WFR distance . . . 5
  - 1.1.4 Probability measures and comparison of differentials . . . 6
- 1.2 Related work . . . 7

2 The Wasserstein–Fisher–Rao geometry **8**
- 2.1 The semicoupling formulation . . . 8
- 2.2 The cone formulation . . . 9
  - 2.2.1 The cone geometry . . . 9
  - 2.2.2 Wasserstein–Fisher–Rao as Wasserstein on the cone . . . 10
- 2.3 Wasserstein–Fisher–Rao geodesics . . . 11

3 Regular logarithms and tangency **15**
3.1 Regular and semicoupling logarithms 15
3.2 Representation by cone geodesics 20
3.3 Endpoint tangency 24
3.4 First-order independence of cut-locus selections 30

4 Regular Wasserstein–Fisher–Rao calculus **31**
4.1 Strength and uniqueness of the gradients 32
4.2 Calculus rules and differentiability criteria 36
4.2.1 A differentiability criterion and chain rules 36
4.2.2 Some differentiable functions 40
4.2.3 Compatibility with finite-dimensional derivatives 41

5 Differentiability of the squared Wasserstein–Fisher–Rao distance **42**
5.1 The differentiability theorem 43
5.2 Stability and first variation 45
5.3 Proof of differentiability and characterization 49

6 Restriction to probability measures **53**
6.1 The Wasserstein component 57
6.2 The reaction component and flat directions 59

A Expansions of costs and logarithms **64**
A.1 Calculations along cone geodesics 64
A.2 Graph semicouplings 66

B Radial dilations and gluing **67**
B.1 Radial dilations 67
B.2 Gluing prescribed cone couplings 68

C Technical results **70**
C.1 Vanishing-measure estimates 70
C.2 Approximation and tangency 71

1 Introduction

Optimal transport has become a fundamental framework for variational problems on spaces of probability measures, with applications ranging from mean-field control and games to data analysis, sampling, and generative modeling. A key ingredient behind many of these developments is the differential structure of the quadratic Wasserstein space $\mathcal{P}_2(\mathbb{R}^d)$. This can be formulated intrinsically, through the Wasserstein tangent structure and the corresponding notions of differential and gradient [2], or through the closely related Lions differential, obtained by lifting functionals of probability measures to a Hilbert space of random variables [8]. These tools make it possible to describe the sensitivity of functionals and formulate first-order optimality conditions directly at the level of probability distributions.

The Wasserstein geometry is, however, intrinsically mass preserving. While this is appropriate for evolutions governed by spatial transport alone (e.g., continuity equations), it becomes restrictive when spatial transport is accompanied by local creation, destruction, or reweighting of mass. Such mechanisms arise in reaction–transport equations, population dynamics, and partially

observed control, where observations induce Bayesian reweighting of conditional distributions. A further motivation comes from mean-field control problems involving weighted particle systems, in which both particle positions and weights evolve over time. These examples motivate a first-order calculus that accounts simultaneously for spatial displacements and weight variations.

Unbalanced optimal transport extends classical transport by incorporating mechanisms for mass variation. Different choices of these mechanisms lead to different geometries, including the generalized Wasserstein distances of Piccoli and Rossi, developed in connection with transport equations with source terms [27]. In this work, we focus on the Wasserstein–Fisher–Rao (WFR) distance, also known as the Hellinger–Kantorovich (HK) distance. In its dynamic formulation, the continuity equation includes a source term, and the action penalizes both spatial velocity and relative mass variation [10, 20]. Equivalent static formulations use semicouplings or relax the marginal constraints through entropy penalties [11, 24]. Infinitesimally, WFR combines the transport geometry of the quadratic Wasserstein metric with the mass variation geometry of the Fisher–Rao metric. This connection also holds at the metric level: De Ponti, Sodini, and Tamanini [12] characterize the Hellinger–Kantorovich distance as the metric infimal convolution of the Hellinger and Wasserstein distances. The metric and geometric foundations of WFR are well established. Its cone representation provides a detailed description of geodesics and their relation to optimal transport on the underlying cone space [23, 21, 24]. The associated gradient-flow theory has also received considerable attention [18, 22, 26].

Passing from these geometric foundations to a static first-order calculus nevertheless raises several issues. At a general source measure, optimal transport need not be described by a map, and geodesics may be non-unique. In particular, at the critical transport distance, different combinations of transport and reaction can have the same optimal cost. Moreover, creation of mass from the cone vertex is not represented by a transport–reaction field carried by the source measure. Its contribution must nevertheless be controlled in a first-order expansion whose remainder is measured by the WFR distance. A source-based differential calculus therefore requires a precise identification of the regular initial directions, a proof that the barycentric fields representing them belong to the tangent space, and estimates that remain valid in the presence of creation, annihilation, and geodesic non-uniqueness.

We address these issues by developing a regular intrinsic WFR differential at an arbitrary finite nonnegative measure. Its defining expansion pairs a tangent field with barycentric logarithms based at the source, with a remainder of smaller order than the WFR distance. We prove that this expansion extends to non-optimal semicouplings, allowing differentiation along explicit transport and reweighting perturbations. The calculus also applies to functionals defined on probability measures: its transport component recovers the Wasserstein gradient whenever the reference measure has finite second moment, while one quarter of its reaction component gives the centered flat derivative whenever the latter exists.

1.1 Main contributions

The main results concern endpoint tangency, strong first-order expansions, differentiability of the squared WFR distance, and comparison with Wasserstein and flat differentiability.

1.1.1 Regular barycentric logarithms and tangency

Let $\mathcal{M}(\mathbb{R}^d)$ denote the space of finite nonnegative Radon measures on $\mathbb{R}^d$. For $\mu \in \mathcal{M}(\mathbb{R}^d)$, consider $H(\mu) \coloneqq L^2(\mu; \mathbb{R}^d \times \mathbb{R})$ with scalar product

$$\langle (\xi_1, \zeta_1), (\xi_2, \zeta_2) \rangle_\mu \coloneqq \int_{\mathbb{R}^d} \left(\xi_1(x) \cdot \xi_2(x) + \frac{1}{4} \zeta_1(x) \zeta_2(x) \right) d\mu(x),$$

and define the regular tangent space by

$$T_\mu \mathcal{M}(\mathbb{R}^d) \coloneqq \overline{\{(\nabla\varphi, 4\varphi) : \varphi \in C_c^\infty(\mathbb{R}^d)\}}^{H(\mu)}.$$

A regular barycentric logarithm associated with a constant-speed WFR geodesic $(\boldsymbol{\mu}_t)_{t\in[0,1]}$ starting from μ is a field $(\xi, \zeta) \in T_\mu \mathcal{M}(\mathbb{R}^d)$ satisfying

$$\lim_{t \downarrow 0} \frac{\int_{\mathbb{R}^d} \varphi \, d\boldsymbol{\mu}_t - \int_{\mathbb{R}^d} \varphi \, d\mu}{t} = \int_{\mathbb{R}^d} (\nabla\varphi \cdot \xi + \varphi\zeta) \, d\mu \qquad \text{for every } \varphi \in C_c^\infty(\mathbb{R}^d).$$

Thus the logarithm represents the initial variation of smooth observables along the geodesic.

We characterize these fields through the semicoupling formulation of WFR. Writing $\mathrm{OptSC}(\mu, \nu)$ for the optimal semicouplings between μ and ν, we associate with each $\boldsymbol{\gamma} \in \mathrm{OptSC}(\mu, \nu)$ an explicit barycentric logarithm $\overline{\mathrm{Log}}_{\boldsymbol{\gamma}, u}$. Here the measurable selection $u \colon \mathbb{R}^d \times \mathbb{R}^d \to [0, 1]$ accounts for the non-uniqueness of the transport component at the threshold $\|x - y\| = \pi/2$; see Definition 3.2.

Theorem A (Characterization and tangency)**.** *Let $\mu, \nu \in \mathcal{M}(\mathbb{R}^d)$. Every constant-speed WFR geodesic connecting μ to ν admits a unique regular barycentric logarithm. The set of all such logarithms is*

$$\left\{ \overline{\mathrm{Log}}_{\boldsymbol{\gamma}, u} : \boldsymbol{\gamma} \in \mathrm{OptSC}(\mu, \nu), \ u \colon \mathbb{R}^d \times \mathbb{R}^d \to [0, 1] \text{ measurable} \right\}.$$

In particular, every field in this set belongs to $T_\mu \mathcal{M}(\mathbb{R}^d)$ and has $H(\mu)$-norm bounded by $\mathrm{WFR}(\mu, \nu)$.

The tangency statement holds at every prescribed finite source measure, without absolute continuity or a Monge representation, and for every optimal semicoupling and admissible threshold selection. Its proof combines the cone representation with a superposition principle for the continuity equation with reaction.

The explicit semicoupling construction also applies to non-optimal semicouplings. For every $\boldsymbol{\gamma} \in \mathrm{SC}(\mu, \nu)$ and every measurable selection u, its barycentric logarithm belongs to $H(\mu)$ and satisfies

$$\left\| \overline{\mathrm{Log}}_{\boldsymbol{\gamma}, u} \right\|_\mu^2 \leq \mathcal{C}(\boldsymbol{\gamma}),$$

where $\mathcal{C}$ denotes the semicoupling cost. This estimate is fundamental to the first-order expansions developed below.

1.1.2 Strong first-order expansions

In Section 4, we introduce the fundamental notion of the paper. Proposition 3.12 shows that the choice of u at the cut locus is irrelevant to first-order WFR expansions. We therefore fix $u \equiv 1$, choosing the transport logarithm at the threshold, and suppress u from the notation.

We say that a functional $F \colon \mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ is WFR differentiable at μ if there exists a regular tangent field $\mathcal{V} \in T_\mu \mathcal{M}(\mathbb{R}^d)$ such that

$$F(\nu) - F(\mu) = \left\langle \mathcal{V}, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}} \right\rangle_\mu + o\big(\mathrm{WFR}(\mu, \nu)\big) \qquad \text{as } \mathrm{WFR}(\mu, \nu) \to 0,$$

uniformly over optimal semicouplings $\boldsymbol{\gamma}$ between μ and ν.

A fundamental result for applications is that this expansion also holds along non-optimal semicouplings whose costs vanish.

Theorem B (Strength and uniqueness)**.** *Let $F\colon \mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ be WFR differentiable at μ. Its tangent representative is unique and is denoted by*

$$\nabla_{\mathrm{WFR}} F(\mu) = (\nabla^T F(\mu), \nabla^R F(\mu)).$$

For every sequence $\boldsymbol{\gamma}^n \in \mathrm{SC}(\mu, \mu_n)$ satisfying $\mathcal{C}(\boldsymbol{\gamma}^n) \to 0$, one has

$$F(\mu_n) - F(\mu) = \left\langle \nabla_{\mathrm{WFR}} F(\mu), \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \right\rangle_\mu + o\left(\sqrt{\mathcal{C}(\boldsymbol{\gamma}^n)}\right).$$

This result allows one to differentiate along explicit perturbations without identifying an optimal semicoupling. In particular for every $(T, \rho) \in H(\mu)$ the perturbation

$$\mu^\varepsilon := (\mathrm{Id} + \varepsilon T)_\# \left(\left(1 + \frac{\varepsilon}{2}\rho\right)^2 \mu \right)$$

satisfies

$$\lim_{\varepsilon \to 0} \frac{F(\mu^\varepsilon) - F(\mu)}{\varepsilon} = \int_{\mathbb{R}^d} \left(\nabla^T F(\mu) \cdot T + \frac{1}{4} \nabla^R F(\mu) \rho \right) d\mu.$$

We also establish a chain rule and a sufficient criterion for WFR differentiability based on flat derivatives. The criterion assumes an integral representation along linear interpolations, joint continuity of the flat derivative and its spatial gradient, and locally uniform $C_b^{1,1}$ bounds in the spatial variable. It yields WFR differentiability of smooth expectations, cylindrical functionals, and interaction energies.

Finally, the calculus is compatible with weighted empirical measures. If F is WFR differentiable at $\mu = \sum_{i=1}^N e^{\ell_i} \delta_{x_i}$, then $\widetilde{F}(\hat{x}, \hat{\ell}) = F(\mu)$ is Fréchet differentiable at $(\hat{x}, \hat{\ell})$, with

$$D\widetilde{F}(\hat{x}, \hat{\ell})[\hat{\xi}, \hat{\zeta}] = \sum_{i=1}^N e^{\ell_i} \left(\nabla^T F(\mu)(x_i) \cdot \xi_i + \frac{1}{4} \nabla^R F(\mu)(x_i) \zeta_i \right).$$

This formula remains valid when particle positions coincide and gives a finite-dimensional interpretation of both components of the WFR gradient.

1.1.3 Differentiability of the squared WFR distance

In Section 5, we characterize WFR differentiability of the squared-distance functional

$$F_{\hat{\mu}}(\mu) := \frac{1}{2} \mathrm{WFR}^2(\mu, \hat{\mu})$$

for a fixed target measure $\hat{\mu} \in \mathcal{M}(\mathbb{R}^d)$.

Theorem C. *The following conditions are equivalent:*

(i) $F_{\hat{\mu}}$ is WFR differentiable at μ.

(ii) There is exactly one regular barycentric logarithm $\mathcal{L}$ between μ and $\hat{\mu}$, and

$$\|\mathcal{L}\|_\mu = \mathrm{WFR}(\mu, \hat{\mu}).$$

(iii) Every optimal cone coupling between μ and $\hat{\mu}$ has the same Monge transport–growth representation at positive source radii and carries no target mass over the source vertex, as in (93)*.*

Whenever these conditions hold, writing T and q for the corresponding transport map and radial growth factor, the semicoupling

$$\gamma^\star = \big((\mathrm{Id}, T)_\# \mu, (\mathrm{Id}, T)_\# (q^2 \mu)\big)$$

is optimal between μ and $\hat{\mu}$. Its barycentric logarithm is $\overline{\mathrm{Log}}_{\gamma^\star} = \mathcal{L}$, and

$$\nabla_{\mathrm{WFR}} F_{\hat{\mu}}(\mu) = -\overline{\mathrm{Log}}_{\gamma^\star}.$$

Necessity follows from the metric-slope characterization of WFR gradients: differentiability forces equality in the logarithm norm bound, excluding both loss through barycentric averaging and target creation. Sufficiency combines an integrated first-variation estimate on the cone with stability of asymptotically optimal cone couplings.

1.1.4 Probability measures and comparison of differentials

On $\mathcal{P}(\mathbb{R}^d)$, the unit-mass constraint leads to the tangent space

$$T_\mu \mathcal{P}(\mathbb{R}^d) \coloneqq \left\{ (\xi, \zeta) \in T_\mu \mathcal{M}\big(\mathbb{R}^d\big) : \int_{\mathbb{R}^d} \zeta \, d\mu = 0 \right\}.$$

The same static expansion defines WFR differentiability for functionals on $\mathcal{P}(\mathbb{R}^d)$, with a tangent representative in this space. Strength and uniqueness remain valid, and we denote the resulting gradient by $\widetilde{\nabla}_{\mathrm{WFR}} F(\mu)$.

The restriction of an ambient WFR-differentiable functional is differentiable in this sense, with

$$\widetilde{\nabla}_{\mathrm{WFR}}(F_{|\mathcal{P}})(\mu) = \Pi_\mu \nabla_{\mathrm{WFR}} F(\mu), \qquad \Pi_\mu(\xi, \zeta) = \left(\xi, \zeta - \int_{\mathbb{R}^d} \zeta \, d\mu\right).$$

The transport and reaction components can then be compared with Wasserstein and flat derivatives, respectively.

Theorem D (Comparison of differentials)**.** *Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ and let $F \colon \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$ be WFR differentiable at μ, with*

$$\widetilde{\nabla}_{\mathrm{WFR}} F(\mu) = \big(\widetilde{\nabla}^T F(\mu), \widetilde{\nabla}^R F(\mu)\big).$$

Then $F_{|\mathcal{P}_2(\mathbb{R}^d)}$ is Wasserstein differentiable at μ, and

$$\nabla_{W_2} F(\mu) = \widetilde{\nabla}^T F(\mu).$$

If, in addition, F admits a flat derivative $\frac{\delta F}{\delta \mu}(\mu, \cdot) \in C_b(\mathbb{R}^d)$, then

$$\widetilde{\nabla}^R F(\mu) = 4 \left(\frac{\delta F}{\delta \mu}(\mu, \cdot) - \int_{\mathbb{R}^d} \frac{\delta F}{\delta \mu}(\mu, y) \, d\mu(y) \right) \qquad \mu\text{-a.e.}$$

Thus the transport component recovers the Wasserstein gradient, while one quarter of the reaction component gives the zero-mean representative of the flat derivative whenever the latter exists. The probability-space calculus itself requires no second-moment assumption.

Combining this comparison with our characterization of differentiability of the squared WFR distance, we obtain Wasserstein differentiability of $\frac{1}{2}\mathrm{WFR}^2(\cdot, \hat{\mu})$ at every $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ satisfying the equivalent conditions of Theorem 5.1, with Wasserstein gradient equal to minus the transport component of the regular barycentric logarithm from μ to $\hat{\mu}$.

WFR differentiability alone does not imply flat differentiability. We identify a class of absolutely continuous flat directions controlled by the WFR gradient and exhibit a counterexample based on singular mass creation, for which a linear mixture perturbation has WFR size of order $\sqrt{\varepsilon}$.

1.2 Related work

Our approach builds on the intrinsic differential calculus of the quadratic Wasserstein space developed in [2], where tangent fields, subdifferentials, and first-order expansions of functionals are described through optimal transport plans. This provides the model for our static formulation of WFR differentiability. The closely related Lions approach uses a lifting to random variables; see [8]. Differential tools on spaces of measures also play an important role in the derivation of necessary optimality conditions for measure-valued control problems [3]. For the squared Wasserstein distance, Alfonsi and Jourdain [1] characterize Lions differentiability in terms of uniqueness of a map-induced optimal coupling. Our characterization of differentiability of the squared WFR distance provides an unbalanced counterpart, in which the regular logarithm must additionally capture the full metric displacement.

The cone representation and geodesic lifting results of Liero, Mielke, and Savaré [24] provide the starting point for our characterization of regular logarithms. Their earlier work already describes the non-uniqueness of critical-distance Dirac geodesics [23, Section 5.2]. Further geometric properties of cones and HK spaces are established in [21], while [25] develops regularity results for optimal potentials and transport–growth representations from interior points of geodesics. Related regularity and Monge-type results are obtained by Gallouët, Ghezzi, and Vialard [17].

The infinitesimal transport–reaction structure has been extensively used in the study of evolution equations. Kondratyev, Monsaigneon, and Vorotnikov [20] develop a formal Riemannian calculus based on weighted Sobolev potentials. Variational constructions and qualitative properties of WFR gradient flows are studied, among others, in [18, 22, 26]. These works establish the role of the transport–reaction gradient structure in variational evolution problems.

A particularly close reference is Fleißner [16], who proves tangency of minimal velocity fields at almost every time along absolutely continuous curves, establishes a metric linearization theorem, and differentiates the squared HK distance along such curves. Her earlier work [15] also identifies elements of the Fréchet subdifferential at $h = 0$ of $h \mapsto -\frac{1}{2}\mathrm{HK}^2(\mu_h, \nu)$ for transport–reaction perturbations μ_h, and provides a quadratic remainder estimate for smooth test integrals expressed through optimal cone plans. This estimate is closely related to the smooth-observable expansions used here. Our endpoint tangency theorem concerns the specific barycentric fields associated with every optimal semicoupling at an arbitrary prescribed source measure, including all admissible selections at the transport threshold.

Logarithmic maps have also been developed in the linearized optimal transport literature. Cai, Cheng, Schmitzer, and Thorpe [7] introduce HK logarithmic and exponential maps for absolutely continuous reference measures, retaining an additional measure-valued component for singular creation, and discuss barycentric approximations when transport maps are unavailable. Sarrazin and Schmitzer [30] explicitly define barycentric logarithmic maps for nondeterministic soft plans, without requiring optimality in their definition. Our semicoupling construction retains the pairing information at the critical distance and incorporates measurable selections describing the possible regular initial transport components. We use this construction to characterize the full set of regular source logarithms and prove tangency for every optimal choice.

Other recent works develop complementary differential structures on HK spaces. Dello Schiavo and Sodini [13] establish universal infinitesimal Hilbertianity and density of cylinder functions in the corresponding Sobolev spaces. For a canonical reference measure on the space of measures, they also identify a natural Dirichlet form with the Cheeger energy. Saidi, Mena, and Gunsilius [29] relate the local differential geometry of HK to Wasserstein geometry on the cone, obtaining tangent-space isometries and applications to parallel transport under suitable regularity assumptions. In particular, their admissible class yields lifts uniformly separated from the cone vertex. Our results concern pointwise static expansions at arbitrary finite source measures, allowing sin-

gular sources and vertex contributions in the representing plans.

Finally, our comparison with flat derivatives is related to the calculus of functions on spaces of finite measures. Ren and Wang [28] establish relationships between extrinsic, linear functional, intrinsic, and L-derivatives under appropriate regularity assumptions. Our sufficient criterion identifies conditions under which a flat derivative yields the static WFR expansion required by our definition, including uniformity over optimal semicouplings and logarithmic selections.

2 The Wasserstein–Fisher–Rao geometry

In this section we recall some formulations and basic properties of the Wasserstein–Fisher–Rao distance. The material is mainly borrowed from [11, 24].

2.1 The semicoupling formulation

Definition 2.1. Let $\mu, \nu \in \mathcal{M}\big(\mathbb{R}^d\big)$ be given. A *semicoupling* between μ and ν is a pair

$$\boldsymbol{\gamma} = (\gamma_0, \gamma_1) \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^d)^2$$

such that

$$\pi^1_{\#}\gamma_0 = \mu \qquad \text{and} \qquad \pi^2_{\#}\gamma_1 = \nu.$$

We denote the set of semicouplings between μ and ν by $\mathrm{SC}(\mu, \nu)$.

We introduce the cost $c\colon \mathbb{R}^d \times \mathbb{R}^d \times [0,\infty) \times [0,\infty) \to [0,\infty)$ as

$$c(x, y, a, b) \coloneqq a + b - 2\sqrt{ab}\cos\Big(\Big(\|y - x\| \wedge \frac{\pi}{2}\Big)\Big) \tag{1}$$

The cost c is jointly 1-homogeneous in the variables a and b, namely

$$c(x, y, \lambda a, \lambda b) = \lambda c(x, y, a, b) \qquad \text{for every } \lambda \geq 0.$$

Therefore, if $\lambda \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^d)$ dominates both γ_0 and γ_1, and we set

$$a \coloneqq \frac{d\gamma_0}{d\lambda}, \qquad b \coloneqq \frac{d\gamma_1}{d\lambda},$$

then the semicoupling cost

$$\mathcal{C}(\boldsymbol{\gamma}) \coloneqq \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y, a(x, y), b(x, y))\, d\lambda(x, y)$$

is independent of the choice of λ and only depends on $\boldsymbol{\gamma}$. Equivalently,

$$\mathcal{C}(\boldsymbol{\gamma}) = \mu(\mathbb{R}^d) + \nu(\mathbb{R}^d) - 2\int_{\mathbb{R}^d \times \mathbb{R}^d} \sqrt{a(x, y)b(x, y)}\cos\Big(\Big(\|y - x\| \wedge \frac{\pi}{2}\Big)\Big)\, d\lambda(x, y). \tag{2}$$

The Wasserstein–Fisher–Rao distance is defined as follows.

Definition 2.2. Given $\mu, \nu \in \mathcal{M}\big(\mathbb{R}^d\big)$, we define their Wasserstein–Fisher–Rao distance as

$$\mathrm{WFR}^2(\mu, \nu) \coloneqq \inf_{\boldsymbol{\gamma}} \mathcal{C}(\boldsymbol{\gamma}), \tag{3}$$

where the infimum ranges over all semicouplings between μ and ν. We denote the set of minimizers in (3) by $\mathrm{OptSC}(\mu, \nu)$.

Remark 2.3 (Interpretation of semicouplings). Unlike classical transport plans, semicouplings allow the amount of mass leaving a point x to differ from the amount arriving at y, thereby accounting for mass variation during transport.

To interpret the cost, let $a = d\gamma_0/d\lambda$ and $b = d\gamma_1/d\lambda$. The cost integrand can be decomposed as

$$a + b - 2\sqrt{ab}\cos\Big(\|y-x\| \wedge \frac{\pi}{2}\Big) = (\sqrt{a}-\sqrt{b})^2 + 2\sqrt{ab}\big(1 - \cos\Big(\|y-x\| \wedge \frac{\pi}{2}\Big)\big).$$

The first term penalizes mass variation, while the second accounts for spatial displacement. In particular, for small displacements,

$$2\sqrt{ab}\big(1-\cos(\|y-x\|)\big) = \sqrt{ab}\|y-x\|^2 + O\big(\sqrt{ab}\|y-x\|^4\big),$$

recovering the classical quadratic transport cost at leading order when $a = b$. Finally, if either density vanishes, the cost describes pure annihilation or creation of mass, while for $\|y-x\| \geq \pi/2$ the interaction vanishes, so transporting mass is no more advantageous than annihilating it at the source and recreating it at the target.

2.2 The cone formulation

2.2.1 The cone geometry

Let $Y := \mathbb{R}^d \times [0,\infty)$. Consider the function $\mathrm{d}_{\mathfrak{C}} \colon Y \times Y \to [0,\infty)$ defined by

$$\mathrm{d}_{\mathfrak{C}}^2\big((x,r),(y,s)\big) := r^2 + s^2 - 2rs\cos\big(\|x-y\| \wedge \pi\big). \tag{4}$$

The function $\mathrm{d}_{\mathfrak{C}}$ is nonnegative, symmetric and satisfies the triangle inequality. It is only a semidistance on Y, since all the points of the form $(x,0)$ have zero distance from each other. We therefore consider the equivalence relation

$$(x,r) \sim (y,s) \quad \Longleftrightarrow \quad r = s = 0 \quad \text{or} \quad x = y,\ r = s, \tag{5}$$

and define the cone

$$\mathfrak{C} := Y/\sim$$

equipped with the induced distance, which we still denote by $\mathrm{d}_{\mathfrak{C}}$. We denote a generic element of $\mathfrak{C}$ by $\mathfrak{y} = [x,r]$, while the cone vertex is denoted by $\mathfrak{o}$. The projection map is $\mathfrak{p} \colon Y \to \mathfrak{C}$:

$$\mathfrak{p}\,((x,r)) = [x,r],$$

which admits right inverses $\mathfrak{i}_{\bar{x}} \colon \mathfrak{C} \to Y$ defined as

$$\mathfrak{i}_{\bar{x}}(\mathfrak{y}) := \begin{cases} (x,r), & \mathfrak{y} = [x,r],\ r > 0, \\ (\bar{x},0), & \mathfrak{y} = \mathfrak{o}. \end{cases} \tag{6}$$

The radial component of $i_{\bar{x}}$ is continuous, while $i_{\bar{x}}$ itself is in general only Borel measurable due to its behavior at the vertex.

2.2.2 Wasserstein–Fisher–Rao as Wasserstein on the cone

It will also be useful to consider the truncated cone distance

$$\mathrm{d}^2_{\mathfrak{C},\frac{\pi}{2}}\big([x,r],[y,s]\big) \coloneqq r^2+s^2-2rs\cos\big(\|x-y\|\wedge\frac{\pi}{2}\big). \tag{7}$$

Notice that

$$c(x,y,a,b)=\mathrm{d}^2_{\mathfrak{C},\frac{\pi}{2}}\big([x,\sqrt{a}],[y,\sqrt{b}]\big). \tag{8}$$

The distinction between $\mathrm{d}_{\mathfrak{C}}$ and $\mathrm{d}_{\mathfrak{C},\frac{\pi}{2}}$ is useful: the first one is the canonical cone distance and is the natural one for the description of geodesics, whereas the second one is directly related to the semicoupling formulation. Nevertheless, the two distances induce the same Wasserstein–Fisher–Rao distance on $\mathcal{M}\big(\mathbb{R}^d\big)$, as recalled below.

Let $A\in\mathcal{M}(\mathfrak{C}^N)$ and write

$$\mathfrak{y}_i=[x_i,r_i],\qquad i=1,\dots,N.$$

We say that $A\in\mathcal{M}_2(\mathfrak{C}^N)$ if

$$\int_{\mathfrak{C}^N}\sum_{i=1}^N r_i^2\,dA<+\infty.$$

For $i=1,\dots,N$, we define the i-th *homogeneous marginal* of A as

$$\mathfrak{h}_i(A)\coloneqq(x_i)_\#(r_i^2A)\in\mathcal{M}\big(\mathbb{R}^d\big). \tag{9}$$

Here the value assigned to x_i when $\mathfrak{y}_i=\mathfrak{o}$ is irrelevant, since the measure r_i^2A does not charge the vertex. Equivalently, for every bounded and continuous function $\varphi\colon\mathbb{R}^d\to\mathbb{R}$,

$$\int_{\mathbb{R}^d}\varphi(x)\,d\mathfrak{h}_i(A)(x)=\int_{\mathfrak{C}^N}r_i^2\varphi(x_i)\,dA(\mathfrak{y}_1,\dots,\mathfrak{y}_N). \tag{10}$$

When $N=1$ we simply write $\mathfrak{h}(A)$.

For $i=1,\dots,N$, define the coordinate projections on Y^N by

$$\mathsf{x}_i\big((x_j,r_j)_{j=1}^N\big)\coloneqq x_i,\qquad \mathsf{r}_i\big((x_j,r_j)_{j=1}^N\big)\coloneqq r_i.$$

Fix $\bar{x}\in\mathbb{R}^d$ and let $\mathfrak{i}_{\bar{x}}^{\otimes N}$ and $\mathfrak{p}^{\otimes N}$ denote the product maps induced by the right inverse $\mathfrak{i}_{\bar{x}}$ and the quotient map $\mathfrak{p}$. For $A\in\mathcal{M}_2(\mathfrak{C}^N)$, its coordinate lift $\widetilde{A}\coloneqq(\mathfrak{i}_{\bar{x}}^{\otimes N})_\#A$ satisfies

$$(\mathfrak{p}^{\otimes N})_\#\widetilde{A}=A,\qquad h_i(A)=(\mathsf{x}_i)_\#\big(\mathsf{r}_i^2\widetilde{A}\big),\quad i=1,\dots,N.$$

Conversely, for any finite nonnegative measure $\widetilde{A}$ on $(\mathbb{R}^d\times[0,\infty))^N$ satisfying $\int\sum_{i=1}^N\mathsf{r}_i^2\,d\widetilde{A}<\infty$,

$$h_i\big((\mathfrak{p}^{\otimes N})_\#\widetilde{A}\big)=(\mathsf{x}_i)_\#\big(\mathsf{r}_i^2\widetilde{A}\big),\quad i=1,\dots,N.$$

These identities are independent of the spatial representative at the vertex, since the weight r_i^2 vanishes there. Thus passing to the quotient, or choosing a coordinate lift, preserves all homogeneous marginals.

Definition 2.4. Let $\mu,\nu\in\mathcal{M}\big(\mathbb{R}^d\big)$. A measure $A\in\mathcal{M}_2(\mathfrak{C}^2)$ is called a *cone coupling* between μ and ν if

$$\mathfrak{h}_1(A)=\mu\qquad\text{and}\qquad\mathfrak{h}_2(A)=\nu.$$

We denote the set of cone couplings between μ and ν by

$$\Gamma_{\mathfrak{C}}(\mu,\nu)\coloneqq\big\{A\in\mathcal{M}_2(\mathfrak{C}^2):\mathfrak{h}_1(A)=\mu,\ \mathfrak{h}_2(A)=\nu\big\}.$$

We have the following characterization of the Wasserstein–Fisher–Rao distance; see [24, Problem 7.4 and Lemma 7.19].

Theorem 2.5 (Cone formulation)**.** *For every $\mu, \nu \in \mathcal{M}(\mathbb{R}^d)$ it holds*

$$\mathrm{WFR}^2(\mu,\nu) = \min_{A \in \Gamma_{\mathfrak{C}}(\mu,\nu)} \int_{\mathfrak{C}^2} \mathrm{d}^2_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)\, dA(\mathfrak{y}_0,\mathfrak{y}_1). \tag{11}$$

Moreover,

$$\mathrm{WFR}^2(\mu,\nu) = \min_{A \in \Gamma_{\mathfrak{C}}(\mu,\nu)} \int_{\mathfrak{C}^2} \mathrm{d}^2_{\mathfrak{C},\frac{\pi}{2}}(\mathfrak{y}_0,\mathfrak{y}_1)\, dA(\mathfrak{y}_0,\mathfrak{y}_1). \tag{12}$$

Accordingly, we say that a cone coupling is *optimal* if it attains the minimum in (11). We denote by

$$\mathrm{Opt}\Gamma_{\mathfrak{C}}(\mu,\nu)$$

the collection of optimal cone couplings between μ and ν.

From the previous theorem (see again [24, Lemma 7.19]), if we introduce the set of *ineffective pairs*

$$\mathfrak{B} \coloneqq \Big\{ \big([x,r],[y,s]\big) \in \mathfrak{C} \times \mathfrak{C} \colon rs > 0, \|x-y\| > \frac{\pi}{2} \Big\} \tag{13}$$

we get that if $A \in \mathrm{Opt}\Gamma_{\mathfrak{C}}(\mu,\nu)$, then $A(\mathfrak{B}) = 0$, that is, any optimal cone plan does not pair points whose spatial coordinates are more than $\frac{\pi}{2}$ apart. We say that a pair $(\mathfrak{y}_0,\mathfrak{y}_1) \in \mathfrak{C}^2$ is *reduced* if it does not belong to $\mathfrak{B}$.

Remark 2.6. The homogeneous formulation has an important geometric interpretation. The cone variable r should be thought of as the square root of the mass: a cone point $[x,r]$ represents the measure $r^2\delta_x$. Hence transport on the cone simultaneously encodes displacement of the position x and variation of the mass through the radial coordinate r. Together with Theorem 2.5, this provides a close geometric relation between WFR and the Wasserstein distance on the cone. This geometric relation also suggests a close connection between the corresponding differential structures. Nonetheless, applying standard Wasserstein differential calculus on the cone requires care because of the singularity at the vertex: the cone is not a smooth Riemannian manifold there. This also prevents a direct application of the usual Lions lifting approach, since $L^2(\Omega;\mathfrak{C})$ is a metric space rather than a Hilbert space with its natural cone-induced distance.

2.3 Wasserstein–Fisher–Rao geodesics

We now recall the characterization of Wasserstein–Fisher–Rao geodesics in terms of geodesics on the cone, following [24, Theorem 8.6], and record corresponding formulas and estimates used in the remainder of the paper.

A constant-speed WFR geodesic connecting μ and ν is a curve $\boldsymbol{\mu}\colon [0,1] \to \mathcal{M}(\mathbb{R}^d)$ satisfying, for every $t_1, t_2 \in [0,1]$,

$$\boldsymbol{\mu}_0 = \mu, \qquad \boldsymbol{\mu}_1 = \nu, \qquad \mathrm{WFR}(\boldsymbol{\mu}_{t_2}, \boldsymbol{\mu}_{t_1}) = |t_2 - t_1| \mathrm{WFR}(\mu,\nu).$$

We denote by

$$\mathrm{Geo}(\mathfrak{C}) \coloneqq \big\{ \mathfrak{y}\colon [0,1] \to \mathfrak{C} \colon \mathrm{d}_{\mathfrak{C}}\left(\mathfrak{y}_{t_1}, \mathfrak{y}_{t_2}\right) = |t_2 - t_1| \mathrm{d}_{\mathfrak{C}}\left(\mathfrak{y}_1, \mathfrak{y}_0\right), \quad t_1, t_2 \in [0,1] \big\}$$

the set of constant-speed cone geodesics. With the uniform topology, this is a closed subset of the Polish space $C([0,1];\mathfrak{C})$, hence itself Polish. Since $\mathbb{R}^d$ is uniquely geodesic, so is $(\mathfrak{C}, \mathrm{d}_{\mathfrak{C}})$; see [6, Chapter I.5, Corollary 5.11]. The endpoint map

$$E\colon \mathrm{Geo}(\mathfrak{C}) \to \mathfrak{C} \times \mathfrak{C}, \qquad E(\mathfrak{y}) \coloneqq (\mathfrak{y}_0, \mathfrak{y}_1),$$

is therefore a continuous bijection between Polish spaces. Its inverse G, which assigns to each pair its unique cone geodesic, is Borel measurable by the Lusin–Souslin theorem [19, Corollary 15.2].

For $t \in [0,1]$, let

$$e_t \colon \mathrm{Geo}(\mathfrak{C}) \to \mathfrak{C}, \qquad e_t(\mathfrak{y}) \coloneqq \mathfrak{y}_t,$$

and define the Borel coordinate evaluation maps

$$\mathsf{x}_t \coloneqq \mathrm{pr}_x \circ \mathfrak{i}_0 \circ e_t, \qquad \mathsf{r}_t \coloneqq \mathrm{pr}_r \circ \mathfrak{i}_0 \circ e_t,$$

where $\mathrm{pr}_x, \mathrm{pr}_r$ are the coordinate projections on $\mathbb{R}^d \times [0,\infty)$. For $\mathfrak{y} \in \mathrm{Geo}(\mathfrak{C})$, write

$$\boldsymbol{x}_t \coloneqq \mathsf{x}_t(\mathfrak{y}), \qquad \boldsymbol{r}_t \coloneqq \mathsf{r}_t(\mathfrak{y}), \qquad \mathfrak{y}_t = [\boldsymbol{x}_t, \boldsymbol{r}_t].$$

Both coordinate evaluation maps are Borel measurable. The radial coordinate is intrinsic, whereas the spatial coordinate depends on the chosen representative only at the vertex. In particular, homogeneous expressions such as $\boldsymbol{r}_t^2 \varphi(\boldsymbol{x}_t)$ are independent of this choice.

For a finite measure $\boldsymbol{\Pi} \in \mathcal{M}(\mathrm{Geo}(\mathfrak{C}))$ satisfying

$$(e_t)_{\#}\boldsymbol{\Pi} \in \mathcal{M}_2(\mathfrak{C}) \qquad t \in [0,1], \tag{14}$$

we define its homogeneous marginal at time t by

$$\mathfrak{h}_t(\boldsymbol{\Pi}) \coloneqq \mathfrak{h}\big((e_t)_{\#}\boldsymbol{\Pi}\big). \tag{15}$$

Equivalently, for every $\varphi \in C_b(\mathbb{R}^d)$,

$$\int_{\mathbb{R}^d} \varphi(x)\, d\mathfrak{h}_t(\boldsymbol{\Pi})(x) = \int_{\mathrm{Geo}(\mathfrak{C})} \boldsymbol{r}_t^2 \varphi(\boldsymbol{x}_t)\, d\boldsymbol{\Pi}(\mathfrak{y}). \tag{16}$$

The following characterization, which states that every Wasserstein–Fisher–Rao geodesic is the homogeneous marginal of a superposition of geodesics on the cone, can be found in [24, Theorem 8.6].

Theorem 2.7 (Characterization of WFR geodesics)**.** *Let $\mu_0, \mu_1 \in \mathcal{M}\big(\mathbb{R}^d\big)$. A curve $\boldsymbol{\mu} \colon [0,1] \to \mathcal{M}\big(\mathbb{R}^d\big)$ is a constant-speed WFR geodesic connecting μ_0 to μ_1 if and only if there exists*

$$\boldsymbol{\Pi} \in \mathcal{P}(\mathrm{Geo}(\mathfrak{C}))$$

such that

$$\boldsymbol{\mu}_t = \mathfrak{h}_t(\boldsymbol{\Pi}) \qquad \text{for every } t \in [0,1], \tag{17}$$

and

$$(e_0, e_1)_{\#}\boldsymbol{\Pi} \in \mathrm{Opt}\Gamma_{\mathfrak{C}}(\mu_0, \mu_1).$$

For every such representation and every $0 \le t_1 < t_2 \le 1$, one also has

$$(e_{t_1}, e_{t_2})_{\#}\boldsymbol{\Pi} \in \mathrm{Opt}\Gamma_{\mathfrak{C}}(\boldsymbol{\mu}_{t_1}, \boldsymbol{\mu}_{t_2}). \tag{18}$$

Moreover, every $A \in \mathrm{Opt}\Gamma_{\mathfrak{C}}(\mu_0, \mu_1)$ admits a representation

$$A = (e_0, e_1)_{\#}\boldsymbol{\Pi}_A, \qquad \boldsymbol{\Pi}_A \in \mathcal{M}(\mathrm{Geo}(\mathfrak{C})).$$

The curve $\boldsymbol{\mu}_t \coloneqq \mathfrak{h}_t(\boldsymbol{\Pi}_A)$ is a constant-speed WFR geodesic connecting μ_0 to μ_1, and (18) holds. If $A \in \mathcal{P}(\mathfrak{C}^2)$, then $\boldsymbol{\Pi}_A$ is a probability measure.

We next record the cone-geodesic formulas and estimates used to construct and characterize regular barycentric logarithms in Section 3.

The set of reduced cone geodesics is

$$\mathrm{Geo}^*(\mathfrak{C}) \coloneqq \{\mathfrak{y} \in \mathrm{Geo}(\mathfrak{C})\colon (\mathfrak{y}_0, \mathfrak{y}_1) \notin \mathfrak{B}\}\,.$$

Every probability representation $\mathbf{\Pi}$ in Theorem 2.7 is concentrated on this set: its optimal endpoint plan $A \coloneqq (e_0, e_1)_{\#}\mathbf{\Pi}$ satisfies $A(\mathfrak{B}) = 0$, hence

$$\mathbf{\Pi}\big(\mathrm{Geo}^*(\mathfrak{C})\big) = 1.$$

Define the vector-valued sine function

$$\mathbf{sin}(z) \coloneqq \begin{cases} \sin(\|z\|)\dfrac{z}{\|z\|}, & z \neq 0, \\ 0, & z = 0. \end{cases}$$

and, for $y \in \mathbb{R}^d$, the kernel

$$K_y(x) \coloneqq \cos\left(\|x - y\| \wedge \pi\right).$$

Then $K_y \in C_b^{1,1}(\mathbb{R}^d)$ and, uniformly in y,

$$|K_y(x)|^2 + \|\nabla K_y(x)\|^2 = 1, \qquad \mathrm{Lip}(\nabla K_y) \leq 1. \tag{19}$$

Indeed, on $0 < \|x - y\| < \pi$, the radial and tangential eigenvalues of the Hessian are respectively $-\cos(\|x - y\|)$ and $-\sin(\|x - y\|)/\|x - y\|$, both bounded in absolute value by one. The gradient extends continuously across $\|x - y\| = 0, \pi$ and vanishes on $\|x - y\| \geq \pi$.

For the cone geodesic $\mathfrak{y}_t = [\boldsymbol{x}_t, \boldsymbol{r}_t]$ joining $\mathfrak{y}_0 = [x, r]$ to $\mathfrak{y}_1 = [y, s]$, the explicit formulas are as follows; see [25, Section 2.1.2, Eq. (2.7)].

When $r, s > 0$ and $\|x - y\| < \pi$, they follow by identifying the cone over the segment joining x and y with a Euclidean sector, in which the geodesic is a straight line segment. When $\|x - y\| \geq \pi$, the geodesic passes through the vertex.

The radial component satisfies

$$\boldsymbol{r}_t^2 = \boldsymbol{m}_t(\mathfrak{y}_0, \mathfrak{y}_1) \coloneqq (1-t)^2 r^2 + t^2 s^2 + 2rst(1-t)K_y(x).$$

For the spatial component we need instead to distinguish different behaviors depending on the endpoints.

If $r, s > 0$, set

$$u \coloneqq \frac{s}{r}K_y(x) - 1, \qquad v \coloneqq \frac{s}{r}\nabla K_y(x).$$

If $\|y - x\| = 0$, then $\boldsymbol{x}_t \equiv x$. If $0 < \|y - x\| < \pi$, then

$$\boldsymbol{x}_t = x + \mathrm{Arg}\left((1-t)r + tse^{\mathrm{i}\|y-x\|}\right)\frac{y - x}{\|y - x\|}, \qquad t \in [0, 1],$$

where Arg denotes the principal argument, which belongs to $[0, \pi]$ for the displayed complex numbers. If $\|y - x\| \geq \pi$, setting $\tau \coloneqq r/(r + s)$, we have

$$\mathfrak{y}_t = \begin{cases} [x, r - t(r+s)], & 0 \leq t < \tau, \\ \mathfrak{o}, & t = \tau, \\ [y, t(r+s) - r], & \tau < t \leq 1. \end{cases}$$

Finally, if one of the endpoints is the vertex, the geodesic is simply radial:

$$\mathfrak{y}_t([x,r],\mathfrak{o}) = [x,\boldsymbol{r}_t] = [x,(1-t)r], \qquad \mathfrak{y}_t(\mathfrak{o},[y,s]) = [y,\boldsymbol{r}_t] = [y,ts].$$

If both endpoints coincide with $\mathfrak{o}$, the geodesic is constant: $\mathfrak{y}_t \equiv \mathfrak{o}$.

Therefore, apart from the trivial geodesic $\mathfrak{y}_t \equiv \mathfrak{o}$, no reduced geodesic hits the vertex in $(0,1)$.

The next lemma records the velocities along cone geodesics with nonzero initial radius. Its proof and that of Lemma 2.9 are given in Appendix A.1.

Lemma 2.8 (Velocities along cone geodesics)**.** *Let $\mathfrak{y}_t = [\boldsymbol{x}_t, \boldsymbol{r}_t] \in \mathrm{Geo}(\mathfrak{C})$ be a cone geodesic with $\mathfrak{y}_0 = [x,r] \neq \mathfrak{o}$ and $t \in [0,1)$ such that $\boldsymbol{r}_t > 0$. Derivatives at $t = 0$ are understood as right derivatives.*

If $\mathfrak{y}_1 = [y,s] \neq \mathfrak{o}$, then, with u, v defined above,

$$\dot{\boldsymbol{x}}_t = \frac{v}{(1+tu)^2 + t^2\|v\|^2}, \qquad \dot{\boldsymbol{r}}_t = r\frac{(1+tu)u + t\|v\|^2}{\sqrt{(1+tu)^2 + t^2\|v\|^2}},$$

and

$$\dot{\boldsymbol{m}}_t = -2(1-t)r^2 + 2ts^2 + 2(1-2t)rsK_y(x).$$

If $\mathfrak{y}_1 = \mathfrak{o}$, then the geodesic is radial and, for $t \in [0,1)$, $\dot{\boldsymbol{x}}_t = 0$ and $\dot{\boldsymbol{r}}_t = -r$. In both cases,

$$\boldsymbol{r}_t^2\|\dot{\boldsymbol{x}}_t\|^2 + |\dot{\boldsymbol{r}}_t|^2 = \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0,\mathfrak{y}_1), \qquad t \in [0,1) \text{ with } \boldsymbol{r}_t > 0. \tag{20}$$

For $\mathfrak{y}_0 = [x,r]$ and $\mathfrak{y}_1 = [y,s]$ with $r,s > 0$, Lemma 2.8 identifies the cone logarithm in spatial and radial coordinates as

$$\mathrm{Log}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1) \coloneqq (\dot{\boldsymbol{x}}_0, \dot{\boldsymbol{r}}_0) = \left(\frac{s}{r}\nabla K_y(x), sK_y(x) - r\right).$$

If $\mathfrak{y}_1 = \mathfrak{o}$, the radial geodesic instead gives

$$\mathrm{Log}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{o}) = (0,-r).$$

We denote the components of the cone logarithm as follows:

$$\mathrm{Log}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1) = (\mathrm{Log}^T_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1), \mathrm{Log}^R_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)).$$

The next lemma gives the initial derivative and a quadratic expansion of homogeneous observables along cone geodesics.

Lemma 2.9. *Let $\varphi \in C^{1,1}_b(\mathbb{R}^d)$ and define*

$$\Phi_\varphi([x,r]) \coloneqq r^2\varphi(x), \qquad \Phi_\varphi(\mathfrak{o}) \coloneqq 0.$$

Let $\mathfrak{y}_t \in \mathrm{Geo}(\mathfrak{C})$ be a cone geodesic joining $\mathfrak{y}_0 = [x,r]$ and $\mathfrak{y}_1 = [y,s]$. The map $t \mapsto \Phi_\varphi(\mathfrak{y}_t)$ has a right derivative at $t = 0$. If $r > 0$, then

$$\frac{d}{dt}\Phi_\varphi(\mathfrak{y}_t)\Big|_{t=0^+} = r^2\nabla\varphi(x)\cdot\mathrm{Log}^T_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1) + 2r\varphi(x)\,\mathrm{Log}^R_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1),$$

while, if $\mathfrak{y}_0 = \mathfrak{o}$, the right derivative is zero.

Moreover, the map $t \mapsto \Phi_\varphi(\mathfrak{y}_t)$ is differentiable in $(0,1)$ with

$$\left|\frac{d}{dt}\Phi_\varphi(\mathfrak{y}_t)\right| \le B_\varphi(r^2+s^2)$$

at every interior time, where $B_\varphi \coloneqq 3\|\varphi\|_\infty + \frac{1}{2}\|\nabla\varphi\|_\infty$, and consequently

$$\left|\frac{\Phi_\varphi(\mathfrak{y}_t)-\Phi_\varphi(\mathfrak{y}_0)}{t}\right| \le B_\varphi(r^2+s^2), \qquad t\in(0,1]. \tag{21}$$

Finally,

$$\left|\Phi_\varphi(\mathfrak{y}_t)-\Phi_\varphi(\mathfrak{y}_0)-t\,\frac{d}{d\tau}\Big|_{\tau=0^+}\Phi_\varphi(\mathfrak{y}_\tau)\right| \le C_\varphi t^2 \mathrm{d}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)^2, \qquad t\in[0,1],$$

where one may take

$$C_\varphi = \|\varphi\|_\infty + \frac{1}{2}\|\nabla\varphi\|_\infty + \frac{1}{2}\operatorname{Lip}(\nabla\varphi). \tag{22}$$

3 Regular logarithms and tangency

In this section we study the notion of regular Wasserstein–Fisher–Rao logarithms and characterize them in terms of optimal semicouplings and measurable selections at the cut locus $\|x-y\| = \frac{\pi}{2}$. The key ingredient is the tangency result of Theorem 3.11, which shows that the barycentric logarithm associated with any optimal semicoupling, and any admissible selection at the cut locus, belongs to the WFR tangent space at the source measure. Combining this with the representation of WFR geodesics through probabilities supported on $\mathrm{Geo}(\mathfrak{C})$ yields the characterization of regular logarithms stated in Theorem 3.7. We end the section by showing that the choice at the cut locus is irrelevant for first-order expansions.

3.1 Regular and semicoupling logarithms

We first recall the definition of the tangent space [20]. On $\mathbb{R}^d\times\mathbb{R}$, we consider the scalar product

$$\langle(\xi_1,\zeta_1),(\xi_2,\zeta_2)\rangle_{\mathrm{WFR}} \coloneqq \xi_1\cdot\xi_2 + \frac{1}{4}\zeta_1\zeta_2, \tag{23}$$

with associated norm $\|\cdot\|_{\mathrm{WFR}}$. Then, for $\mu\in\mathcal{M}(\mathbb{R}^d)$, we introduce the Hilbert space

$$H(\mu) \coloneqq L^2(\mu;\mathbb{R}^d\times\mathbb{R})$$

equipped with the scalar product

$$\langle\mathcal{V}_1,\mathcal{V}_2\rangle_\mu \coloneqq \int_{\mathbb{R}^d}\langle(\xi_1(x),\zeta_1(x)),(\xi_2(x),\zeta_2(x))\rangle_{\mathrm{WFR}}\,d\mu(x), \tag{24}$$

where $\mathcal{V}_i=(\xi_i,\zeta_i)$, $i=1,2$. The associated norm is denoted by $\|\cdot\|_\mu$. The tangent space to $\mathcal{M}(\mathbb{R}^d)$ at μ is then defined by

$$T_\mu\mathcal{M}(\mathbb{R}^d) \coloneqq \overline{\{(\nabla\varphi,4\varphi)\colon\varphi\in C_c^\infty(\mathbb{R}^d)\}}^{H(\mu)}. \tag{25}$$

We now introduce the fundamental notion of this section.

Definition 3.1 (Regular barycentric logarithm)**.** We say that $\mathcal{V} = (\xi, \zeta) \in T_\mu \mathcal{M}\big(\mathbb{R}^d\big)$ is a *regular barycentric logarithm* between μ and ν if there exists a constant-speed WFR geodesic $\boldsymbol{\mu}\colon [0,1] \to \mathcal{M}\big(\mathbb{R}^d\big)$, with $\boldsymbol{\mu}_0 = \mu$ and $\boldsymbol{\mu}_1 = \nu$, such that, for every $\varphi \in C_c^\infty(\mathbb{R}^d)$,

$$\lim_{t \downarrow 0} \frac{\int_{\mathbb{R}^d} \varphi \, d\boldsymbol{\mu}_t - \int_{\mathbb{R}^d} \varphi \, d\mu}{t} = \int_{\mathbb{R}^d} (\nabla\varphi \cdot \xi + \varphi\zeta) \, d\mu. \tag{26}$$

For a fixed geodesic, there is at most one such field. Indeed, the difference of two representatives belongs to $T_\mu \mathcal{M}\big(\mathbb{R}^d\big)$ and is orthogonal to every $(\nabla\varphi, 4\varphi)$, $\varphi \in C_c^\infty(\mathbb{R}^d)$, and therefore vanishes.

To characterize all regular barycentric logarithms between two measures, we introduce the following definition.

Definition 3.2 (Semicoupling logarithmic map)**.** Let $\boldsymbol{\gamma} = (\gamma_0, \gamma_1)$ be a semicoupling between μ and ν, and let $u\colon \mathbb{R}^d \times \mathbb{R}^d \to [0,1]$ be measurable. Choose a finite positive measure λ dominating γ_0 and γ_1, and set

$$a \coloneqq \frac{d\gamma_0}{d\lambda}, \qquad b \coloneqq \frac{d\gamma_1}{d\lambda}.$$

We define the *transport* and *reaction* components, γ_0-a.e., by

$$\mathrm{Log}^T_{\boldsymbol{\gamma},u}(x,y) \coloneqq \sqrt{\frac{b(x,y)}{a(x,y)}} \begin{cases} \mathbf{sin}(y-x), & \|y-x\| < \pi/2, \\ u(x,y)\dfrac{y-x}{\|y-x\|}, & \|y-x\| = \pi/2, \\ 0, & \|y-x\| > \pi/2, \end{cases} \tag{27a}$$

$$\mathrm{Log}^R_{\boldsymbol{\gamma}}(x,y) \coloneqq 2\left(\sqrt{\frac{b(x,y)}{a(x,y)}} \cos\Big(\|y-x\| \wedge \frac{\pi}{2}\Big) - 1\right). \tag{27b}$$

The *semicoupling logarithmic map* is

$$\mathrm{Log}_{\boldsymbol{\gamma},u} \coloneqq \big(\mathrm{Log}^T_{\boldsymbol{\gamma},u}, \mathrm{Log}^R_{\boldsymbol{\gamma}}\big).$$

Disintegrating the first leg as $\gamma_0(dx,dy) = \mu(dx)\kappa_x(dy)$, we define the corresponding *barycentric components* by

$$\overline{\mathrm{Log}}^T_{\boldsymbol{\gamma},u}(x) \coloneqq \int_{\mathbb{R}^d} \mathrm{Log}^T_{\boldsymbol{\gamma},u}(x,y)\, \kappa_x(dy), \qquad \overline{\mathrm{Log}}^R_{\boldsymbol{\gamma}}(x) \coloneqq \int_{\mathbb{R}^d} \mathrm{Log}^R_{\boldsymbol{\gamma}}(x,y)\, \kappa_x(dy).$$

The *barycentric logarithmic map* is then

$$\overline{\mathrm{Log}}_{\boldsymbol{\gamma},u}(x) \coloneqq \Big(\overline{\mathrm{Log}}^T_{\boldsymbol{\gamma},u}(x), \overline{\mathrm{Log}}^R_{\boldsymbol{\gamma}}(x)\Big) = \int_{\mathbb{R}^d} \mathrm{Log}_{\boldsymbol{\gamma},u}(x,y)\, \kappa_x(dy).$$

Remark 3.3. The semicoupling logarithm is the normalized version of the cone logarithm. For $r > 0$ define

$$N_r(\xi,\zeta) \coloneqq \left(\xi, \frac{2\zeta}{r}\right).$$

For every reduced cone pair $(\mathfrak{y}_0, \mathfrak{y}_1)$, we define the normalized cone logarithm

$$\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1) = (\mathcal{L}^T_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1), \mathcal{L}^R_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)) \coloneqq \begin{cases} N_r\big(\mathrm{Log}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)\big), & \mathfrak{y}_0 = [x,r],\ r > 0, \\ (0,0), & \mathfrak{y}_0 = \mathfrak{o}. \end{cases}$$

By the formulas of Section 2.3, for $r, s > 0$ and $\|y - x\| \le \pi/2$, the normalized cone logarithm satisfies

$$\mathcal{L}_{\mathfrak{C}}([x, r], [y, s]) = \left(\frac{s}{r}\mathbf{sin}(y - x), 2\left(\frac{s}{r}\cos(\|y - x\|) - 1\right)\right), \tag{28}$$

whereas $\mathcal{L}_{\mathfrak{C}}([x, r], \mathfrak{o}) = (0, -2)$. Then, on $\{a > 0,\ \|x - y\| < \pi/2\}$,

$$\mathrm{Log}_{\boldsymbol{\gamma},u}(x, y) = \mathcal{L}_{\mathfrak{C}}\left([x, \sqrt{a(x, y)}], [y, \sqrt{b(x, y)}]\right), \tag{29}$$

and the same identity holds at the cutoff $\|y - x\| = \pi/2$ when $u(x, y) = 1$.

The definition is independent of the dominating measure λ, since the ratio b/a is unchanged γ_0-a.e.; this follows by comparing any two choices through a common dominating measure. Moreover, the normalized cone logarithm is jointly 0-homogeneous in the radial variables:

$$\mathcal{L}_{\mathfrak{C}}([x, \theta r], [y, \theta s]) = \mathcal{L}_{\mathfrak{C}}([x, r], [y, s]), \qquad \theta > 0.$$

The dilation identities and the resulting normalization of finite cone-geodesic measures are collected in Lemma B.1.

Finally, the definition is only needed γ_0-a.e. Since $d\gamma_0 = a\, d\lambda$, one has $\gamma_0(\{a = 0\}) = 0$; on this set the source cone point $[x, \sqrt{a}]$ is the vertex $\mathfrak{o}$, and any value assigned to the logarithm is therefore irrelevant.

Remark 3.4 (On the role of the selection u). Let $a, b > 0$ and $x, y \in \mathbb{R}^d$ satisfy $\|y - x\| = \pi/2$. By the cone-geodesic formulas in Section 2.3, the curves

$$\boldsymbol{\mu}_t^T \coloneqq m_t \delta_{x_t}, \qquad \boldsymbol{\mu}_t^R \coloneqq (1 - t)^2 a \delta_x + t^2 b \delta_y,$$

where

$$m_t \coloneqq (1 - t)^2 a + t^2 b, \qquad x_t \coloneqq x + \mathrm{Arg}\big((1 - t)\sqrt{a} + \mathrm{i}t\sqrt{b}\big)\frac{(y - x)}{\|y - x\|},$$

are constant-speed WFR geodesics between $\mu \coloneqq a\delta_x$ and $\nu \coloneqq b\delta_y$. Indeed, they arise from the optimal cone couplings

$$A^T \coloneqq \delta_{([x,\sqrt{a}],[y,\sqrt{b}])}, \qquad A^R \coloneqq \delta_{([x,\sqrt{a}],\mathfrak{o})} + \delta_{(\mathfrak{o},[y,\sqrt{b}])},$$

both of cost $a + b = \mathrm{WFR}^2(\mu, \nu)$, by Theorem 2.7. The first curve is a traveling Dirac mass, whereas the second consists of two static Dirac masses with changing weights.

For each $u \in [0, 1]$, the coupling $uA^T + (1 - u)A^R$ is optimal as well, so the same theorem shows that $\boldsymbol{\mu}_t^u \coloneqq u\boldsymbol{\mu}_t^T + (1 - u)\boldsymbol{\mu}_t^R$ is a WFR geodesic. Differentiating smooth expectations at $t = 0$ gives its regular barycentric logarithm

$$\mathcal{V}^u(x) = \left(u\sqrt{\frac{b}{a}}\frac{(y - x)}{\|y - x\|}, -2\right).$$

Thus u specifies the proportion assigned to the traveling geodesic: it scales the transport component, while the reaction component is unchanged. The measurable selection in Definition 3.2 makes this choice separately at each pair on the cut locus.

Barycentric semicoupling logarithms belong to $H(\mu)$, and their squared norm is bounded by the semicoupling cost.

Lemma 3.5. *Let $\boldsymbol{\gamma} \in \mathrm{SC}(\mu, \nu)$ and let $u\colon \mathbb{R}^d \times \mathbb{R}^d \to [0, 1]$ be measurable. Then $\overline{\mathrm{Log}}_{\boldsymbol{\gamma},u} \in H(\mu)$ and*

$$\big\|\overline{\mathrm{Log}}_{\boldsymbol{\gamma},u}\big\|_\mu^2 \le \int_{\mathbb{R}^d \times \mathbb{R}^d} \big\|\mathrm{Log}_{\boldsymbol{\gamma},u}(x, y)\big\|_{\mathrm{WFR}}^2\, d\gamma_0(x, y) \le \mathcal{C}(\boldsymbol{\gamma}). \tag{30}$$

Proof. Let λ dominate γ_0 and γ_1, and write $a = d\gamma_0/d\lambda$ and $b = d\gamma_1/d\lambda$. On $\{a > 0\}$, a direct computation gives

$$a\big\|\mathrm{Log}_{\boldsymbol{\gamma},u}(x,y)\big\|^2_{\mathrm{WFR}} = \begin{cases} a + b - 2\sqrt{ab}\cos(\|x-y\|), & \|x-y\| < \pi/2, \\ a + u(x,y)^2 b, & \|x-y\| = \pi/2, \\ a, & \|x-y\| > \pi/2. \end{cases}$$

Since $0 \le u \le 1$, these expressions are bounded above by $c(x, y, a, b)$. Extending the logarithm by zero on $\{a = 0\}$ and integrating against λ proves the second inequality in (30). The first follows from Jensen's inequality applied to the disintegration $\gamma_0(dx, dy) = \mu(dx)\kappa_x(dy)$. □

The first inequality in (30) reflects barycentric averaging. The second reflects mass creation and, at the cutoff, the attenuation of the transport component by u. The following example isolates the first two effects and shows that both inequalities can be strict even for an optimal semicoupling.

Example 3.6. Work in dimension one. Fix $0 < \rho < \pi/2$, $R > \pi/2$, and $m > 0$, and set

$$\mu = \delta_0, \qquad \nu = \frac{1}{2}\delta_{-\rho} + \frac{1}{2}\delta_{\rho} + m\delta_R.$$

Consider the semicoupling $\boldsymbol{\gamma}^\star := (\gamma_0^\star, \gamma_1^\star)$ with

$$\gamma_0^\star = \frac{1}{2}\delta_{(0,-\rho)} + \frac{1}{2}\delta_{(0,\rho)}, \qquad \gamma_1^\star = \gamma_0^\star + m\delta_{(0,R)}.$$

This semicoupling is optimal. Indeed, let $\boldsymbol{\gamma} = (\gamma_0, \gamma_1) \in \mathrm{SC}(\mu, \nu)$ and set

$$a_\pm := \gamma_0(\{(0, \pm\rho)\}), \qquad b_\pm := \gamma_1(\{(0, \pm\rho)\}).$$

The atom at R has zero interaction with the source, while the marginal constraints yield $a_- + a_+ \le 1$ and $b_\pm \le 1/2$. By (2), minimizing the cost amounts to maximizing the interaction term

$$\int_{\mathbb{R}\times\mathbb{R}} \sqrt{a(x,y)b(x,y)}\cos\left(|y-x| \wedge \frac{\pi}{2}\right) d\lambda(x,y) = \cos(\rho)\left(\sqrt{a_-b_-} + \sqrt{a_+b_+}\right),$$

where $a = d\gamma_0/d\lambda$ and $b = d\gamma_1/d\lambda$ for a common dominating measure λ. By Cauchy–Schwarz,

$$\cos(\rho)\left(\sqrt{a_-b_-} + \sqrt{a_+b_+}\right) \le \cos(\rho)\sqrt{(a_- + a_+)(b_- + b_+)} \le \cos(\rho),$$

with equality attained by $\boldsymbol{\gamma}^\star$. Consequently,

$$\mathrm{WFR}^2(\mu, \nu) = \mathcal{C}(\boldsymbol{\gamma}^\star) = 2(1 - \cos(\rho)) + m.$$

The selection u plays no role in this example, and the logarithm at the two atoms of $\gamma_0^\star$ is

$$\mathrm{Log}_{\boldsymbol{\gamma}^\star,u}(0, \pm\rho) = \big(\pm\sin(\rho),\, 2(\cos(\rho) - 1)\big).$$

Its barycenter is therefore

$$\overline{\mathrm{Log}}_{\boldsymbol{\gamma}^\star,u}(0) = \big(0,\, 2(\cos(\rho) - 1)\big).$$

In particular,

$$\big\|\overline{\mathrm{Log}}_{\boldsymbol{\gamma}^\star,u}\big\|^2_\mu = (1 - \cos(\rho))^2, \qquad \int_{\mathbb{R}\times\mathbb{R}} \big\|\mathrm{Log}_{\boldsymbol{\gamma}^\star,u}\big\|^2_{\mathrm{WFR}}\, d\gamma_0^\star = 2(1 - \cos(\rho)).$$

Thus the first inequality in (30) loses exactly $\sin^2(\rho)$ through the cancellation of the two transport components. The second loses exactly m, the cost of creating the atom at R, which is invisible to the source-supported logarithm.

We can now state our main characterization theorem. For this purpose, we introduce the barycentric field associated with an optimal cone-geodesic representation.

Let $\boldsymbol{\mu}$ be a constant-speed WFR geodesic connecting μ to ν, and let $\mathbf{\Pi} \in \mathcal{P}(\mathrm{Geo}(\mathfrak{C}))$ be an optimal representation as in Theorem 2.7. Writing $e_t(\mathfrak{y}) = [\boldsymbol{x}_t, \boldsymbol{r}_t]$, define the finite measure

$$dQ^{\mathbf{\Pi}}(\mathfrak{y}) \coloneqq \boldsymbol{r}_0^2 \, d\mathbf{\Pi}(\mathfrak{y}).$$

Since $(\mathsf{x}_0)_{\#} Q^{\mathbf{\Pi}} = \mu$, we may disintegrate

$$Q^{\mathbf{\Pi}} = \int_{\mathbb{R}^d} Q_x^{\mathbf{\Pi}} \, d\mu(x),$$

where $Q_x^{\mathbf{\Pi}}$ is a probability measure concentrated on $\mathsf{x}_0^{-1}(\{x\})$ for μ-a.e. x. We define the corresponding barycentric field by

$$\mathcal{V}^{\mathbf{\Pi}}(x) \coloneqq \int_{\mathrm{Geo}(\mathfrak{C})} \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_1) \, dQ_x^{\mathbf{\Pi}}(\mathfrak{y}). \tag{31}$$

Theorem 3.7 (Characterization of regular barycentric logarithms)**.** *Let $\mu, \nu \in \mathcal{M}\big(\mathbb{R}^d\big)$. The set of regular barycentric logarithms between μ and ν is*

$$\left\{ \overline{\mathrm{Log}}_{\boldsymbol{\gamma},u} : \boldsymbol{\gamma} \in \mathrm{OptSC}(\mu,\nu),\ u : \mathbb{R}^d \times \mathbb{R}^d \to [0,1] \text{ measurable} \right\}. \tag{32}$$

Moreover, every constant-speed WFR geodesic connecting μ to ν admits a unique regular barycentric logarithm. For every optimal cone-geodesic representation $\mathbf{\Pi}$ of that geodesic, this logarithm equals $\mathcal{V}^{\mathbf{\Pi}}$. In particular, $\mathcal{V}^{\mathbf{\Pi}}$ is independent, up to μ-a.e. equality, of the optimal representation.

Every regular barycentric logarithm $\mathcal{V}$ between μ and ν satisfies

$$\|\mathcal{V}\|_{\mu} \le \mathrm{WFR}(\mu,\nu).$$

Proof. Let $\boldsymbol{\gamma} \in \mathrm{OptSC}(\mu,\nu)$ and let $u : \mathbb{R}^d \times \mathbb{R}^d \to [0,1]$ be measurable. By Proposition 3.9, there exists an optimal cone-geodesic representation $\mathbf{\Pi}$ such that

$$\mathcal{V}^{\mathbf{\Pi}} = \overline{\mathrm{Log}}_{\boldsymbol{\gamma},u}.$$

Its homogeneous marginals form a constant-speed WFR geodesic connecting μ to ν, and Proposition 3.8 gives the initial derivative identity (26). By Theorem 3.11, the field $\overline{\mathrm{Log}}_{\boldsymbol{\gamma},u}$ belongs to $T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$. Hence it is a regular barycentric logarithm.

Conversely, let $\mathcal{V} \in T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$ be a regular barycentric logarithm associated with a constant-speed WFR geodesic $\boldsymbol{\mu}$. Choose an optimal cone-geodesic representation $\mathbf{\Pi}$ of $\boldsymbol{\mu}$ using Theorem 2.7. By Proposition 3.10, there exist $\boldsymbol{\gamma} \in \mathrm{OptSC}(\mu,\nu)$ and a measurable selection $u\colon \mathbb{R}^d \times \mathbb{R}^d \to [0,1]$ such that

$$\mathcal{V}^{\mathbf{\Pi}} = \overline{\mathrm{Log}}_{\boldsymbol{\gamma},u}.$$

Theorem 3.11 implies that $\mathcal{V}^{\mathbf{\Pi}} \in T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$. Moreover, by Proposition 3.8, both $\mathcal{V}$ and $\mathcal{V}^{\mathbf{\Pi}}$ represent the same initial derivative. Therefore

$$\big\langle \mathcal{V} - \mathcal{V}^{\mathbf{\Pi}}, (\nabla\varphi, 4\varphi) \big\rangle_{\mu} = 0 \qquad \text{for every } \varphi \in C_c^\infty(\mathbb{R}^d).$$

Since $\mathcal{V} - \mathcal{V}^{\mathbf{\Pi}} \in T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$, the definition of the tangent space gives $\mathcal{V} = \mathcal{V}^{\mathbf{\Pi}}$ in $H(\mu)$. This proves (32).

The same argument applies to every constant-speed WFR geodesic connecting μ to ν and to each of its optimal cone-geodesic representations. It proves both uniqueness of the associated regular barycentric logarithm and independence of $\mathcal{V}^{\mathbf{\Pi}}$ from the representation.

Finally, the norm bound follows from Proposition 3.8. □

3.2 Representation by cone geodesics

We prove the representation results used in the preceding Theorem 3.7, starting with the initial derivative along a WFR geodesic.

Proposition 3.8. *Let $\boldsymbol{\mu}$ be a constant-speed WFR geodesic connecting μ and ν, and let $\mathbf{\Pi} \in \mathcal{P}(\mathrm{Geo}(\mathfrak{C}))$ be an optimal representation as in Theorem 2.7. Then $\mathcal{V}^{\mathbf{\Pi}} := (\xi^{\mathbf{\Pi}}, \zeta^{\mathbf{\Pi}}) \in H(\mu)$ satisfies*

$$\left\|\mathcal{V}^{\mathbf{\Pi}}\right\|_{\mu}^2 \leq \mathrm{WFR}^2(\mu,\nu)$$

and, for every $\varphi \in C_c^\infty(\mathbb{R}^d)$,

$$\lim_{t\downarrow 0} \frac{\int_{\mathbb{R}^d} \varphi \, d\boldsymbol{\mu}_t - \int_{\mathbb{R}^d} \varphi \, d\mu}{t} = \int_{\mathbb{R}^d} \left(\nabla\varphi \cdot \xi^{\mathbf{\Pi}} + \varphi\zeta^{\mathbf{\Pi}}\right) d\mu. \tag{33}$$

In particular, if $\mathcal{V}^{\mathbf{\Pi}} \in T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$, then it is the regular barycentric logarithm associated with $\boldsymbol{\mu}$.

Proof. We first verify square integrability. By the cone logarithmic energy identity (20), on $\{\boldsymbol{r}_0 > 0\}$,

$$\boldsymbol{r}_0^2 \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)\|_{\mathrm{WFR}}^2 = d_{\mathfrak{C}}^2(\mathfrak{y}_0,\mathfrak{y}_1).$$

Therefore,

$$\int_{\mathrm{Geo}(\mathfrak{C})} \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)\|_{\mathrm{WFR}}^2 \, dQ^{\mathbf{\Pi}} = \int_{\{\boldsymbol{r}_0>0\}} d_{\mathfrak{C}}^2(\mathfrak{y}_0,\mathfrak{y}_1) \, d\mathbf{\Pi} \leq \mathrm{WFR}^2(\mu,\nu).$$

Conditional Jensen's inequality proves that $\mathcal{V}^{\mathbf{\Pi}} \in H(\mu)$ and gives the stated norm bound.

Fix $\varphi \in C_c^\infty(\mathbb{R}^d)$ and write $\mathfrak{y}_t = [\boldsymbol{x}_t, \boldsymbol{r}_t]$. By (16),

$$\frac{\int_{\mathbb{R}^d} \varphi \, d\boldsymbol{\mu}_t - \int_{\mathbb{R}^d} \varphi \, d\mu}{t} = \int_{\mathrm{Geo}^*(\mathfrak{C})} \frac{\boldsymbol{r}_t^2\varphi(\boldsymbol{x}_t) - \boldsymbol{r}_0^2\varphi(\boldsymbol{x}_0)}{t} \, d\mathbf{\Pi}(\mathfrak{y}). \tag{34}$$

By (21), the absolute value of the integrand is bounded by $B_\varphi(\boldsymbol{r}_0^2 + \boldsymbol{r}_1^2)$. This bound is integrable: since $(e_0, e_1)_{\#}\mathbf{\Pi} \in \Gamma_{\mathfrak{C}}(\mu,\nu)$,

$$\int (\boldsymbol{r}_0^2 + \boldsymbol{r}_1^2) \, d\mathbf{\Pi}(\mathfrak{y}) = \mu(\mathbb{R}^d) + \nu(\mathbb{R}^d) < \infty.$$

From Lemma 2.9, the pointwise limit of the difference quotient is

$$\boldsymbol{r}_0^2 \left(\nabla\varphi(\boldsymbol{x}_0) \cdot \mathcal{L}_{\mathfrak{C}}^T(\mathfrak{y}_0,\mathfrak{y}_1) + \varphi(\boldsymbol{x}_0)\mathcal{L}_{\mathfrak{C}}^R(\mathfrak{y}_0,\mathfrak{y}_1)\right)$$

on $\{\boldsymbol{r}_0 > 0\}$, and zero on $\{\boldsymbol{r}_0 = 0\}$. Dominated convergence therefore yields

$$\begin{aligned}\lim_{t\downarrow 0} \frac{\int_{\mathbb{R}^d} \varphi \, d\boldsymbol{\mu}_t - \int_{\mathbb{R}^d} \varphi \, d\mu}{t} &= \int_{\mathrm{Geo}(\mathfrak{C})} \boldsymbol{r}_0^2 \left(\nabla\varphi(\boldsymbol{x}_0) \cdot \mathcal{L}_{\mathfrak{C}}^T(\mathfrak{y}_0,\mathfrak{y}_1) + \varphi(\boldsymbol{x}_0)\mathcal{L}_{\mathfrak{C}}^R(\mathfrak{y}_0,\mathfrak{y}_1)\right) d\mathbf{\Pi}(\mathfrak{y}) \\ &= \int_{\mathbb{R}^d} \left(\nabla\varphi \cdot \xi^{\mathbf{\Pi}} + \varphi\zeta^{\mathbf{\Pi}}\right) d\mu,\end{aligned}$$

where the last equality follows from disintegration. □

If $\mathbf{\Pi}$ is an optimal cone-geodesic representation and $\mu = \mathfrak{h}_0(\mathbf{\Pi})$, its barycentric field $\mathcal{V}^{\mathbf{\Pi}}$ satisfies

$$\int_{\mathbb{R}^d} \psi(x)\mathcal{V}^{\mathbf{\Pi}}(x)\, d\mu(x) = \int_{\mathrm{Geo}^*(\mathfrak{C})} \psi(\boldsymbol{x}_0)\boldsymbol{r}_0^2 \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_1)\, d\mathbf{\Pi}(\mathfrak{y}) \tag{35}$$

for every bounded Borel function $\psi : \mathbb{R}^d \to \mathbb{R}$. The integrals are vector-valued. This identity follows directly from the source-weighted disintegration defining $\mathcal{V}^{\mathbf{\Pi}}$.

We now compare the semicoupling construction with the barycentric fields associated with optimal cone-geodesic representations.

Both constructions below use the identity

$$r^2\mathcal{L}_{\mathfrak{C}}([x,r],[y,s]) = \Big(rs\mathbf{sin}(y-x),\, 2rs\cos\Big(\|y-x\| \wedge \frac{\pi}{2}\Big) - 2r^2\Big) \tag{36}$$

for reduced cone pairs. It follows from (28) when both radii are positive, and also holds for annihilation and creation, with the source-weighted expression set to zero when $r = 0$. In particular, above the cutoff reducedness gives $rs = 0$.

Proposition 3.9 (Realization of semicoupling logarithms)**.** *Let $\mu, \nu \in \mathcal{M}\big(\mathbb{R}^d\big)$, let $\boldsymbol{\gamma} \in \mathrm{OptSC}(\mu,\nu)$, and let $u\colon \mathbb{R}^d \times \mathbb{R}^d \to [0,1]$ be measurable. There exists*

$$\mathbf{\Pi} \in \mathcal{P}(\mathrm{Geo}^*(\mathfrak{C}))$$

such that

$$(e_0, e_1)_{\#}\mathbf{\Pi} \in \mathrm{Opt}\Gamma_{\mathfrak{C}}(\mu,\nu)$$

and

$$\mathcal{V}^{\mathbf{\Pi}} = \overline{\mathrm{Log}}_{\boldsymbol{\gamma},u} \qquad \mu\text{-a.e.} \tag{37}$$

In particular, $\overline{\mathrm{Log}}_{\boldsymbol{\gamma},u}$ satisfies (26) along $\boldsymbol{\mu}_t \coloneqq \mathfrak{h}_t(\mathbf{\Pi})$.

Proof. Choose a finite positive measure λ dominating both legs of $\boldsymbol{\gamma}$, and write

$$\gamma_0 = a\lambda, \qquad \gamma_1 = b\lambda.$$

We fix finite nonnegative measurable versions of a, b.

For each (x,y), define a finite measure on $[0,\infty)^2$ by

$$\eta_{x,y} \coloneqq \begin{cases} \delta_{(\sqrt{a},\sqrt{b})}, & \|x-y\| < \pi/2, \\ u\,\delta_{(\sqrt{a},\sqrt{b})} + (1-u)\big[\delta_{(\sqrt{a},0)} + \delta_{(0,\sqrt{b})}\big], & \|x-y\| = \pi/2, \\ \delta_{(\sqrt{a},0)} + \delta_{(0,\sqrt{b})}, & \|x-y\| > \pi/2, \end{cases} \tag{38}$$

where a, b, u are evaluated at (x,y). This is a measurable kernel whose total mass is finite. Its radial moments satisfy

$$\int r^2\, d\eta_{x,y}(r,s) = a(x,y), \qquad \int s^2\, d\eta_{x,y}(r,s) = b(x,y). \tag{39}$$

Define a finite measure $\widetilde{A}$ on $\big(\mathbb{R}^d \times [0,\infty)\big)^2$ by

$$\widetilde{A}(dx, dr, dy, ds) \coloneqq \lambda(dx,dy)\eta_{x,y}(dr,ds),$$

and let $A \coloneqq (\mathfrak{p}^{\otimes 2})_{\#}\widetilde{A}$. By (39), its homogeneous marginals are μ, ν, and it is concentrated on reduced pairs. The mixed radial moment is

$$m(x,y) \coloneqq \int rs\, d\eta_{x,y}(r,s) = \begin{cases} \sqrt{ab}, & \|x-y\| < \pi/2, \\ u\sqrt{ab}, & \|x-y\| = \pi/2, \\ 0, & \|x-y\| > \pi/2. \end{cases} \tag{40}$$

Set $k(x,y) \coloneqq \cos(\|y-x\| \wedge \pi/2)$. Expanding the cone cost and integrating in the radial variables gives

$$\int_{\mathfrak{C}^2} \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_1)\, dA = \int_{\mathbb{R}^{2d}} (a+b-2mk)\, d\lambda = \mathcal{C}(\boldsymbol{\gamma}) = \mathrm{WFR}^2(\mu,\nu), \tag{41}$$

where we used (40) and the fact that $k = 0$ at and above the cutoff. Thus $A \in \mathrm{Opt}\Gamma_{\mathfrak{C}}(\mu,\nu)$.

Similarly, integrating (36) gives

$$\int r^2 \mathcal{L}_{\mathfrak{C}}([x,r],[y,s])\, d\eta_{x,y}(r,s) = \big(m\mathbf{sin}(y-x), 2(mk-a)\big) = a(x,y)\,\mathrm{Log}_{\boldsymbol{\gamma},u}(x,y), \tag{42}$$

where the right-hand side is set to zero when $a = 0$. Indeed, below the cutoff $m = \sqrt{ab}$, at the cutoff $m = u\sqrt{ab}$ and $k = 0$, and above it $m = k = 0$. This also accounts for the radial annihilation and creation contributions. Set $\widetilde{\boldsymbol{\Pi}} \coloneqq G_{\#}A$. By (42) and the disintegration of γ_0, for every bounded Borel function ψ we obtain

$$\int_{\mathrm{Geo}(\mathfrak{C})} \psi(\boldsymbol{x}_0)\boldsymbol{r}_0^2 \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)\, d\widetilde{\boldsymbol{\Pi}} = \int_{\mathbb{R}^{2d}} \psi(x)a(x,y)\,\mathrm{Log}_{\boldsymbol{\gamma},u}(x,y)\, d\lambda = \int_{\mathbb{R}^d} \psi\, \overline{\mathrm{Log}}_{\boldsymbol{\gamma},u}\, d\mu. \tag{43}$$

All integrals are finite, since $m \le \sqrt{ab}$ and $\int (a+b)\, d\lambda < \infty$. By Lemma B.1, we may normalize $\widetilde{\boldsymbol{\Pi}}$ to a probability measure $\boldsymbol{\Pi}$ without changing its homogeneous time marginals, endpoint cost, or the source-weighted integrals in (43). It remains concentrated on $\mathrm{Geo}^*(\mathfrak{C})$, and its endpoint coupling is optimal. Identity (35) then proves (37). Theorem 2.7 and Proposition 3.8 give the asserted geodesic and initial derivative identity. □

Conversely, we recover an optimal semicoupling and a selection u from an optimal cone-geodesic representation. The same radial moment identities apply, but now optimality forces saturation of the mixed-moment bound below the cutoff.

Proposition 3.10 (Semicoupling representation of cone barycenters)**.** *Let $\mu, \nu \in \mathcal{M}(\mathbb{R}^d)$ and let $\boldsymbol{\Pi} \in \mathcal{P}(\mathrm{Geo}(\mathfrak{C}))$ satisfy*

$$A \coloneqq (e_0, e_1)_{\#}\boldsymbol{\Pi} \in \mathrm{Opt}\Gamma_{\mathfrak{C}}(\mu,\nu).$$

Then there exist $\boldsymbol{\gamma} \in \mathrm{OptSC}(\mu,\nu)$ and a measurable function $u\colon \mathbb{R}^d \times \mathbb{R}^d \to [0,1]$ such that

$$\mathcal{V}^{\boldsymbol{\Pi}} = \overline{\mathrm{Log}}_{\boldsymbol{\gamma},u} \qquad \mu\text{-a.e.} \tag{44}$$

Proof. The endpoint coupling A is concentrated on reduced pairs. We lift A to a probability measure $\widetilde{A} \coloneqq (\mathfrak{i}_0^{\otimes 2})_{\#}A$ on $\big(\mathbb{R}^d \times [0,\infty)\big)^2$.

Let λ be the (x,y)-marginal of $\widetilde{A}$. Disintegrate

$$\widetilde{A}(dx,dr,dy,ds) = \lambda(dx,dy)\eta_{x,y}(dr,ds), \tag{45}$$

where $\eta_{x,y}$ is a probability measure for λ-a.e. (x,y). Define

$$a \coloneqq \int r^2\, d\eta_{x,y}, \qquad b \coloneqq \int s^2\, d\eta_{x,y}, \qquad m \coloneqq \int rs\, d\eta_{x,y}. \tag{46}$$

These functions are measurable and finite λ-a.e., since

$$\int (a+b)\, d\lambda = \mu(\mathbb{R}^d) + \nu(\mathbb{R}^d) < \infty.$$

Cauchy–Schwarz in $L^2(\eta_{x,y})$ gives

$$0 \le m \le \sqrt{ab} \qquad \lambda\text{-a.e.} \tag{47}$$

Set $\gamma_0 := a\lambda$ and $\gamma_1 := b\lambda$. The homogeneous marginal constraints on A show that $\boldsymbol{\gamma} = (\gamma_0, \gamma_1) \in \mathrm{SC}(\mu, \nu)$.

For brevity, write

$$k(x,y) := \cos\left(\|y-x\| \wedge \pi/2\right).$$

Since A is reduced, its coordinate lift satisfies $rs = 0$ almost everywhere on $\{\|y-x\| > \pi/2\}$. Consequently, $m = 0$ λ-a.e. on $\{\|y-x\| > \pi/2\}$ and, using (45), its optimal cone cost can be written as

$$\mathrm{WFR}^2(\mu,\nu) = \int (a+b-2mk)\, d\lambda.$$

Using (47) and the semicoupling formulation of WFR, we obtain

$$\mathrm{WFR}^2(\mu,\nu) \le \mathcal{C}(\boldsymbol{\gamma}) = \int (a+b-2\sqrt{ab}\,k)\, d\lambda \le \int (a+b-2mk)\, d\lambda = \mathrm{WFR}^2(\mu,\nu).$$

Thus $\boldsymbol{\gamma} \in \mathrm{OptSC}(\mu,\nu)$ and

$$\int k(\sqrt{ab} - m)\, d\lambda = 0.$$

The integrand is nonnegative. Since $k > 0$ below the cutoff, we conclude that

$$m = \sqrt{ab} \qquad \lambda\text{-a.e. on } \{\|y-x\| < \pi/2\}. \tag{48}$$

Define the measurable selection

$$u(x,y) := \begin{cases} \dfrac{m(x,y)}{\sqrt{a(x,y)b(x,y)}}, & \|y-x\| = \pi/2,\ a(x,y)b(x,y) > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{49}$$

By (47), it takes values in $[0,1]$.

For $a(x,y) > 0$, integrating (36) gives

$$\begin{aligned} \ell(x,y) &:= \frac{1}{a(x,y)} \int r^2 \mathcal{L}_{\mathfrak{C}}([x,r],[y,s])\, d\eta_{x,y}(r,s) \\ &= \left(\frac{m}{a}\mathbf{sin}(y-x),\, 2\left(\frac{m}{a}k - 1\right)\right). \end{aligned} \tag{50}$$

Below the cutoff, (48) identifies this with $\mathrm{Log}_{\boldsymbol{\gamma},u}$. At the cutoff, the same follows from (49); if $b = 0$, then $m = 0$ and both expressions equal $(0,-2)$. Above the cutoff, $m = k = 0$, so they again equal $(0,-2)$. Thus $\ell = \mathrm{Log}_{\boldsymbol{\gamma},u}$ γ_0-a.e. Set $\ell = 0$ on $\{a = 0\}$.

By (35) and the disintegrations of $\widetilde{A}$ and γ_0, for every bounded Borel function ψ,

$$\begin{aligned} \int_{\mathbb{R}^d} \psi\, \mathcal{V}^{\boldsymbol{\Pi}}\, d\mu &= \int_{\mathbb{R}^{2d}} \psi(x) a(x,y) \ell(x,y)\, d\lambda \\ &= \int_{\mathbb{R}^d} \psi\, \overline{\mathrm{Log}}_{\boldsymbol{\gamma},u}\, d\mu \end{aligned}$$

and all integrals are finite by the logarithmic energy bounds. This proves (44). □

3.3 Endpoint tangency

Theorem 3.11 (Tangency of optimal semicoupling logarithms)**.** *Let $\mu, \nu \in \mathcal{M}(\mathbb{R}^d)$, $\boldsymbol{\gamma} \in \mathrm{OptSC}(\mu, \nu)$, and let $u\colon \mathbb{R}^d \times \mathbb{R}^d \to [0,1]$ be measurable. Then*

$$\overline{\mathrm{Log}}_{\boldsymbol{\gamma},u} \in T_\mu \mathcal{M}(\mathbb{R}^d).$$

Proof. Set $\mathcal{L} \coloneqq \overline{\mathrm{Log}}_{\boldsymbol{\gamma},u}$. By Proposition 3.9, there exists $\boldsymbol{\Pi} \in \mathcal{P}(\mathrm{Geo}^*(\mathfrak{C}))$ such that

$$A \coloneqq (e_0, e_1)_{\#}\boldsymbol{\Pi} \in \mathrm{Opt}\Gamma_{\mathfrak{C}}(\mu, \nu), \qquad \mathcal{V}^{\boldsymbol{\Pi}} = \mathcal{L}.$$

Proposition 3.8 gives $\mathcal{L} \in H(\mu)$. Moreover, (35) identifies $\mathcal{L}$ as the source-weighted barycenter of the normalized cone logarithm $\mathcal{L}_{\mathfrak{C}}$.

Let $\mathcal{W} = (w, \eta) \in T_\mu \mathcal{M}(\mathbb{R}^d)^\perp$. Thus

$$\int_{\mathbb{R}^d} (w \cdot \nabla\varphi + \eta\varphi)\, d\mu = 0 \qquad \text{for every } \varphi \in C_c^\infty(\mathbb{R}^d). \tag{51}$$

It suffices to prove $\langle \mathcal{L}, \mathcal{W} \rangle_\mu = 0$.

The orthogonality relation (51) means that $\mathcal{W}$ generates a stationary solution of the inhomogeneous continuity equation. After restricting to a bounded domain, we use the superposition principle of [4] to represent this solution by characteristics and choose an interior time τ at which almost every characteristic with positive radius has the prescribed derivatives. We then normalize the positive first radii of A to one and glue its source coordinate to the characteristics at time τ. Moving along these characteristics perturbs the first cone endpoint; a correction accounts for paths whose radius vanishes at τ and restores the first homogeneous marginal exactly. This yields competitors $A_\varepsilon \in \Gamma_{\mathfrak{C}}(\mu, \nu)$ for both signs of ε, with $A_0 = A$ after normalization.

Optimality gives a local minimum of the competitor cost at zero. To differentiate it, we use the stationary energy identity to obtain a uniform L^2 bound on the interaction difference quotients, then truncate the possibly unbounded target radius. Finally, we identify the derivative with $-2\langle \mathcal{L}, \mathcal{W} \rangle_\mu$, which proves the required orthogonality.

Restriction to bounded domains: By Lemma C.4 and continuity of the pairing in $H(\mu)$, we may assume that both components of $\mathcal{W}$ vanish μ-a.e. outside B_R, for some $R > 0$. We therefore choose finite-valued Borel representatives that vanish everywhere outside B_R.

Let $\Omega = \overline{B}_{2R}$ and $\widetilde{\mu} \coloneqq \mu \llcorner \Omega$. Since

$$w\widetilde{\mu} = w\mu, \qquad \eta\widetilde{\mu} = \eta\mu,$$

equation (51) gives

$$\mathrm{div}(w\widetilde{\mu}) = \eta\widetilde{\mu} \qquad \text{in } \mathcal{D}'(\mathbb{R}^d). \tag{52}$$

Consequently, the constant curve $\widetilde{\boldsymbol{\mu}}_t \equiv \widetilde{\mu}$ solves the inhomogeneous continuity equation on Ω. The equation holds with no boundary flux, since both coefficients vanish on a neighborhood of $\partial\Omega$. Testing (51) with a smooth function equal to one on $\overline{B}_R$ also gives

$$\int_{\mathbb{R}^d} \eta\, d\mu = \int_\Omega \eta\, d\widetilde{\mu} = 0. \tag{53}$$

If $\widetilde{\mu} = 0$, then $\mathcal{W} = 0$ in $H(\mu)$ and there is nothing to prove. Henceforth assume $\widetilde{\mu} \neq 0$.

Let

$$\mathcal{A}_\Omega \coloneqq \{r^2\delta_x : r \geq 0,\ x \in \Omega\} \subset \mathcal{M}^+(\Omega) \quad \text{and} \quad \mathbf{C}_\Omega \coloneqq C([0,1]; \mathcal{A}_\Omega),$$

where $\mathcal{A}_\Omega$ is endowed with the bounded-Lipschitz distance and $\mathbf{C}_\Omega$ with the corresponding uniform distance. For $\boldsymbol{\omega} \in \mathbf{C}_\Omega$, write

$$\boldsymbol{\omega}_t = \boldsymbol{r}_t(\boldsymbol{\omega})^2 \delta_{\boldsymbol{x}_t(\boldsymbol{\omega})},$$

where $\boldsymbol{r}_t(\boldsymbol{\omega}) \geq 0$, $\boldsymbol{x}_t(\boldsymbol{\omega}) \in \Omega$ whenever $\boldsymbol{r}_t(\boldsymbol{\omega}) > 0$, and we set $\boldsymbol{x}_t(\boldsymbol{\omega}) = 0$ when $\boldsymbol{r}_t(\boldsymbol{\omega}) = 0$, consistently with the choice of $\mathfrak{i}_0$ as a right inverse of $\mathfrak{p}$.

We further set

$$\mathbf{S}_\Omega \coloneqq \left\{ \boldsymbol{\omega} \in \mathbf{C}_\Omega : \boldsymbol{r}_\cdot(\boldsymbol{\omega}) \in AC^2([0,1]),\ \boldsymbol{r}_\cdot(\boldsymbol{\omega})\boldsymbol{x}_\cdot(\boldsymbol{\omega}) \in AC^2([0,1];\mathbb{R}^d) \right\}.$$

For $\boldsymbol{\omega} \in \mathbf{S}_\Omega$, consider the characteristic equations

$$\dot{\boldsymbol{x}}_t(\boldsymbol{\omega}) = w(\boldsymbol{x}_t(\boldsymbol{\omega})), \qquad \dot{\boldsymbol{r}}_t(\boldsymbol{\omega}) = \frac{1}{2}\eta(\boldsymbol{x}_t(\boldsymbol{\omega}))\boldsymbol{r}_t(\boldsymbol{\omega}), \tag{54}$$

and define

$$\mathbf{S}_\Omega^{w,\eta} \coloneqq \{ \boldsymbol{\omega} \in \mathbf{S}_\Omega : (54) \text{ holds for a.e. } t \in \{\boldsymbol{r}_\cdot(\boldsymbol{\omega}) > 0\} \},$$

together with

$$\mathcal{S} \coloneqq \left\{ \boldsymbol{\omega} \in \mathbf{S}_\Omega^{w,\eta} : \int_0^1 \boldsymbol{r}_t(\boldsymbol{\omega})^2 \, dt = 1 \right\}.$$

By [4, Theorem 4.3 and definition (32)], there exists a finite positive Borel measure σ on $\mathcal{S}$ such that

$$\widetilde{\mu} = \int_{\mathcal{S}} \boldsymbol{r}_t(\boldsymbol{\omega})^2 \delta_{\boldsymbol{x}_t(\boldsymbol{\omega})} \, d\sigma(\boldsymbol{\omega}) \qquad \text{for every } t \in [0,1]. \tag{55}$$

Moreover, for σ-a.e. $\boldsymbol{\omega}$, $\boldsymbol{x}_\cdot(\boldsymbol{\omega})$ is locally absolutely continuous on $\{\boldsymbol{r}_\cdot(\boldsymbol{\omega}) > 0\}$.

Let E be the set of pairs $(t, \boldsymbol{\omega}) \in (0,1) \times \mathcal{S}$ with $\boldsymbol{r}_t(\boldsymbol{\omega}) > 0$ for which at least one of the two derivatives in (54) does not exist or does not have the prescribed value. To verify measurability, set

$$Z(t, \boldsymbol{\omega}) \coloneqq (\boldsymbol{x}_t(\boldsymbol{\omega}), \boldsymbol{r}_t(\boldsymbol{\omega})),$$
$$G(t, \boldsymbol{\omega}) \coloneqq \left(w(\boldsymbol{x}_t(\boldsymbol{\omega})), \tfrac{1}{2}\eta(\boldsymbol{x}_t(\boldsymbol{\omega}))\boldsymbol{r}_t(\boldsymbol{\omega}) \right).$$

The radius map is continuous, and the spatial coordinate map is continuous where the radius is positive and Borel under our convention at zero. Thus Z and G are jointly Borel measurable. For each nonzero rational h, the difference quotient

$$\frac{Z(t+h, \boldsymbol{\omega}) - Z(t, \boldsymbol{\omega})}{h}$$

is Borel on $\{(t, \boldsymbol{\omega}) : 0 < t, t+h < 1\}$. Convergence of these quotients to $G(t, \boldsymbol{\omega})$ as rational $h \to 0$ is a Borel condition: for every $k \geq 1$, there must exist $n \geq 1$ such that the error is at most $1/k$ for every admissible rational h with $0 < |h| < 1/n$. At a point with $\boldsymbol{r}_t > 0$, both coordinate paths are continuous in a neighborhood of t, so this rational limit exists with the prescribed value if and only if the ordinary derivative does. Hence E is Borel measurable. For σ-a.e. $\boldsymbol{\omega}$, its time section is Lebesgue-null. Therefore, Tonelli's theorem gives

$$0 = \int_{\mathcal{S}} \int_0^1 \mathbf{1}_E(t, \boldsymbol{\omega}) \, dt \, d\sigma(\boldsymbol{\omega}) = \int_0^1 \sigma(E_t) \, dt,$$

where

$$E_t \coloneqq \{ \boldsymbol{\omega} \in \mathcal{S} \colon (t, \boldsymbol{\omega}) \in E \}.$$

Thus $\sigma(E_t) = 0$ for a.e. $t \in (0,1)$.

For $\boldsymbol{\omega} \in \mathcal{S}$, define the Borel energy density

$$\ell_t(\boldsymbol{\omega}) \coloneqq \boldsymbol{r}_t(\boldsymbol{\omega})^2 \left(\|w(\boldsymbol{x}_t(\boldsymbol{\omega}))\|^2 + \frac{1}{4}|\eta(\boldsymbol{x}_t(\boldsymbol{\omega}))|^2 \right).$$

By (54), this agrees a.e. in time on $\{\boldsymbol{r}_t > 0\}$ with the cone energy $|\dot{\boldsymbol{r}}_t|^2 + \boldsymbol{r}_t^2 \|w(\boldsymbol{x}_t)\|^2$. The same holds on $\{\boldsymbol{r}_t = 0\}$, since an absolutely continuous function has derivative zero a.e. on each of its level sets. By (55),

$$\int_{\mathcal{S}} \ell_t \, d\sigma = \int_{\Omega} \left(\|w\|^2 + \frac{1}{4}|\eta|^2 \right) d\widetilde{\mu} = \|\mathcal{W}\|_{\mu}^2 =: \mathcal{E} \qquad (t \in [0,1]). \tag{56}$$

Choose $\tau \in (0,1)$ such that $\sigma(E_\tau) = 0$. Then, for σ-a.e. $\boldsymbol{\omega}$ with $\boldsymbol{r}_\tau(\boldsymbol{\omega}) > 0$, both coordinate paths are differentiable at τ and satisfy (54). This gives the pointwise derivatives needed below; the energy identity will provide a uniform L^2 bound on the corresponding difference quotients.

Building the competitors A_ε: Write $\mathfrak{y}_0 = [x, r]$ and $\mathfrak{y}_1 = [y, s]$. We may assume without loss of generality that $r = 1$ for A-a.e. pair with $r > 0$. Indeed, consider the dilation

$$\vartheta(\mathfrak{y}_0, \mathfrak{y}_1) \coloneqq \begin{cases} r, & r > 0, \\ 1, & r = 0, \end{cases}$$

which is admissible, since

$$\int \vartheta^2 \, dA = \mu(\mathbb{R}^d) + A(\{r = 0\}) < \infty.$$

It sends $([x, r], [y, s])$ to $([x, 1], [y, s/r])$ on $\{r > 0\}$ and leaves the source-vertex part unchanged, and by Lemma B.1, it preserves both homogeneous marginals, the optimal cost, reducedness, and the source-weighted barycenter $\mathcal{L}$.

For $-\tau < \varepsilon < 1 - \tau$, we perturb the first endpoint along the characteristics (54). Set

$$\mathfrak{C}_\Omega \coloneqq \{(\mathfrak{y}_0, \mathfrak{y}_1) \in \mathfrak{C}^2 : \mathfrak{y}_0 = [x, r],\ r > 0,\ x \in \Omega\}, \qquad A_\Omega \coloneqq A \llcorner \mathfrak{C}_\Omega.$$

Since its first radius equals one, $(\mathsf{x})_{\#} A_\Omega = \widetilde{\mu}$. Disintegrate

$$A_\Omega(d\mathfrak{y}_0, d\mathfrak{y}_1) = \widetilde{\mu}(dx) Q_x(d\mathfrak{y}_0, d\mathfrak{y}_1).$$

For $\widetilde{\mu}$-a.e. x, the kernel Q_x is a probability measure concentrated on pairs with $\mathfrak{y}_0 = [x, 1]$. On $\mathcal{S} \times \mathfrak{C}^2$, define

$$\Lambda(d\boldsymbol{\omega}, d\mathfrak{y}_0, d\mathfrak{y}_1) \coloneqq \boldsymbol{r}_\tau(\boldsymbol{\omega})^2 \, \sigma(d\boldsymbol{\omega}) Q_{\boldsymbol{x}_\tau(\boldsymbol{\omega})}(d\mathfrak{y}_0, d\mathfrak{y}_1). \tag{57}$$

This measure is concentrated on triples satisfying

$$\mathfrak{y}_0 = [\boldsymbol{x}_\tau(\boldsymbol{\omega}), 1], \qquad \boldsymbol{r}_\tau(\boldsymbol{\omega}) > 0,$$

and its marginals on $\mathcal{S}$ and $\mathfrak{C}^2$ are respectively $\boldsymbol{r}_\tau^2 \sigma$ and A_Ω. In particular,

$$\Lambda(\mathcal{S} \times \mathfrak{C}^2) = \widetilde{\mu}(\Omega) < \infty, \qquad \int s^2 \, d\Lambda = \int s^2 \, dA_\Omega \le \nu(\mathbb{R}^d). \tag{58}$$

For $-\tau < \varepsilon < 1 - \tau$, set

$$a_\varepsilon(\boldsymbol{\omega}) \coloneqq \frac{\boldsymbol{r}_{\tau+\varepsilon}(\boldsymbol{\omega})}{\boldsymbol{r}_\tau(\boldsymbol{\omega})} \qquad \text{on } \{\boldsymbol{r}_\tau > 0\},$$

and define, Λ-a.e.,

$$F_\varepsilon(\boldsymbol{\omega}, \mathfrak{y}_0, \mathfrak{y}_1) \coloneqq ([\boldsymbol{x}_{\tau+\varepsilon}(\boldsymbol{\omega}), a_\varepsilon(\boldsymbol{\omega})], \mathfrak{y}_1)\,.$$

Then F_ε leaves the second cone endpoint fixed and moves the first one according to the characteristic indexed by $\boldsymbol{\omega}$.

By (58), Λ is finite, so we may push it forward to a finite measure on $\mathfrak{C}^2$ through F_ε. Its first homogeneous marginal is characterized, for every bounded measurable φ, by

$$\begin{aligned}\int \varphi(z)\, d\mathfrak{h}_1\big((F_\varepsilon)_\#\Lambda\big)(z) &= \int_{\mathcal{S}\times\mathfrak{C}^2} a_\varepsilon(\boldsymbol{\omega})^2 \varphi(\boldsymbol{x}_{\tau+\varepsilon}(\boldsymbol{\omega}))\, d\Lambda \\ &= \int_{\{\boldsymbol{r}_\tau>0\}} \boldsymbol{r}_{\tau+\varepsilon}(\boldsymbol{\omega})^2 \varphi(\boldsymbol{x}_{\tau+\varepsilon}(\boldsymbol{\omega}))\, d\sigma(\boldsymbol{\omega}).\end{aligned}$$

Therefore, by (55),

$$\mathfrak{h}_1\big((F_\varepsilon)_\#\Lambda\big) = \widetilde{\mu} - \beta_\varepsilon, \tag{59}$$

where

$$\beta_\varepsilon \coloneqq \int_{\{\boldsymbol{r}_\tau=0\}} \boldsymbol{r}_{\tau+\varepsilon}(\boldsymbol{\omega})^2 \delta_{\boldsymbol{x}_{\tau+\varepsilon}(\boldsymbol{\omega})}\, d\sigma(\boldsymbol{\omega}) \in \mathcal{M}^+(\Omega)$$

is the contribution at time $\tau+\varepsilon$ of characteristics whose radius vanishes at time τ.

The second homogeneous marginal is unchanged. Indeed, for every bounded measurable φ,

$$\begin{aligned}\int \varphi(z)\, d\mathfrak{h}_2\big((F_\varepsilon)_\#\Lambda\big)(z) &= \int_{\mathcal{S}\times\mathfrak{C}^2} s^2 \varphi(y)\, d\Lambda \\ &= \int_{\mathfrak{C}^2} s^2 \varphi(y)\, dA_\Omega.\end{aligned}$$

Hence

$$\mathfrak{h}_2\big((F_\varepsilon)_\#\Lambda\big) = \mathfrak{h}_2(A_\Omega).$$

Define

$$\mathfrak{c}(z) \coloneqq ([z,1], \mathfrak{o}).$$

Since

$$\mathfrak{h}_1(\mathfrak{c}_\#\beta_\varepsilon) = \beta_\varepsilon, \qquad \mathfrak{h}_2(\mathfrak{c}_\#\beta_\varepsilon) = 0,$$

set

$$A_\varepsilon \coloneqq (F_\varepsilon)_\#\Lambda + (A - A_\Omega) + \mathfrak{c}_\#\beta_\varepsilon. \tag{60}$$

Using (59) and the preceding identity for the second homogeneous marginal, we obtain

$$\begin{aligned}\mathfrak{h}_1(A_\varepsilon) &= (\widetilde{\mu} - \beta_\varepsilon) + \mathfrak{h}_1(A - A_\Omega) + \beta_\varepsilon = \mu, \\ \mathfrak{h}_2(A_\varepsilon) &= \mathfrak{h}_2(A_\Omega) + \mathfrak{h}_2(A - A_\Omega) = \nu.\end{aligned}$$

Moreover,

$$\int r^2\, dA_\varepsilon = \mu(\mathbb{R}^d), \qquad \int s^2\, dA_\varepsilon = \nu(\mathbb{R}^d),$$

so that $A_\varepsilon \in \Gamma_{\mathfrak{C}}(\mu,\nu)$.

Notice also that $\beta_0 = 0$ and F_0 leaves both cone endpoints unchanged on the support of Λ. Since the $\mathfrak{C}^2$-marginal of Λ is A_Ω,

$$(F_0)_\#\Lambda = A_\Omega.$$

Consequently $A_0 = A$.

Differentiation: Set

$$C(\varepsilon) \coloneqq \int_{\mathfrak{C}^2} \mathrm{d}_{\mathfrak{C}}^2 \, dA_\varepsilon.$$

Since $A_\varepsilon \in \Gamma_{\mathfrak{C}}(\mu,\nu)$ and $A = A_0$ is optimal,

$$C(\varepsilon) \geq C(0) = \mathrm{WFR}^2(\mu,\nu) \tag{61}$$

for every $-\tau < \varepsilon < 1-\tau$.

Expanding the cone cost (4) and using the fixed homogeneous marginals of A_ε, we obtain

$$C(\varepsilon) - C(0) = -2 \int_{\mathcal{S}\times\mathfrak{C}^2} s\,[a_\varepsilon K_y(\boldsymbol{x}_{\tau+\varepsilon}) - K_y(\boldsymbol{x}_\tau)] \, d\Lambda. \tag{62}$$

Indeed, $A - A_\Omega$ is unchanged, while the correction term has target radius zero and hence gives no contribution to the interaction term. On the moving part, the interaction term is

$$-2 \int_{\mathcal{S}\times\mathfrak{C}^2} s a_\varepsilon K_y(\boldsymbol{x}_{\tau+\varepsilon}) \, d\Lambda,$$

which yields (62) after subtracting the value at $\varepsilon = 0$.

We now justify differentiation in (62). For fixed y, set

$$\psi_y(t,\boldsymbol{\omega}) \coloneqq \boldsymbol{r}_t(\boldsymbol{\omega}) K_y(\boldsymbol{x}_t(\boldsymbol{\omega})).$$

The function $\psi_y(\cdot,\boldsymbol{\omega})$ is absolutely continuous, also across times at which $\boldsymbol{r}_t = 0$. Indeed, using $|K_y| \leq 1$, $\mathrm{Lip}(K_y) \leq 1$, and $\|\boldsymbol{x}_t\| \leq 2R$, we have

$$\begin{aligned} |\psi_y(t_2,\boldsymbol{\omega}) - \psi_y(t_1,\boldsymbol{\omega})| &\leq |\boldsymbol{r}_{t_2} - \boldsymbol{r}_{t_1}| + \boldsymbol{r}_{t_1}\|\boldsymbol{x}_{t_2} - \boldsymbol{x}_{t_1}\| \\ &\leq (1+2R)|\boldsymbol{r}_{t_2} - \boldsymbol{r}_{t_1}| + \|\boldsymbol{r}_{t_2}\boldsymbol{x}_{t_2} - \boldsymbol{r}_{t_1}\boldsymbol{x}_{t_1}\|. \end{aligned}$$

Since both $\boldsymbol{r}_\cdot$ and $\boldsymbol{r}_\cdot\boldsymbol{x}_\cdot$ are absolutely continuous, the same holds for ψ_y.

On $\{\boldsymbol{r}_t > 0\}$, the chain rule and (54) give

$$\partial_t \psi_y = \dot{\boldsymbol{r}}_t K_y(\boldsymbol{x}_t) + \boldsymbol{r}_t \nabla K_y(\boldsymbol{x}_t) \cdot w(\boldsymbol{x}_t).$$

Hence, by Cauchy–Schwarz and (19),

$$|\partial_t \psi_y| \leq \left(|\dot{\boldsymbol{r}}_t|^2 + \boldsymbol{r}_t^2 \|w(\boldsymbol{x}_t)\|^2\right)^{1/2} \left(|K_y(\boldsymbol{x}_t)|^2 + \|\nabla K_y(\boldsymbol{x}_t)\|^2\right)^{1/2} = \sqrt{\ell_t}.$$

On $\{\boldsymbol{r}_t = 0\}$, $\psi_y = 0$, and therefore $\partial_t \psi_y = 0$ for a.e. such t. Consequently,

$$|\psi_y(t_2,\boldsymbol{\omega}) - \psi_y(t_1,\boldsymbol{\omega})| \leq \int_{\min\{t_1,t_2\}}^{\max\{t_1,t_2\}} \sqrt{\ell_\theta(\boldsymbol{\omega})} \, d\theta. \tag{63}$$

For $\varepsilon \neq 0$, define

$$U_\varepsilon \coloneqq \frac{a_\varepsilon K_y(\boldsymbol{x}_{\tau+\varepsilon}) - K_y(\boldsymbol{x}_\tau)}{\varepsilon}.$$

Since Λ is concentrated on $\{\boldsymbol{r}_\tau > 0\}$,

$$U_\varepsilon = \frac{\psi_y(\tau+\varepsilon,\boldsymbol{\omega}) - \psi_y(\tau,\boldsymbol{\omega})}{\varepsilon \boldsymbol{r}_\tau(\boldsymbol{\omega})} \qquad \Lambda\text{-a.e.}$$

By the choice of τ, for Λ-a.e. triple both $\boldsymbol{x}_\cdot$ and $\boldsymbol{r}_\cdot$ are differentiable at τ and satisfy (54). Therefore

$$U_\varepsilon \longrightarrow U_0 \coloneqq \nabla K_y(\boldsymbol{x}_\tau)\cdot w(\boldsymbol{x}_\tau) + \frac{1}{2}K_y(\boldsymbol{x}_\tau)\eta(\boldsymbol{x}_\tau) \qquad \Lambda\text{-a.e.} \tag{64}$$

The exceptional set is independent of y, since once $\boldsymbol{x}_\cdot$ and $\boldsymbol{r}_\cdot$ are differentiable at τ, the ordinary chain rule applies to every C^1 function K_y.

Let I_ε be the interval with endpoints τ and $\tau+\varepsilon$. By (63),

$$|U_\varepsilon| \le \frac{1}{|\varepsilon|\boldsymbol{r}_\tau}\int_{I_\varepsilon}\sqrt{\ell_t}\,dt.$$

Cauchy–Schwarz in time gives

$$|U_\varepsilon|^2 \le \frac{1}{|\varepsilon|^2\boldsymbol{r}_\tau^2}\left(\int_{I_\varepsilon}\sqrt{\ell_t}\,dt\right)^2 \le \frac{1}{|\varepsilon|\boldsymbol{r}_\tau^2}\int_{I_\varepsilon}\ell_t\,dt.$$

Integrating with respect to Λ, the factor $\boldsymbol{r}_\tau^2$ in (57) cancels the denominator. Since Q_x is a probability measure for $\widetilde{\mu}$-a.e. x, we obtain

$$\begin{aligned}\int |U_\varepsilon|^2\,d\Lambda &\le \frac{1}{|\varepsilon|}\int_{I_\varepsilon}\int_{\{\boldsymbol{r}_\tau>0\}}\ell_t(\boldsymbol{\omega})\,d\sigma(\boldsymbol{\omega})\,dt\\ &\le \frac{1}{|\varepsilon|}\int_{I_\varepsilon}\int_{\mathcal{S}}\ell_t(\boldsymbol{\omega})\,d\sigma(\boldsymbol{\omega})\,dt\\ &\le \mathcal{E},\end{aligned}$$

where we used (56). Thus

$$\sup_\varepsilon \|U_\varepsilon\|_{L^2(\Lambda)} \le \sqrt{\mathcal{E}}. \tag{65}$$

Together with (64), Fatou's lemma also gives

$$\|U_0\|_{L^2(\Lambda)} \le \sqrt{\mathcal{E}}.$$

Since Λ is finite, (65) implies that $\{U_\varepsilon\}$ is uniformly integrable in $L^1(\Lambda)$. Indeed, for every measurable set B,

$$\int_B |U_\varepsilon|\,d\Lambda \le \sqrt{\mathcal{E}}\,\Lambda(B)^{1/2}.$$

Hence Vitali's theorem and (64) yield

$$U_\varepsilon \longrightarrow U_0 \qquad \text{in } L^1(\Lambda). \tag{66}$$

It remains to account for the possibly unbounded target radius $s \in L^2(\Lambda)$; see (58). For every $N>0$,

$$\begin{aligned}\left|\int_{\mathcal{S}\times\mathfrak{C}^2} s(U_\varepsilon-U_0)\,d\Lambda\right| &\le N\|U_\varepsilon-U_0\|_{L^1(\Lambda)} + \left\|s\mathbf{1}_{\{s>N\}}\right\|_{L^2(\Lambda)}\|U_\varepsilon-U_0\|_{L^2(\Lambda)}\\ &\le N\|U_\varepsilon-U_0\|_{L^1(\Lambda)} + 2\sqrt{\mathcal{E}}\left\|s\mathbf{1}_{\{s>N\}}\right\|_{L^2(\Lambda)}.\end{aligned}$$

Letting first $\varepsilon\to 0$ and using (66), and then $N\to\infty$, gives

$$\int_{\mathcal{S}\times\mathfrak{C}^2} sU_\varepsilon\,d\Lambda \longrightarrow \int_{\mathcal{S}\times\mathfrak{C}^2} sU_0\,d\Lambda.$$

Dividing (62) by ε and passing to the limit therefore yields the two-sided derivative

$$C'(0) = -2 \int_{\mathcal{S}\times\mathfrak{C}^2} s \left[\nabla K_y(\boldsymbol{x}_\tau) \cdot w(\boldsymbol{x}_\tau) + \frac{1}{2} K_y(\boldsymbol{x}_\tau)\eta(\boldsymbol{x}_\tau) \right] d\Lambda. \tag{67}$$

Conclusion: It remains to identify this expression with the normal pairing. Since A is concentrated on reduced pairs and has first radius one on $\mathfrak{C}_\Omega$, the normalized cone logarithm there is

$$\mathcal{L}_{\mathfrak{C}}([x,1],[y,s]) = \left(s\nabla K_y(x),\, 2(sK_y(x) - 1)\right). \tag{68}$$

The formula also covers $s = 0$. Using (35), which is preserved by the dilation, and the fact that $\mathcal{W}$ vanishes outside Ω, we obtain

$$\langle \mathcal{L}, \mathcal{W} \rangle_\mu = \int_{\mathfrak{C}^2} \left[s\nabla K_y(x) \cdot w(x) + \frac{1}{2} sK_y(x)\eta(x) - \frac{1}{2}\eta(x) \right] dA_\Omega.$$

The last term vanishes, since the first spatial marginal of A_Ω is $\widetilde{\mu}$ and

$$\int_{\mathfrak{C}^2} \eta(x)\, dA_\Omega = \int_\Omega \eta\, d\widetilde{\mu} = 0$$

by (53). Since the $\mathfrak{C}^2$-marginal of Λ is A_Ω, it follows that

$$\langle \mathcal{L}, \mathcal{W} \rangle_\mu = \int s \left[\nabla K_y(\boldsymbol{x}_\tau) \cdot w(\boldsymbol{x}_\tau) + \frac{1}{2} K_y(\boldsymbol{x}_\tau)\eta(\boldsymbol{x}_\tau) \right] d\Lambda = -\frac{1}{2} C'(0).$$

Since (61) shows that C has a local minimum at zero and $C'(0)$ exists as a two-sided derivative, $C'(0) = 0$. Hence

$$\langle \mathcal{L}, \mathcal{W} \rangle_\mu = 0.$$

We have proved the desired orthogonality for every compactly supported normal field. For a general $\mathcal{W} \in T_\mu\mathcal{M}\big(\mathbb{R}^d\big)^\perp$, Lemma C.4 provides compactly supported normal fields $\mathcal{W}_n \to \mathcal{W}$ in $H(\mu)$. Since $\mathcal{L} \in H(\mu)$, continuity of the pairing gives $\langle \mathcal{L}, \mathcal{W} \rangle_\mu = 0$. Finally, since $T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$ is a closed subspace of $H(\mu)$,

$$\mathcal{L} \in (T_\mu\mathcal{M}\big(\mathbb{R}^d\big)^\perp)^\perp = T_\mu\mathcal{M}\big(\mathbb{R}^d\big). \qquad \square$$

3.4 First-order independence of cut-locus selections

For a fixed semicoupling $\boldsymbol{\gamma}$, the choice of u affects only the transport component of the logarithm on the cut locus

$$\mathrm{Cut} \coloneqq \Big\{ (x,y) \in \mathbb{R}^d \times \mathbb{R}^d \colon \|x - y\| = \frac{\pi}{2} \Big\}. \tag{69}$$

The following estimate shows that the difference between two selections, paired with any fixed $L^2(\mu;\mathbb{R}^d)$ field, is negligible at first order in the WFR distance.

Proposition 3.12. *Assume* $\mathrm{WFR}(\mu,\mu_n) \to 0$. *Let* $\boldsymbol{\gamma}^n = (\gamma_0^n, \gamma_1^n)$ *be an optimal semicoupling between* μ *and* μ_n, *and let*

$$u_n^1, u_n^2 \colon \mathbb{R}^d \times \mathbb{R}^d \to [0,1]$$

be measurable functions. Write

$$\gamma_0^n = a_n \lambda_n, \qquad \gamma_1^n = b_n \lambda_n,$$

for some common dominating measure λ_n. Set also $d_n \coloneqq \mathrm{WFR}(\mu, \mu_n)$ and

$$\mathrm{Log}_{\boldsymbol{\gamma}_n, u_n^i} \coloneqq (\mathrm{Log}_{n,i}^T, \mathrm{Log}_{n,i}^R) \qquad i = 1, 2.$$

Then, for every $\xi \in L^2(\mu; \mathbb{R}^d)$,

$$\int_{\mathrm{Cut}} \xi(x) \cdot \big(\mathrm{Log}_{n,2}^T(x,y) - \mathrm{Log}_{n,1}^T(x,y)\big)\, d\gamma_0^n(x,y) = o(d_n). \tag{70}$$

Proof. Define

$$\Delta_n(x,y) \coloneqq \big\|\mathrm{Log}_{n,2}^T(x,y) - \mathrm{Log}_{n,1}^T(x,y)\big\|$$

By definition, since $u_n^1, u_n^2 \in [0,1]$ and the logarithms coincide outside of Cut we have

$$\Delta_n(x,y) \le \mathbb{1}_{\mathrm{Cut}}(x,y) |u_n^2(x,y) - u_n^1(x,y)| \sqrt{\frac{b_n(x,y)}{a_n(x,y)}} \le \mathbb{1}_{\mathrm{Cut}}(x,y) \sqrt{\frac{b_n(x,y)}{a_n(x,y)}}.$$

Moreover, on Cut it holds

$$c(x, y, a_n(x,y), b_n(x,y)) = a_n(x,y) + b_n(x,y)$$

and thus

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} \Delta_n^2 \, d\gamma_0^n(x,y) \le \int_{\mathrm{Cut}} b_n(x,y)\, d\lambda_n(x,y) \le \int_{\mathrm{Cut}} c(x, y, a_n(x,y), b_n(x,y))\, d\lambda_n(x,y) \le d_n^2$$

Using the Cauchy–Schwarz inequality we get

$$\left| \int_{\mathrm{Cut}} \xi(x) \cdot \big(\mathrm{Log}_{n,2}^T - \mathrm{Log}_{n,1}^T\big)(x,y)\, d\gamma_0^n(x,y) \right| \le 2 d_n \left(\int_{\mathrm{Cut}} \|\xi(x)\|^2 \, d\gamma_0^n(x,y) \right)^{1/2} \tag{71}$$

It remains to prove that

$$\int_{\mathrm{Cut}} |\xi(x)|^2 \, d\gamma_0^n(x,y) \longrightarrow 0. \tag{72}$$

To this end, define $\eta_n \coloneqq \pi^1_{\#}\big(\mathbb{1}_{\mathrm{Cut}} \gamma_0^n\big)$. Since $\pi^1_{\#}\gamma_0^n = \mu$, one has $\eta_n \le \mu$. Moreover,

$$\eta_n(\mathbb{R}^d) = \gamma_0^n(\mathrm{Cut}) = \int_{\mathrm{Cut}} a_n \, d\lambda_n \ \le \int_{\mathrm{Cut}} (a_n + b_n)\, d\lambda_n \le d_n^2,$$

and therefore $\eta_n(\mathbb{R}^d) \to 0$, hence Corollary C.1.1 proves (72).

Combining (71) and (72) yields (70). □

We henceforth set $u \equiv 1$ and write $\mathrm{Log}_{\boldsymbol{\gamma}}$ and $\overline{\mathrm{Log}}_{\boldsymbol{\gamma}}$ for the resulting logarithms. Proposition 3.12 ensures that first-order pairings are independent of this convention.

4 Regular Wasserstein–Fisher–Rao calculus

We can now introduce our notion of differentiability.

Definition 4.1 (WFR differentiability)**.** Let $F\colon \mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ and $\mu \in \mathcal{M}(\mathbb{R}^d)$. We say that F is WFR differentiable at μ if there exists

$$\mathcal{V} = (\xi, \zeta) \in T_\mu \mathcal{M}(\mathbb{R}^d)$$

such that, for every sequence $(\mu_n)_{n\in\mathbb{N}} \subset \mathcal{M}(\mathbb{R}^d)$ satisfying $\mathrm{WFR}(\mu_n, \mu) \to 0$, and every sequence of optimal semicouplings

$$\boldsymbol{\gamma}^n = (\gamma_0^n, \gamma_1^n) \in \mathrm{OptSC}(\mu, \mu_n),$$

one has

$$F(\mu_n) = F(\mu) + \langle \mathcal{V}, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \rangle_\mu + o\big(\mathrm{WFR}(\mu, \mu_n)\big) \tag{73}$$

as $n \to +\infty$.

We recall that we have the following equality due to the definition of the barycentric logarithm

$$\langle \mathcal{V}, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \rangle_\mu = \int_{\mathbb{R}^d\times\mathbb{R}^d} \xi(x) \cdot \mathrm{Log}^T_{\boldsymbol{\gamma}^n}(x,y)\, d\gamma_0^n(x,y) + \frac{1}{4}\int_{\mathbb{R}^d\times\mathbb{R}^d} \zeta(x)\, \mathrm{Log}^R_{\boldsymbol{\gamma}^n}(x,y)\, d\gamma_0^n(x,y).$$

4.1 Strength and uniqueness of the gradients

Before presenting differentiability results for specific functionals, we show that Wasserstein–Fisher–Rao gradients are strong, meaning that, for WFR differentiable functionals, first-order expansions also hold along non-optimal semicouplings. As a byproduct, we obtain uniqueness of the gradients.

The main ingredient is the following expansion estimate for smooth expectations.

Proposition 4.2. *Let $\varphi \in C_b^{1,1}(\mathbb{R}^d)$ and define*

$$F(\mu) \coloneqq \int_{\mathbb{R}^d} \varphi\, d\mu.$$

Then there exists a constant $C_\varphi > 0$ such that, for every $\mu, \nu \in \mathcal{M}(\mathbb{R}^d)$ and every semicoupling $\boldsymbol{\gamma} \in \mathrm{SC}(\mu,\nu)$, one has

$$\Big| F(\nu) - F(\mu) - \big\langle (\nabla\varphi, 4\varphi), \overline{\mathrm{Log}}_{\boldsymbol{\gamma}} \big\rangle_\mu \Big| \le C_\varphi\, \mathcal{C}(\boldsymbol{\gamma}),$$

with C_φ as in (22)*.*

Proof. Let λ be a finite positive measure dominating γ_0 and γ_1, and set as usual

$$a = \frac{d\gamma_0}{d\lambda}, \qquad b = \frac{d\gamma_1}{d\lambda}.$$

Throughout, terms involving the logarithm are set to zero on $\{a = 0\}$. Since $(\pi^1)_\#\gamma_0 = \mu$ and $(\pi^2)_\#\gamma_1 = \nu$, we have

$$F(\nu) - F(\mu) = \int_{\mathbb{R}^d\times\mathbb{R}^d} \big(b\varphi(y) - a\varphi(x)\big)\, d\lambda(x,y).$$

By the definition of the barycentric logarithm and of the WFR inner product,

$$\big\langle (\nabla\varphi, 4\varphi), \overline{\mathrm{Log}}_{\boldsymbol{\gamma}} \big\rangle_\mu = \int_{\mathbb{R}^d\times\mathbb{R}^d} a\,\big(\nabla\varphi(x)\cdot \mathrm{Log}^T_{\boldsymbol{\gamma}}(x,y) + \varphi(x)\,\mathrm{Log}^R_{\boldsymbol{\gamma}}(x,y)\big)\, d\lambda(x,y).$$

Define

$$R_\varphi(x,y) \coloneqq b\varphi(y) - a\varphi(x) - a\left(\nabla\varphi(x)\cdot \mathrm{Log}^T_\gamma(x,y) + \varphi(x)\,\mathrm{Log}^R_\gamma(x,y)\right).$$

We first restrict to $\{a,b>0,\ \|y-x\|\le \pi/2\}$. Consider the cone points

$$\mathfrak{y}_0 = [x,\sqrt{a(x,y)}], \qquad \mathfrak{y}_1 = [y,\sqrt{b(x,y)}].$$

Neither is the vertex, and $(\mathfrak{y}_0,\mathfrak{y}_1)\notin\mathfrak{B}$. Recalling the choice $u\equiv 1$, we have

$$\mathrm{Log}^T_\gamma(x,y) = \mathrm{Log}^T_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1), \qquad \mathrm{Log}^R_\gamma(x,y) = \frac{2}{\sqrt{a(x,y)}}\,\mathrm{Log}^R_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1).$$

Let $\mathfrak{y}_t$ be the reduced cone geodesic joining these points and set $h(t)\coloneqq \Phi_\varphi(\mathfrak{y}_t)$. Lemma 2.9 therefore gives

$$|R_\varphi(x,y)| = |h(1)-h(0)-h'(0^+)| \le C_\varphi \mathrm{d}^2_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1) = C_\varphi\, c(x,y,a,b),$$

where we used $\mathrm{d}^2_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1) = \mathrm{d}^2_{\mathfrak{C},\frac{\pi}{2}}(\mathfrak{y}_0,\mathfrak{y}_1)$ since $\|y-x\|\le\pi/2$.

On $\{a,b>0,\ \|y-x\|>\pi/2\}$, one has

$$\mathrm{Log}^T_\gamma(x,y) = 0, \qquad \mathrm{Log}^R_\gamma(x,y) = -2.$$

Hence $R_\varphi(x,y) = b\varphi(y)+a\varphi(x)$, and therefore

$$|R_\varphi(x,y)| \le \|\varphi\|_\infty (a+b) \le C_\varphi\, c(x,y,a,b),$$

since $c(x,y,a,b) = a+b$ on this set. Similarly,

$$R_\varphi(x,y) = b\varphi(y) \quad \text{on } \{a=0,\ b>0\}, \qquad R_\varphi(x,y) = a\varphi(x) \quad \text{on } \{a>0,\ b=0\},$$

while $R_\varphi(x,y)=0$ on $\{a=b=0\}$. Since $c(x,y,a,b)=a+b$ whenever $ab=0$, we conclude in every case that

$$|R_\varphi(x,y)| \le C_\varphi\, c(x,y,a,b).$$

Integrating with respect to λ and using (1) proves the claimed estimate. □

Theorem 4.3. *Let $F\colon \mathcal{M}(\mathbb{R}^d)\to\mathbb{R}$ be WFR differentiable at $\mu\in\mathcal{M}(\mathbb{R}^d)$ and $\mathcal{V}\in T_\mu\mathcal{M}(\mathbb{R}^d)$ a tangent vector satisfying the expansion in Definition 4.1. Then for every sequence $(\mu_n)_{n\in\mathbb{N}}\subset\mathcal{M}(\mathbb{R}^d)$ and every sequence of semicouplings*

$$\boldsymbol{\gamma}^n = (\gamma_0^n,\gamma_1^n)\in \mathrm{SC}(\mu,\mu_n),$$

with $\varepsilon_n^2 \coloneqq \mathcal{C}(\boldsymbol{\gamma}^n)\to 0$ one has

$$F(\mu_n) = F(\mu) + \langle \mathcal{V}, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n}\rangle_\mu + o\big(\varepsilon_n\big) \tag{74}$$

Proof. For every n, choose $\tilde{\boldsymbol{\gamma}}^n\in\mathrm{OptSC}(\mu,\mu_n)$, and set $d_n\coloneqq \mathrm{WFR}(\mu,\mu_n)$. By optimality, $d_n\le\varepsilon_n$. The assumption $\varepsilon_n\to 0$ therefore implies $\mu_n\to\mu$ in WFR.

If $\varepsilon_n=0$, then $\mu_n=\mu$ and the barycentric logarithm associated with $\boldsymbol{\gamma}^n$ vanishes. Thus, (74) holds trivially for these indices, and we may restrict to those for which $\varepsilon_n>0$.

By WFR differentiability,

$$F(\mu_n) = F(\mu) + \big\langle \mathcal{V}, \overline{\mathrm{Log}}_{\tilde{\boldsymbol{\gamma}}^n}\big\rangle_\mu + r_n, \qquad r_n = o(d_n). \tag{75}$$

It remains to compare the two barycentric logarithms. Define $T_n\colon T_\mu\mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ by

$$T_n\mathcal{W} \coloneqq \frac{\langle \mathcal{W}, \overline{\mathrm{Log}}_{\tilde{\boldsymbol{\gamma}}^n} - \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n}\rangle_\mu}{\varepsilon_n}.$$

By the energy estimate for barycentric logarithms (30),

$$\big\|\overline{\mathrm{Log}}_{\tilde{\boldsymbol{\gamma}}^n}\big\|_\mu \le d_n, \qquad \big\|\overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n}\big\|_\mu \le \varepsilon_n.$$

Thus

$$|T_n\mathcal{W}| \le \frac{d_n + \varepsilon_n}{\varepsilon_n}\|\mathcal{W}\|_\mu \le 2\|\mathcal{W}\|_\mu.$$

In particular,

$$\sup_n \|T_n\| \le 2. \tag{76}$$

Consider the dense subspace

$$D \coloneqq \big\{(\nabla\varphi, 4\varphi) : \varphi \in C_c^\infty(\mathbb{R}^d)\big\} \subset T_\mu\mathcal{M}(\mathbb{R}^d).$$

Fix $\mathcal{W}_\varphi = (\nabla\varphi, 4\varphi) \in D$. Applying Proposition 4.2 first to $\boldsymbol{\gamma}^n$ and then to $\tilde{\boldsymbol{\gamma}}^n$, we obtain

$$\int \varphi\, d\mu_n - \int \varphi\, d\mu = \langle \mathcal{W}_\varphi, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n}\rangle_\mu + R_n,$$

and

$$\int \varphi\, d\mu_n - \int \varphi\, d\mu = \langle \mathcal{W}_\varphi, \overline{\mathrm{Log}}_{\tilde{\boldsymbol{\gamma}}^n}\rangle_\mu + \tilde{R}_n$$

with $|R_n| \le C_\varphi \varepsilon_n^2$ and $|\tilde{R}_n| \le C_\varphi d_n^2$ respectively. Subtracting yields

$$|T_n\mathcal{W}_\varphi| \le C_\varphi \frac{\varepsilon_n^2 + d_n^2}{\varepsilon_n} \longrightarrow 0.$$

Hence $T_n \to 0$ pointwise on D. Using the density of D and the uniform bound (76) one readily extends the convergence to the full tangent space $T_\mu\mathcal{M}(\mathbb{R}^d)$.

Taking $\mathcal{W} = \mathcal{V}$ gives

$$\langle \mathcal{V}, \overline{\mathrm{Log}}_{\tilde{\boldsymbol{\gamma}}^n} - \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n}\rangle_\mu = \varepsilon_n T_n\mathcal{V} = o(\varepsilon_n).$$

Finally, $r_n = o(d_n)$ and $d_n \le \varepsilon_n$ imply $r_n = o(\varepsilon_n)$. Substituting the preceding estimate into (75) proves (74). □

As a first consequence, we obtain first-order expansions along the natural family of perturbations

$$\mu^\varepsilon \coloneqq (\mathrm{Id} + \varepsilon T)_\# \left(1 + \frac{\varepsilon}{2}\rho\right)^2 \mu. \tag{77}$$

Corollary 4.3.1. *Let $F\colon \mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ be WFR differentiable at μ, and let $\mathcal{V} = (\xi, \zeta) \in T_\mu\mathcal{M}(\mathbb{R}^d)$ be a tangent vector realizing its WFR differential. Then, for every*

$$T \in L^2(\mu; \mathbb{R}^d), \qquad \rho \in L^2(\mu),$$

one has

$$\lim_{\varepsilon\to 0} \frac{F(\mu^\varepsilon) - F(\mu)}{\varepsilon} = \int_{\mathbb{R}^d} \xi \cdot T\, d\mu + \frac{1}{4}\int_{\mathbb{R}^d} \zeta\rho\, d\mu, \tag{78}$$

where μ^ε is defined by (77).

Proof. Apply Lemma A.1 with $T_\varepsilon = T$ and $\rho_\varepsilon = \rho$. The resulting semicouplings satisfy

$$\overline{\mathrm{Log}}_{\boldsymbol{\gamma}^\varepsilon} = \varepsilon(T,\rho) + o_{H(\mu)}(|\varepsilon|), \qquad \sqrt{\mathcal{C}(\boldsymbol{\gamma}^\varepsilon)} = O(|\varepsilon|).$$

The strong expansion consequently gives

$$F(\mu^\varepsilon) - F(\mu) = \varepsilon\langle \mathcal{V}, (T,\rho)\rangle_\mu + o(|\varepsilon|),$$

which proves the claim. □

Uniqueness of the tangent vector representing the WFR differential now follows readily.

Theorem 4.4 (Uniqueness of the WFR gradient)**.** *Assume that $F\colon \mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ is WFR differentiable at $\mu \in \mathcal{M}(\mathbb{R}^d)$. Then there exists a unique $\mathcal{V} \in T_\mu\mathcal{M}(\mathbb{R}^d)$ for which the first-order expansion in the definition of WFR differentiability holds. We denote this vector by*

$$\nabla_{\mathrm{WFR}} F(\mu) = (\nabla^T F(\mu), \nabla^R F(\mu))$$

and call it the WFR gradient of F at μ.

Proof. Suppose that $\mathcal{V}_1, \mathcal{V}_2 \in T_\mu\mathcal{M}(\mathbb{R}^d)$ both satisfy the definition of WFR differentiability. Let $\varphi \in C_c^\infty(\mathbb{R}^d)$ and consider the perturbation

$$\mu^\varepsilon = (\mathrm{Id} + \varepsilon\nabla\varphi)_\# (1 + 2\varepsilon\varphi)^2 \mu,$$

corresponding to $T = \nabla\varphi$ and $\rho = 4\varphi$. Applying Corollary 4.3.1 to each $\mathcal{V}_i$ gives

$$\lim_{\varepsilon\to 0} \frac{F(\mu^\varepsilon) - F(\mu)}{\varepsilon} = \langle \mathcal{V}_i, (\nabla\varphi, 4\varphi)\rangle_\mu .$$

Since the left-hand side is independent of i,

$$\langle \mathcal{V}_1 - \mathcal{V}_2, (\nabla\varphi, 4\varphi)\rangle_\mu = 0$$

for every $\varphi \in C_c^\infty(\mathbb{R}^d)$. By definition,

$$\big\{(\nabla\varphi, 4\varphi) : \varphi \in C_c^\infty(\mathbb{R}^d)\big\}$$

is dense in $T_\mu\mathcal{M}(\mathbb{R}^d)$. Since $\mathcal{V}_1 - \mathcal{V}_2 \in T_\mu\mathcal{M}(\mathbb{R}^d)$, it follows that $\mathcal{V}_1 - \mathcal{V}_2 = 0$ in $T_\mu\mathcal{M}(\mathbb{R}^d)$. Therefore $\mathcal{V}_1 = \mathcal{V}_2$. □

We conclude this subsection by identifying the norm of the WFR gradient with the descending metric slope. Unlike the WFR differential, the latter is defined entirely in terms of the metric and quantifies the maximal local rate of decrease of a functional. Their equality shows that the differential structure developed here recovers the intrinsic metric notion of steepest descent, without regularity assumptions on the reference measure.

Proposition 4.5. *Let $F\colon \mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ be WFR differentiable at $\mu \in \mathcal{M}(\mathbb{R}^d)$. Then*

$$|\partial^- F|(\mu) = \|\nabla_{\mathrm{WFR}} F(\mu)\|_\mu,$$

where the descending metric slope is defined by

$$|\partial^- F|(\mu) \coloneqq \limsup_{\substack{\nu\to\mu \\ \nu\neq\mu}} \frac{(F(\mu) - F(\nu))_+}{\mathrm{WFR}(\mu,\nu)}.$$

Proof. For any $\nu \to \mu$ and $\boldsymbol{\gamma} \in \mathrm{OptSC}(\mu,\nu)$, WFR differentiability and Lemma 3.5 give

$$F(\mu) - F(\nu) \le \|\nabla_{\mathrm{WFR}} F(\mu)\|_\mu \mathrm{WFR}(\mu,\nu) + o(\mathrm{WFR}(\mu,\nu)).$$

Hence $|\partial^- F|(\mu) \le \|\nabla_{\mathrm{WFR}} F(\mu)\|_\mu$.

If $\nabla_{\mathrm{WFR}} F(\mu) = 0$, this proves the claim. Otherwise, set

$$(T,\rho) \coloneqq -\frac{\nabla_{\mathrm{WFR}} F(\mu)}{\|\nabla_{\mathrm{WFR}} F(\mu)\|_\mu}$$

and consider

$$\mu^\varepsilon \coloneqq (\mathrm{Id} + \varepsilon T)_\# \left(\left(1 + \frac{\varepsilon}{2}\rho\right)^2 \mu \right), \qquad \varepsilon > 0.$$

By Corollary 4.3.1,

$$F(\mu) - F(\mu^\varepsilon) = \varepsilon \|\nabla_{\mathrm{WFR}} F(\mu)\|_\mu + o(\varepsilon).$$

Moreover, Lemma A.1 provides semicouplings $\boldsymbol{\gamma}^\varepsilon$ between μ and μ^ε such that

$$\mathcal{C}(\boldsymbol{\gamma}^\varepsilon) = \varepsilon^2 + o(\varepsilon^2).$$

Therefore,

$$|\partial^- F|(\mu) \ge \limsup_{\varepsilon \downarrow 0} \frac{F(\mu) - F(\mu^\varepsilon)}{\mathrm{WFR}(\mu,\mu^\varepsilon)} \ge \lim_{\varepsilon \downarrow 0} \frac{\varepsilon \|\nabla_{\mathrm{WFR}} F(\mu)\|_\mu + o(\varepsilon)}{\sqrt{\mathcal{C}(\boldsymbol{\gamma}^\varepsilon)}} = \|\nabla_{\mathrm{WFR}} F(\mu)\|_\mu. \qquad \square$$

4.2 Calculus rules and differentiability criteria

4.2.1 A differentiability criterion and chain rules

We present a Wasserstein–Fisher–Rao differentiability criterion based on flat derivatives.

Theorem 4.6 (From flat to WFR differentiability)**.** *Let $F\colon \mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ and fix $\mu \in \mathcal{M}(\mathbb{R}^d)$. Assume that there exist a convex WFR-open neighborhood U of μ and a function $f\colon U \times \mathbb{R}^d \to \mathbb{R}$ satisfying the following properties:*

1. *For every $\mu_0, \mu_1 \in U$,*

$$F(\mu_1) - F(\mu_0) = \int_0^1 \int_{\mathbb{R}^d} f\big((1-t)\mu_0 + t\mu_1, x\big)\, d(\mu_1 - \mu_0)(x)\, dt. \tag{79}$$

2. *For every $\nu \in U$, the function $f(\nu,\cdot)$ belongs to $C_b^{1,1}(\mathbb{R}^d)$, and there exists $K < +\infty$ such that*

$$\sup_{\nu \in U} \left(\|f(\nu,\cdot)\|_\infty + \frac{1}{2}\|\nabla_x f(\nu,\cdot)\|_\infty + \frac{1}{2}\,\mathrm{Lip}\big(\nabla_x f(\nu,\cdot)\big) \right) \le K. \tag{80}$$

3. *The maps*

$$(\nu,x) \longmapsto f(\nu,x), \qquad (\nu,x) \longmapsto \nabla_x f(\nu,x)$$

are continuous on $U \times \mathbb{R}^d$, where U is endowed with the WFR topology.

Here $C_b^{1,1}(\mathbb{R}^d)$ denotes the space of bounded continuously differentiable functions with bounded, globally Lipschitz gradient.

Then F is WFR differentiable at μ and

$$\nabla_{\mathrm{WFR}} F(\mu) = \big(\nabla_x f(\mu,\cdot), 4f(\mu,\cdot)\big) \in T_\mu\mathcal{M}\big(\mathbb{R}^d\big). \tag{81}$$

More precisely, there exist $r_0 > 0$ and a nondecreasing function $\omega_\mu\colon [0,r_0) \to [0,+\infty)$, with $\omega_\mu(r) \to 0$ as $r \downarrow 0$, such that

$$\Big|F(\nu) - F(\mu) - \big\langle \nabla_{\mathrm{WFR}} F(\mu), \overline{\mathrm{Log}}_{\boldsymbol{\gamma}}\big\rangle_\mu\Big| \le K\,\mathcal{C}(\boldsymbol{\gamma}) + \omega_\mu\big(\mathrm{WFR}(\mu,\nu)\big)\sqrt{\mathcal{C}(\boldsymbol{\gamma})} \tag{82}$$

for every $\nu \in \mathcal{M}\big(\mathbb{R}^d\big)$ with $\mathrm{WFR}(\mu,\nu) < r_0$ and every $\boldsymbol{\gamma} \in \mathrm{SC}(\mu,\nu)$.

Proof. For $\nu \in U$, write

$$\mathcal{V}_\nu \coloneqq \big(\nabla f(\nu,\cdot), 4f(\nu,\cdot)\big).$$

We have

$$\|\mathcal{V}_\nu - \mathcal{V}_\mu\|_\mu^2 = \int_{\mathbb{R}^d} \big(\|\nabla f(\nu,x) - \nabla f(\mu,x)\|^2 + 4|f(\nu,x) - f(\mu,x)|^2\big)\, d\mu(x).$$

We notice that $\mathcal{V}_\mu \in T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$ due to Lemma C.3. Next, the continuity assumptions and (80) imply

$$\|\mathcal{V}_\nu - \mathcal{V}_\mu\|_\mu \longrightarrow 0 \qquad \text{as } \mathrm{WFR}(\nu,\mu) \to 0. \tag{83}$$

Indeed, for any sequence $\nu_n \to \mu$ in WFR, both $f(\nu_n,x) \to f(\mu,x)$ and $\nabla f(\nu_n,x) \to \nabla f(\mu,x)$ hold for every x. The integrand defining the squared norm is uniformly bounded by (80), so (83) follows from dominated convergence.

Choose $r_0 > 0$ such that $B_{\mathrm{WFR}}(\mu,r_0) \subset U$, and, for $0 \le r < r_0$, define

$$\omega_\mu(r) \coloneqq \sup_{\mathrm{WFR}(\nu,\mu)\le r} \|\mathcal{V}_\nu - \mathcal{V}_\mu\|_\mu. \tag{84}$$

This function is finite and nondecreasing, and (83) implies $\omega_\mu(r) \to 0$ as $r \downarrow 0$.

Fix now ν with $\mathrm{WFR}(\mu,\nu) < r_0$ and set

$$\mu_t \coloneqq (1-t)\mu + t\nu, \qquad t \in [0,1].$$

Convexity of the squared WFR distance [24, Theorem 6.2 and Theorem 7.20] under linear interpolation gives

$$\mathrm{WFR}^2(\mu,\mu_t) \le (1-t)\mathrm{WFR}^2(\mu,\mu) + t\mathrm{WFR}^2(\mu,\nu) = t\mathrm{WFR}^2(\mu,\nu). \tag{85}$$

In particular, $\mu_t \in U$ and for every $t \in [0,1]$

$$\|\mathcal{V}_{\mu_t} - \mathcal{V}_\mu\|_\mu \le \omega_\mu(\mathrm{WFR}(\mu_t,\mu)) \le \omega_\mu\big(\mathrm{WFR}(\mu,\nu)\big) \tag{86}$$

Let $\boldsymbol{\gamma} \in \mathrm{SC}(\mu,\nu)$. By Proposition 4.2, applied to $f(\mu_t)$, and the uniform bound (80), we have

$$\int_{\mathbb{R}^d} f(\mu_t)\, d(\nu-\mu) = \big\langle \mathcal{V}_{\mu_t}, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}}\big\rangle_\mu + R_t, \qquad |R_t| \le K\,\mathcal{C}(\boldsymbol{\gamma}). \tag{87}$$

Here we use the quantitative dependence of the constant in (22). Notice that the same semicoupling $\boldsymbol{\gamma}$ is used for every t. By (83), $t \mapsto \mathcal{V}_{\mu_t}$ is continuous in $H(\mu_)$, hence measurable. Therefore, (87) yields measurability of $t \mapsto R_t$ as well. Integrating (87) and using (79), we obtain

$$F(\nu) - F(\mu) - \big\langle \mathcal{V}_\mu, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}} \big\rangle_\mu = \int_0^1 \big\langle \mathcal{V}_{\mu_t} - \mathcal{V}_\mu, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}} \big\rangle_\mu \, dt + \int_0^1 R_t \, dt.$$

These integrals are well defined by the continuity assumptions and the uniform bounds. By Cauchy–Schwarz and the logarithm energy bound (30) we deduce

$$\Big| F(\nu) - F(\mu) - \big\langle \mathcal{V}_\mu, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}} \big\rangle_\mu \Big| \le \sqrt{\mathcal{C}(\boldsymbol{\gamma})} \int_0^1 \|\mathcal{V}_{\mu_t} - \mathcal{V}_\mu\|_\mu \, dt + K\, \mathcal{C}(\boldsymbol{\gamma}).$$

Together with (86), this proves (82).

Finally, if $\boldsymbol{\gamma} \in \mathrm{OptSC}(\mu,\nu)$, then $\mathcal{C}(\boldsymbol{\gamma}) = \mathrm{WFR}^2(\mu,\nu)$, and hence

$$\Big| F(\nu) - F(\mu) - \big\langle \mathcal{V}_\mu, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}} \big\rangle_\mu \Big| \le K\, \mathrm{WFR}^2(\mu,\nu) + \omega_\mu\big(\mathrm{WFR}(\mu,\nu)\big)\mathrm{WFR}(\mu,\nu) = o\big(\mathrm{WFR}(\mu,\nu)\big).$$

The estimate is uniform over all optimal semicouplings, which proves WFR differentiability at μ. □

Nonetheless, as we show in Section 6.2, there are functionals that are WFR differentiable but do not admit flat derivatives. This reflects the fact that certain flat perturbations of order ε have WFR size of order $\sqrt{\varepsilon}$.

We end this section with two chain rules.

Proposition 4.7 (Chain rule along WFR geodesics)**.** *Let $(\boldsymbol{\mu}_t)_{t\in[0,1]}$ be a constant-speed WFR geodesic and let $F\colon \mathcal{M}\big(\mathbb{R}^d\big) \to \mathbb{R}$. For $s \in [0,1)$, denote by $\mathcal{L}_s$ the regular barycentric logarithm associated with the rescaled geodesic*

$$\tau \mapsto \boldsymbol{\mu}_{s+(1-s)\tau}, \qquad \tau \in [0,1],$$

and set $\mathcal{V}_s \coloneqq \mathcal{L}_s/(1-s)$.

If F is WFR differentiable at μ_s, then

$$\lim_{\delta\to 0} \frac{F(\boldsymbol{\mu}_{s+\delta}) - F(\boldsymbol{\mu}_s)}{\delta} = \langle \nabla_{\mathrm{WFR}} F(\boldsymbol{\mu}_s), \mathcal{V}_s \rangle_{\boldsymbol{\mu}_s}.$$

At $s = 0$, the limit is understood as a right derivative. In particular,

$$F(\mu_t) = F(\mu_0) + t\langle \nabla_{\mathrm{WFR}} F(\boldsymbol{\mu}_0), \mathcal{L}_0 \rangle_{\boldsymbol{\mu}_0} + o(t)$$

whenever F is WFR differentiable at μ_0.

If, moreover, F is WFR differentiable at every μ_t and

$$C_F \coloneqq \sup_{t\in[0,1]} \big\|\nabla_{\mathrm{WFR}} F(\boldsymbol{\mu}_t)\big\|_{\boldsymbol{\mu}_t} < \infty,$$

then $t \mapsto F(\boldsymbol{\mu}_t)$ is Lipschitz, with Lipschitz constant at most $C_F \mathrm{WFR}(\boldsymbol{\mu}_0, \boldsymbol{\mu}_1)$, and

$$F(\boldsymbol{\mu}_{t_2}) - F(\boldsymbol{\mu}_{t_1}) = \int_{t_1}^{t_2} \langle \nabla_{\mathrm{WFR}} F(\boldsymbol{\mu}_t), \mathcal{V}_t \rangle_{\mu_t} \, dt, \qquad 0 \le t_1 \le t_2 \le 1.$$

Proof. Set $D \coloneqq \mathrm{WFR}(\mu_0,\mu_1)^2$. If $D=0$, the geodesic is constant and the conclusions are immediate. Otherwise, let $\mathbf{\Pi}$ be an optimal cone-geodesic representation of $(\boldsymbol{\mu}_t)_{t\in[0,1]}$. Then, for $\varphi \in C_c^\infty(\mathbb{R}^d)$, we have

$$f_\varphi(t) \coloneqq \int_{\mathbb{R}^d} \varphi \, d\boldsymbol{\mu}_t = \int_{\mathrm{Geo}^*(\mathfrak{C})} \boldsymbol{r}_t^2 \varphi(\boldsymbol{x}_t)\, d\mathbf{\Pi}(\mathfrak{y}),$$

where we recall $\mathfrak{y}_t = [\boldsymbol{x}_t, \boldsymbol{r}_t]$. Along each reduced cone geodesic, the integrand has a continuous derivative, with one-sided derivatives at the endpoints. By Lemma 2.9, its derivative is bounded in absolute value by $B_\varphi(\boldsymbol{r}_0^2+\boldsymbol{r}_1^2)$, an integrable function. Differentiation under the integral and dominated convergence therefore gives $f_\varphi \in C^1([0,1])$, with one-sided derivatives at the endpoints.

By the definition of $\mathcal{L}_s$,

$$\langle (\nabla\varphi, 4\varphi), \mathcal{L}_s\rangle_{\boldsymbol{\mu}_s} = \lim_{\tau\downarrow 0} \frac{f_\varphi(s+(1-s)\tau)-f_\varphi(s)}{\tau} = (1-s) f_\varphi'(s).$$

Consequently,

$$f_\varphi'(s) = \langle (\nabla\varphi, 4\varphi), \mathcal{V}_s\rangle_{\boldsymbol{\mu}_s}.$$

Moreover, the logarithm norm bound gives

$$\|\mathcal{V}_s\|_{\boldsymbol{\mu}_s} \le \frac{\mathrm{WFR}(\boldsymbol{\mu}_s,\boldsymbol{\mu}_1)}{1-s} = \sqrt{D}.$$

Fix $s\in[0,1)$ at which F is WFR differentiable. For $\delta\neq 0$ sufficiently small with $s+\delta\in[0,1]$, choose $\gamma^\delta \in \mathrm{OptSC}(\boldsymbol{\mu}_s,\boldsymbol{\mu}_{s+\delta})$ and set

$$Z_\delta \coloneqq \frac{1}{\delta}\overline{\mathrm{Log}}_{\gamma^\delta}.$$

Since $\mathrm{WFR}(\boldsymbol{\mu}_s,\boldsymbol{\mu}_{s+\delta}) = |\delta|\sqrt{D}$, we have $\|Z_\delta\|_{\boldsymbol{\mu}_s} \le \sqrt{D}$.

For every $\varphi\in C_c^\infty(\mathbb{R}^d)$, Proposition 4.2 gives

$$\left| \langle (\nabla\varphi, 4\varphi), Z_\delta\rangle_{\boldsymbol{\mu}_s} - \frac{f_\varphi(s+\delta)-f_\varphi(s)}{\delta}\right| \le C_\varphi |\delta| D.$$

Hence

$$\langle (\nabla\varphi, 4\varphi), Z_\delta - \mathcal{V}_s\rangle_{\boldsymbol{\mu}_s} \longrightarrow 0.$$

The uniform bounds on Z_δ and $\mathcal{V}_s$, together with density of the fields $(\nabla\varphi,4\varphi)$ in $\mathrm{T}_{\boldsymbol{\mu}_s}\mathcal{M}(\mathbb{R}^d)$, imply

$$\langle \nabla_{\mathrm{WFR}} F(\boldsymbol{\mu}_s), Z_\delta - \mathcal{V}_s\rangle_{\boldsymbol{\mu}_s} \longrightarrow 0.$$

WFR differentiability at $\boldsymbol{\mu}_s$ now yields

$$F(\boldsymbol{\mu}_{s+\delta}) - F(\boldsymbol{\mu}_s) = \left\langle \nabla_{\mathrm{WFR}} F(\boldsymbol{\mu}_s), \overline{\mathrm{Log}}_{\gamma^\delta}\right\rangle_{\boldsymbol{\mu}_s} + o(|\delta|) = \delta \langle \nabla_{\mathrm{WFR}} F(\boldsymbol{\mu}_s), \mathcal{V}_s\rangle_{\boldsymbol{\mu}_s} + o(|\delta|).$$

This proves the pointwise chain rule, including the right derivative at $s=0$.

Under the additional assumptions, $g(t) \coloneqq F(\boldsymbol{\mu}_t)$ is continuous on $[0,1]$ and differentiable on $(0,1)$, with

$$|g'(t)| \le \|\nabla_{\mathrm{WFR}} F(\boldsymbol{\mu}_t)\|_{\boldsymbol{\mu}_t} \|\mathcal{V}_t\|_{\boldsymbol{\mu}_t} \le C_F\sqrt{D}.$$

The mean value theorem shows that g is Lipschitz. The integral identity follows from the fundamental theorem of calculus. □

Proposition 4.8 (Chain rule). *Let $F\colon \mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ be WFR differentiable at μ and let $G\colon \mathbb{R} \to \mathbb{R}$ be differentiable at $F(\mu)$. Then $G \circ F$ is WFR differentiable at μ and*

$$\nabla_{\mathrm{WFR}}(G \circ F)(\mu) = G'(F(\mu))\, \nabla_{\mathrm{WFR}} F(\mu).$$

Proof. For any sequence $\mu_n \to \mu$ in WFR and any $\boldsymbol{\gamma}^n \in \mathrm{OptSC}(\mu, \mu_n)$, we have

$$F(\mu_n) - F(\mu) = \big\langle \nabla_{\mathrm{WFR}} F(\mu), \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \big\rangle_\mu + o\big(\mathrm{WFR}(\mu, \mu_n)\big).$$

Since by Cauchy-Schwarz and (30)

$$\Big| \big\langle \nabla_{\mathrm{WFR}} F(\mu), \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \big\rangle_\mu \Big| \le \|\nabla_{\mathrm{WFR}} F(\mu)\|_\mu \mathrm{WFR}(\mu, \mu_n),$$

it follows that

$$|F(\mu_n) - F(\mu)| = O\big(\mathrm{WFR}(\mu, \mu_n)\big).$$

By differentiability of G at $F(\mu)$,

$$G(F(\mu_n)) - G(F(\mu)) = G'(F(\mu))\big(F(\mu_n) - F(\mu)\big) + o\big(|F(\mu_n) - F(\mu)|\big).$$

Combining the previous estimates gives

$$G(F(\mu_n)) - G(F(\mu)) = \big\langle G'(F(\mu)) \nabla_{\mathrm{WFR}} F(\mu), \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \big\rangle_\mu + o\big(\mathrm{WFR}(\mu, \mu_n)\big),$$

and the conclusion follows. □

4.2.2 Some differentiable functions

As a consequence of the criterion developed in Theorem 4.6 we obtain WFR differentiability for smooth enough expectations, cylinder functionals and interaction energies.

Corollary 4.8.1 (Linear observables). *Let $f \in C_b^{1,1}(\mathbb{R}^d)$ and define $F\colon \mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ by*

$$F(\mu) \coloneqq \int_{\mathbb{R}^d} f(x)\, d\mu(x).$$

Then F is everywhere WFR differentiable and

$$\nabla_{\mathrm{WFR}} F(\mu) = \big(\nabla f, 4f\big).$$

Corollary 4.8.2 (Cylinder functionals). *Let $N \in \mathbb{N}$, let $f_1, \dots, f_N \in C_b^{1,1}(\mathbb{R}^d)$, and let $G\colon \mathbb{R}^N \to \mathbb{R}$ be C^1. Define $F\colon \mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ by*

$$F(\mu) \coloneqq G\left(\int_{\mathbb{R}^d} f_1\, d\mu, \dots, \int_{\mathbb{R}^d} f_N\, d\mu \right).$$

Then F is everywhere WFR differentiable and

$$\nabla_{\mathrm{WFR}} F(\mu) = \left(\sum_{i=1}^N \partial_i G \left(\int f_1\, d\mu, \dots, \int f_N\, d\mu \right) \nabla f_i,\ 4 \sum_{i=1}^N \partial_i G \left(\int f_1\, d\mu, \dots, \int f_N\, d\mu \right) f_i \right).$$

Corollary 4.8.3 (Interaction energies). *Let $W \in C_b^{1,1}(\mathbb{R}^d \times \mathbb{R}^d)$ be symmetric, namely*

$$W(x, y) = W(y, x) \qquad \text{for every } x, y \in \mathbb{R}^d,$$

and define

$$F(\mu) \coloneqq \frac{1}{2} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} W(x, y)\, d\mu(x)\, d\mu(y).$$

Then F is everywhere WFR differentiable and

$$\nabla_{\mathrm{WFR}} F(\mu)(x) = \left(\int_{\mathbb{R}^d} \nabla_x W(x, y)\, d\mu(y),\ 4 \int_{\mathbb{R}^d} W(x, y)\, d\mu(y) \right).$$

4.2.3 Compatibility with finite-dimensional derivatives

A classical property of Wasserstein differentiability is its compatibility with empirical measures. If F is Wasserstein differentiable at

$$\mu_{(x_1,\dots,x_N)} \coloneqq \frac{1}{N}\sum_{i=1}^{N}\delta_{x_i},$$

then the map $F_N(\boldsymbol{x}) \coloneqq F(\mu_{\boldsymbol{x}})$ is Fréchet differentiable at $(x_1,\dots,x_N)$, with

$$\nabla_{x_i} F_N(x_1,\dots,x_N) = \frac{1}{N}\nabla_{W_2} F(\mu_{(x_1,\dots,x_N)})(x_i), \qquad i = 1,\dots,N.$$

See, for example, [9, Proposition 3.1].

The WFR differential extends this compatibility to weighted empirical measures, permitting simultaneous perturbations of the particle positions and weights. This provides a finite-dimensional interpretation of both components of the WFR gradient.

To formulate this property, we consider the space

$$M_N \coloneqq (\mathbb{R}^d)^N \times \mathbb{R}^N.$$

For $(\hat{x},\hat{\ell}) \coloneqq (x_1,\dots,x_N,\ell_1,\dots,\ell_N) \in M_N$, define the empirical-measure map

$$\iota_N(\hat{x},\hat{\ell}) \coloneqq \sum_{i=1}^{N} e^{\ell_i}\delta_{x_i}. \tag{88}$$

We write increments as

$$\hat{\xi} \coloneqq (\xi_1,\dots,\xi_N), \qquad \hat{\zeta} \coloneqq (\zeta_1,\dots,\zeta_N).$$

Theorem 4.9 (Compatibility with weighted empirical measures)**.** *Let $F\colon \mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ be WFR differentiable at $\mu = \iota_N(\hat{x},\hat{\ell})$. Then the map $\widetilde{F} \coloneqq F \circ \iota_N\colon M_N \to \mathbb{R}$ is Fréchet differentiable at $(\hat{x},\hat{\ell})$, with*

$$D\widetilde{F}(\hat{x},\hat{\ell})[\hat{\xi},\hat{\zeta}] = \sum_{i=1}^{N} e^{\ell_i}\left(\nabla^T F(\mu)(x_i)\cdot\xi_i + \frac{1}{4}\nabla^R F(\mu)(x_i)\zeta_i\right). \tag{89}$$

In particular, its Euclidean partial derivatives are

$$\nabla_{x_i}\widetilde{F}(\hat{x},\hat{\ell}) = e^{\ell_i}\nabla^T F(\mu)(x_i), \qquad \partial_{\ell_i}\widetilde{F}(\hat{x},\hat{\ell}) = \frac{1}{4}e^{\ell_i}\nabla^R F(\mu)(x_i).$$

Proof. Since μ has finite support, we can choose $\varphi \in C_c^\infty(\mathbb{R}^d)$ such that

$$\nabla\varphi(x) = \nabla^T F(\mu)(x), \qquad 4\varphi(x) = \nabla^R F(\mu)(x), \qquad x \in \operatorname{supp}\mu.$$

Indeed, the value and gradient of a smooth function can be prescribed independently at finitely many distinct points. The smooth expectation $F_\varphi(\nu) \coloneqq \int_{\mathbb{R}^d}\varphi\,d\nu$ therefore has the same WFR gradient as F at μ. As a consequence, for every $\nu \in \mathcal{M}(\mathbb{R}^d)$ and optimal semicoupling $\boldsymbol{\gamma} \in \operatorname{OptSC}(\mu,\nu)$,

$$\begin{aligned}\left\langle \nabla_{\mathrm{WFR}} F(\mu), \overline{\operatorname{Log}}_{\boldsymbol{\gamma}}\right\rangle_\mu &= \left\langle (\nabla\varphi, 4\varphi), \overline{\operatorname{Log}}_{\boldsymbol{\gamma}}\right\rangle_\mu \\ &= \int_{\mathbb{R}^d}\varphi\,d(\nu-\mu) + o\big(\mathrm{WFR}(\mu,\nu)\big)\end{aligned}$$

as $\nu \to \mu$ in WFR, due to Proposition 4.2. Hence, WFR differentiability of F yields

$$F(\nu) - F(\mu) = \int_{\mathbb{R}^d} \varphi \, d(\nu - \mu) + o\big(\mathrm{WFR}(\mu, \nu)\big).$$

For an increment $(\hat{\xi}, \hat{\zeta})$, set

$$\nu \coloneqq \iota_N(\hat{x} + \hat{\xi}, \hat{\ell} + \hat{\zeta}) \quad \text{and} \quad \delta \coloneqq \left(\sum_{i=1}^N \|\xi_i\|^2 + \sum_{i=1}^N \zeta_i^2 \right)^{1/2}.$$

For sufficiently small increments, the cone coupling matching $[x_i, e^{\ell_i/2}]$ with $[x_i + \xi_i, e^{(\ell_i+\zeta_i)/2}]$ yields

$$\mathrm{WFR}^2(\mu, \nu) \le \sum_{i=1}^N e^{\ell_i} \left(1 + e^{\zeta_i} - 2e^{\zeta_i/2} \cos(\|\xi_i\|)\right) = O(\delta^2).$$

Consequently, Taylor expansion gives

$$\begin{aligned} \widetilde{F}(\hat{x} + \hat{\xi}, \hat{\ell} + \hat{\zeta}) - \widetilde{F}(\hat{x}, \hat{\ell}) &= \sum_{i=1}^N e^{\ell_i} \left(e^{\zeta_i} \varphi(x_i + \xi_i) - \varphi(x_i)\right) + o(\delta) \\ &= \sum_{i=1}^N e^{\ell_i} \left(\nabla^T F(\mu)(x_i) \cdot \xi_i + \frac{1}{4} \nabla^R F(\mu)(x_i) \zeta_i \right) + o(\delta). \end{aligned}$$

This proves the claimed Fréchet differentiability. □

The preceding formula also admits a Riemannian interpretation. Writing $w_i = e^{\ell_i}$, the identity $\dot{\ell}_i = \dot{w}_i / w_i$ identifies logarithmic weight velocities with relative mass variations. We therefore endow M_N with the Riemannian metric defined, for $\hat{\xi}^a = (\xi_i^a)_{i=1}^N$ and $\hat{\zeta}^a = (\zeta_i^a)_{i=1}^N$, $a = 1, 2$, by

$$g_{(\hat{x},\hat{\ell})}\big((\hat{\xi}^1, \hat{\zeta}^1), (\hat{\xi}^2, \hat{\zeta}^2)\big) \coloneqq \sum_{i=1}^N e^{\ell_i} \left(\xi_i^1 \cdot \xi_i^2 + \frac{1}{4} \zeta_i^1 \zeta_i^2 \right). \tag{90}$$

By (89), the corresponding Riemannian gradient is

$$\mathrm{grad}_g \, \widetilde{F}(\hat{x}, \hat{\ell}) = \left(\big(\nabla^T F(\mu)(x_i)\big)_{i=1}^N, \big(\nabla^R F(\mu)(x_i)\big)_{i=1}^N \right), \tag{91}$$

where $\mu \coloneqq \iota_N(\hat{x}, \hat{\ell})$.

When the positions are pairwise distinct, the map assigning to each tangent vector $(\hat{\xi}, \hat{\zeta})$ the field taking the value (ξ_i, ζ_i) at x_i identifies this metric with the scalar product of $H(\mu)$.

5 Differentiability of the squared Wasserstein–Fisher–Rao distance

We apply the preceding calculus to the functional

$$F_{\hat{\mu}}(\mu) \coloneqq \frac{1}{2} \mathrm{WFR}^2(\mu, \hat{\mu}), \qquad \hat{\mu} \in \mathcal{M}\big(\mathbb{R}^d\big).$$

The main obstructions are source splitting, which entails a loss of energy under barycentric averaging; target creation, whose cost is invisible to source tangent fields; and non-uniqueness at the transport cutoff. The proof adapts the two-competitor argument of Alfonsi and Jourdain for the squared Wasserstein distance [1], gluing cone couplings after suitable radial dilations. Its main additional ingredient is the stability of source logarithms and creation mass for asymptotically optimal cone couplings.

5.1 The differentiability theorem

We formulate the Monge condition after source normalization by radial dilation. Let

$$\vartheta(\mathfrak{y}_0,\mathfrak{y}_2) \coloneqq \begin{cases} r_0, & r_0>0,\\ 1, & r_0=0. \end{cases}$$

We say that $(\mu,\hat\mu)$ admits a unique source-normalized optimal Monge representation without target creation if there exist Borel maps $T\colon \mathbb{R}^d\to\mathbb{R}^d$ and $q\colon \mathbb{R}^d\to[0,\infty)$ such that, writing

$$A^* \coloneqq \big(x\mapsto([x,1],[T(x),q(x)])\big)_{\#}\mu, \tag{92}$$

every $A\in \mathrm{Opt}\Gamma_{\mathfrak{C}}(\mu,\hat\mu)$ satisfies

$$\mathrm{dil}_{\vartheta,2}\big(A\llcorner\{r_0>0\}\big)=A^*, \qquad \int_{\{r_0=0\}} r_2^2\,dA=0. \tag{93}$$

The value of T on $\{q=0\}$ is irrelevant. The coupling A^* is optimal, and $T_{\#}(q^2\mu)=\hat\mu$. In particular, $q\in L^2(\mu)$.

Moreover, the set

$$R_{\mathrm{Cut}} \coloneqq \Big\{x : q(x)>0,\ \|T(x)-x\|=\frac{\pi}{2}\Big\}$$

satisfies $\mu(R_{\mathrm{Cut}})=0$. Indeed, optimal cone couplings are concentrated on reduced pairs by Theorem 2.5. If R_{Cut} had positive μ-measure, replacing each pair $([x,1],[T(x),q(x)])$ over R_{Cut} by $([x,1],\mathfrak{o})$ and $(\mathfrak{o},[T(x),q(x)])$ would preserve the homogeneous marginals and the cost, since

$$\mathrm{d}^2_{\mathfrak{C}}([x,1],[T(x),q(x)])=1+q(x)^2=\mathrm{d}^2_{\mathfrak{C}}([x,1],\mathfrak{o})+\mathrm{d}^2_{\mathfrak{C}}(\mathfrak{o},[T(x),q(x)]).$$

The resulting optimal coupling $\widetilde{A}$ would satisfy

$$\int_{\{r_0=0\}} r_2^2\,d\widetilde{A}=\int_{R_{\mathrm{Cut}}} q(x)^2\,d\mu(x)>0,$$

contradicting (93).

Define

$$\mathcal{L}(x) \coloneqq \mathcal{L}_{\mathfrak{C}}([x,1],[T(x),q(x)]) = \big(q(x)\nabla K_{T(x)}(x),\,2\big(q(x)K_{T(x)}(x)-1\big)\big). \tag{94}$$

The semicoupling

$$\gamma^* \coloneqq \big((\mathrm{Id},T)_{\#}\mu,(\mathrm{Id},T)_{\#}(q^2\mu)\big)$$

is optimal and satisfies $\mathcal{L}=\overline{\mathrm{Log}}_{\gamma^*}$. Hence Theorem 3.11 gives

$$\mathcal{L}\in \mathrm{T}_\mu\mathcal{M}\big(\mathbb{R}^d\big), \qquad \|\mathcal{L}\|_\mu^2=\int_{\mathfrak{C}^2}\mathrm{d}^2_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_2)\,dA^*(y_0,\mathfrak{y}_2)=\mathrm{WFR}^2(\mu,\hat\mu). \tag{95}$$

The energy identity follows from (98) and (92).

Theorem 5.1 (Differentiability of the squared distance)**.** *Let $\mu,\hat\mu\in\mathcal{M}\big(\mathbb{R}^d\big)$. The following conditions are equivalent.*

(i) $F_{\hat\mu}$ is WFR differentiable at μ.

(ii) There is exactly one regular barycentric logarithm $\mathcal{L}$ between μ and $\hat{\mu}$, and

$$\|\mathcal{L}\|_\mu = \mathrm{WFR}(\mu, \hat{\mu}).$$

(iii) The pair $(\mu, \hat{\mu})$ has the unique source-normalized Monge representation without target creation in (93).

Whenever these conditions hold, $\mathcal{L}$ is the field defined in (94)*, and*

$$\nabla_{\mathrm{WFR}} F_{\hat{\mu}}(\mu) = -\mathcal{L}.$$

More precisely, for every $\mu_n \to \mu$ in WFR and every $\gamma^n \in \mathrm{OptSC}(\mu, \mu_n)$,

$$\frac{1}{2}\mathrm{WFR}^2(\mu_n, \hat{\mu}) = \frac{1}{2}\mathrm{WFR}^2(\mu, \hat{\mu}) - \big\langle \mathcal{L}, \overline{\mathrm{Log}}_{\gamma^n} \big\rangle_\mu + o\big(\mathrm{WFR}(\mu, \mu_n)\big). \tag{96}$$

Example 5.2 (Dirac configurations). Let $\mu = \delta_0 \in \mathcal{M}(\mathbb{R})$ and fix $0 < \rho < \pi/2 < R$. We consider one differentiable case and three obstructions. All uniqueness statements concern source-normalized cone couplings (vertex–vertex mass being immaterial).

(a) *Transport of a single atom.* Let $\hat{\mu} = \delta_\rho$. The unique source-normalized optimal cone coupling is $\delta_{([0,1],[\rho,1])}$, and its barycentric logarithm is

$$\mathcal{L} = (\sin(\rho), 2(\cos(\rho) - 1)).$$

In particular,

$$\|\mathcal{L}\|_\mu^2 = 2(1 - \cos(\rho)) = \mathrm{WFR}^2(\mu, \hat{\mu}).$$

Thus $F_{\hat{\mu}}$ is differentiable at μ.

(b) *Splitting of the source mass.* Let $\hat{\mu} = \frac{1}{2}(\delta_{-\rho} + \delta_\rho)$. The unique source-normalized optimal cone coupling is

$$\frac{1}{2}\delta_{([0,1],[-\rho,1])} + \frac{1}{2}\delta_{([0,1],[\rho,1])}.$$

Indeed, strict concavity of the square root in the correlation term forces the source mass to be divided equally between the two target atoms. The transport components cancel under barycentric averaging, giving

$$\mathcal{L} = (0, 2(\cos(\rho) - 1)).$$

Consequently,

$$\mathrm{WFR}^2(\mu, \hat{\mu}) - \|\mathcal{L}\|_\mu^2 = 2(1 - \cos(\rho)) - (1 - \cos(\rho))^2 = \sin^2(\rho) > 0.$$

Hence $F_{\hat{\mu}}$ is not differentiable at μ: uniqueness after source normalization does not exclude energy loss through barycentric averaging.

(c) *Non-uniqueness at the transport threshold.* Let $\hat{\mu} = \delta_{\pi/2}$. Both direct transport and separate annihilation and creation are optimal, as represented by the cone couplings

$$A^T = \delta_{([0,1],[\pi/2,1])}, \qquad A^R = \delta_{([0,1],\mathfrak{o})} + \delta_{(\mathfrak{o},[\pi/2,1])}.$$

Both have cost 2, while their barycentric logarithms are

$$\mathcal{L}^T = (1, -2), \qquad \mathcal{L}^R = (0, -2).$$

Thus $F_{\hat{\mu}}$ is not differentiable at μ. In particular, the equality $\big\|\mathcal{L}^T\big\|_\mu^2 = \mathrm{WFR}^2(\mu, \hat{\mu})$ for one optimal logarithm does not suffice.

(d) *Creation at the target.* Let $\hat{\mu} = \delta_0 + \delta_R$. An optimal cone coupling is

$$\delta_{([0,1],[0,1])} + \delta_{(\mathfrak{o},[R,1])}.$$

Its source-normalized part is unique, and its barycentric logarithm is $\mathcal{L} = 0$, whereas

$$\mathrm{WFR}^2(\mu, \hat{\mu}) - \|\mathcal{L}\|_\mu^2 = 1.$$

The energy defect is entirely due to creation at R, and $F_{\hat{\mu}}$ is not differentiable at μ.

Remark 5.3 (Absolutely continuous sources: a sufficient condition). The normalized Monge condition can hold for absolutely continuous sources. In this setting, the optimality and uniqueness results of [24, Theorems 6.3, 6.6 and 7.20] yield this representation when the target has no pure-creation part; see also [25, Corollary 3.5]. In particular, a sufficient condition is that μ is absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}^d$ and

$$\hat{\mu}\left(\left\{y \in \mathbb{R}^d : \mathrm{dist}(y, \operatorname{supp}\mu) < \frac{\pi}{2}\right\}\right) = \hat{\mu}(\mathbb{R}^d).$$

The unique active transport map and marginal densities then determine q, while the remaining source part is sent to the vertex. No positive lower bound on q or uniform gap from the transport cutoff is required.

Remark 5.4 (Absolutely continuous sources: target creation). Absolute continuity of the source alone does not ensure differentiability. Let μ be a nonzero absolutely continuous measure with compact support, choose $z \in \mathbb{R}^d$ such that

$$\mathrm{dist}(z, \operatorname{supp}\mu) > \frac{\pi}{2},$$

and set $\hat{\mu} = \mu + m\delta_z$, with $m > 0$. For $\mu^\varepsilon = \mu + \varepsilon^2\delta_z$, $\varepsilon > 0$, separation of the supports gives

$$\mathrm{WFR}(\mu, \mu^\varepsilon) = \varepsilon, \qquad \mathrm{WFR}^2(\mu^\varepsilon, \hat{\mu}) - \mathrm{WFR}^2(\mu, \hat{\mu}) = -2\sqrt{m}\,\varepsilon + \varepsilon^2.$$

The semicoupling

$$\gamma^\varepsilon := \big((\mathrm{Id}, \mathrm{Id})_\#\mu, (\mathrm{Id}, \mathrm{Id})_\#\mu + \varepsilon^2\delta_{(z,z)}\big)$$

is optimal and satisfies $\overline{\mathrm{Log}}_{\gamma^\varepsilon} = 0$. Hence the displayed linear variation cannot be represented by a source tangent field.

The no-creation condition concerns only the fixed pair $(\mu, \hat{\mu})$: the perturbations μ_n in Theorem 5.1 may contain creation mass, whose contribution is controlled by Lemma 5.6.

5.2 Stability and first variation

We first show that, under the source-normalized Monge condition, asymptotically optimal cone couplings have source logarithms converging to $\mathcal{L}$, with vanishing target creation. We then use this stability to obtain an integrated first-variation formula for small perturbations. These two lemmas are the main ingredients in the sufficiency part of Theorem 5.1.

The stability argument also involves non-reduced cone pairs. We therefore use the kernel K_y from Section 2.3 to extend the normalized cone logarithm of Remark 3.3 to arbitrary cone pairs. By Lemma 2.8, for $\mathfrak{y}_i = [x_i, r_i]$ with $r_i > 0$ and $\mathfrak{y}_j = [x_j, r_j]$,

$$\mathcal{L}_{\mathfrak{e}}(\mathfrak{y}_i, \mathfrak{y}_j) = \left(\frac{r_j}{r_i}\nabla K_{x_j}(x_i), 2\left(\frac{r_j}{r_i}K_{x_j}(x_i) - 1\right)\right). \tag{97}$$

When $\mathfrak{y}_j = \mathfrak{o}$, this means $\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_i, \mathfrak{o}) = (0, -2)$. At a source vertex we retain the convention $\mathcal{L}_{\mathfrak{C}}(\mathfrak{o}, \mathfrak{y}_j) = 0$. The identity $|K_y|^2 + \|\nabla K_y\|^2 = 1$ gives, for all cone pairs,

$$\mathrm{d}^2_{\mathfrak{C}}(\mathfrak{y}_i, \mathfrak{y}_j) = r_i^2 \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_i, \mathfrak{y}_j)\|^2_{\mathrm{WFR}} + \mathbf{1}_{\{r_i=0\}} r_j^2, \qquad \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_i, \mathfrak{y}_j) + (0, 2)\|_{\mathrm{WFR}} = \frac{r_j}{r_i} \quad (r_i > 0). \tag{98}$$

Thus the source logarithmic energy is bounded by the cone cost; its deficit at the source vertex is exactly the target-creation cost. Barycentric averaging can produce a further loss.

The identification with semicoupling logarithms remains restricted to reduced pairs, with the choice $u = 1$ at the cutoff; the identities above also hold for non-reduced pairs.

Lemma 5.5 (Stability of asymptotically optimal cone couplings)**.** *Assume* (93)*, and let* $E_n \in \Gamma_{\mathfrak{C}}(\mu, \hat{\mu})$ *satisfy*

$$c_n := \int_{\mathfrak{C}^2} \mathrm{d}^2_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_2)\, dE_n \longrightarrow \mathrm{WFR}^2(\mu, \hat{\mu}).$$

Then

$$\int_{\{r_0>0\}} r_0^2 \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_2) - \mathcal{L}(x_0)\|^2_{\mathrm{WFR}}\, dE_n \longrightarrow 0. \tag{99}$$

and

$$\int_{\{r_0=0\}} r_2^2\, dE_n \longrightarrow 0. \tag{100}$$

Proof. Discard vertex–vertex mass and apply the dilation

$$\vartheta([x_0, r_0], [x_2, r_2]) = (r_0^2 + r_2^2)^{1/2}.$$

All quantities in the statement are unchanged due to Lemma B.1, so we may assume without loss of generality that

$$r_0^2 + r_2^2 = 1 \quad E_n\text{-a.e.}, \qquad E_n(\mathfrak{C}^2) = \mu(\mathbb{R}^d) + \hat{\mu}(\mathbb{R}^d).$$

If both measures vanish, the conclusion is immediate.

The homogeneous-marginal tightness criterion [24, Lemma 7.3] and the fixed total mass give narrow relative compactness. Every narrow limit is concentrated on $\{r_0^2 + r_2^2 = 1\}$ and has homogeneous marginals μ and $\hat{\mu}$. Indeed, for $\varphi \in C_b(\mathbb{R}^d)$, the functions $(r_i^2 \wedge 1)\varphi(x_i)$ extend continuously by zero at the corresponding vertex and agree with $r_i^2\varphi(x_i)$ on this set. The cone cost is also continuous and bounded there. Consequently, every narrow limit belongs to $\mathrm{Opt}\Gamma_{\mathfrak{C}}(\mu, \hat{\mu})$.

For $\mathcal{V} = (\xi, \zeta) \in H(\mu)$, set

$$J_n(\mathcal{V}) := \int_{\{r_0>0\}} r_0^2\, \langle \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_2), \mathcal{V}(x_0)\rangle_{\mathrm{WFR}}\, dE_n.$$

If $\mathcal{V}$ is bounded and continuous, the integrand equals

$$r_0 r_2 \nabla K_{x_2}(x_0) \cdot \xi(x_0) + \frac{1}{2}\big(r_0 r_2 K_{x_2}(x_0) - r_0^2\big)\zeta(x_0).$$

It is bounded and continuous on the normalized set, including at the vertices. Every narrow limit is optimal, so (93) identifies its normalized source logarithm with $\mathcal{L}(x_0)$. It follows that

$$J_n(\mathcal{V}) \longrightarrow \langle \mathcal{L}, \mathcal{V}\rangle_\mu.$$

Here convergence holds for the whole sequence because every subsequence admits a further narrowly convergent subsequence giving the same limit.

By Cauchy–Schwarz and (98),

$$|J_n(\mathcal{V})| \le \sqrt{c_n}\,\|\mathcal{V}\|_\mu,$$

in particular, the linear mappings J_n are uniformly bounded. Since bounded continuous fields are dense in $H(\mu)$, the preceding convergence extends to every $\mathcal{V} \in H(\mu)$. In particular,

$$J_n(\mathcal{L}) \longrightarrow \|\mathcal{L}\|_\mu^2.$$

Finally, using the fixed source homogeneous marginal and (98), we obtain the exact identity

$$\int_{\{r_0>0\}} r_0^2 \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_2) - \mathcal{L}(x_0)\|_{\mathrm{WFR}}^2\, dE_n + \int_{\{r_0=0\}} r_2^2\, dE_n = c_n + \|\mathcal{L}\|_\mu^2 - 2J_n(\mathcal{L}).$$

The right-hand side converges to zero by (95). Since both terms on the left hand side are nonnegative the claim follows. □

We next establish the first-variation estimate used for both competitors. Only the perturbation pairs are assumed to be reduced; the pairs connecting the source to the fixed target may be non-reduced.

Lemma 5.6 (Integrated first variation)**.** *Let $\mathcal{L}$ be defined by* (94)*, and let $A_n \in \mathcal{M}_2(\mathfrak{C}^3)$ have homogeneous marginals $\mu, \mu_n, \hat{\mu}$ at endpoints $0, 1, 2$, respectively. Assume that the $(0,1)$-pairs are reduced and*

$$\varepsilon_n^2 := \int_{\mathfrak{C}^3} \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0,\mathfrak{y}_1)\, dA_n \longrightarrow 0.$$

Suppose also that

$$\begin{aligned} \delta_n^2 &:= \int_{\{r_0>0\}} r_0^2 \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_2) - \mathcal{L}(x_0)\|_{\mathrm{WFR}}^2\, dA_n \longrightarrow 0, \\ m_n &:= \int_{\{r_0=0\}} r_2^2\, dA_n \longrightarrow 0. \end{aligned} \tag{101}$$

Then

$$\int_{\mathfrak{C}^3} \big(\mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_1,\mathfrak{y}_2) - \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0,\mathfrak{y}_2)\big)\, dA_n = -2\int_{\{r_0>0\}} r_0^2\, \langle \mathcal{L}(x_0), \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)\rangle_{\mathrm{WFR}}\, dA_n + o(\varepsilon_n). \tag{102}$$

Proof. If $\varepsilon_n = 0$, then $\mathfrak{y}_0 = \mathfrak{y}_1$ for A_n-a.e. triple, so both the cost difference and the logarithmic pairing vanish. We may therefore restrict to $\varepsilon_n > 0$.

On $\{r_0 > 0\}$, put

$$q_1 := \frac{r_1}{r_0}, \qquad q_2 := \frac{r_2}{r_0}, \qquad \alpha := \frac{\mathrm{d}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)}{r_0}.$$

If $r_i = 0$, we choose the spatial representative $x_i = x_0$. Reducedness then gives $\|x_1 - x_0\| \le \pi/2$ and

$$\begin{aligned} \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1) &= \big(q_1 \mathbf{sin}(x_1 - x_0), 2(q_1\cos(\|x_1 - x_0\|) - 1)\big), \\ \alpha^2 &= (q_1 - 1)^2 + 2q_1(1 - \cos(\|x_1 - x_0\|)) = \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)\|_{\mathrm{WFR}}^2. \end{aligned}$$

Expanding the cone costs and the scalar product yields

$$\frac{\mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_1,\mathfrak{y}_2) - \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0,\mathfrak{y}_2)}{r_0^2} + 2\,\langle \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_2), \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)\rangle_{\mathrm{WFR}} = \alpha^2 - 2q_1q_2B, \tag{103}$$

where

$$B \coloneqq \cos(\|x_2 - x_1\| \wedge \pi) - K_{x_1}(x_0)K_{x_2}(x_0) - \nabla K_{x_1}(x_0) \cdot \nabla K_{x_2}(x_0).$$

To estimate B, write

$$\begin{aligned} B = {} & K_{x_2}(x_1) - K_{x_2}(x_0) - \nabla K_{x_2}(x_0) \cdot (x_1 - x_0) \\ & + \big(1 - K_{x_1}(x_0)\big) K_{x_2}(x_0) \\ & + \nabla K_{x_2}(x_0) \cdot \big(x_1 - x_0 - \nabla K_{x_1}(x_0)\big). \end{aligned}$$

Since $\mathrm{Lip}(\nabla K_{x_2}) \le 1$, the Taylor expansion of K_{x_2} bounds the first line in absolute value by $\frac{1}{2}\|x_1 - x_0\|^2$. Reducedness gives $\|x_1 - x_0\| \le \pi/2$, so

$$\begin{aligned} 0 \le 1 - K_{x_1}(x_0) = 1 - \cos(\|x_1 - x_0\|) \le \frac{1}{2}\|x_1 - x_0\|^2, \\ \|x_1 - x_0 - \nabla K_{x_1}(x_0)\| = \|x_1 - x_0\| - \sin(\|x_1 - x_0\|) \le \frac{\pi}{12}\|x_1 - x_0\|^2. \end{aligned}$$

Together with $|K_{x_2}| \le 1$ and $\|\nabla K_{x_2}\| \le 1$, these estimates yield $|B| \le C\|x_1 - x_0\|^2$.

Moreover, Cauchy–Schwarz and (19) give

$$|K_{x_1}(x_0)K_{x_2}(x_0) + \nabla K_{x_1}(x_0) \cdot \nabla K_{x_2}(x_0)| \le 1.$$

Therefore, we obtain $|B| \le 2$.

The identity defining α implies

$$\alpha^2 \ge 2q_1(1 - \cos(\|x_1 - x_0\|)) \ge \frac{4q_1}{\pi^2}\|x_1 - x_0\|^2.$$

Hence the quadratic estimate for B gives $q_1|B| \le C\alpha^2$. On the other hand, $\alpha^2 \ge (q_1 - 1)^2$ implies $q_1 \le 1 + \alpha$. Thus, on $\{\alpha \ge 1\}$, the bound $|B| \le 2$ gives

$$q_1|B| \le 2(1 + \alpha) \le 4\alpha.$$

Using the quadratic bound for $\alpha \le 1$ and the linear bound for $\alpha \ge 1$, we conclude that

$$q_1|B| \le C\alpha \min\{\alpha, 1\}.$$

Therefore

$$\left| \frac{\mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_1, \mathfrak{y}_2) - \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_2)}{r_0^2} + 2\,\langle \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_2), \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_1) \rangle_{\mathrm{WFR}} \right| \le \alpha^2 + Cq_2\alpha \min\{\alpha, 1\}. \tag{104}$$

Set

$$d\lambda_n \coloneqq r_0^2 \mathbf{1}_{\{r_0 > 0\}}\, dA_n.$$

Then $(x_0)_{\#}\lambda_n = \mu$ and $\int \alpha^2\, d\lambda_n \le \varepsilon_n^2$. By (98) and the definition of $\mathcal{L}$, the reverse triangle inequality gives

$$\begin{aligned} |q_2 - q(x_0)| &= \big|\|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_2) + (0, 2)\|_{\mathrm{WFR}} - \|\mathcal{L}(x_0) + (0, 2)\|_{\mathrm{WFR}}\big| \\ &\le \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_2) - \mathcal{L}(x_0)\|_{\mathrm{WFR}}. \end{aligned}$$

Define

$$\eta_n \coloneqq (x_0)_{\#}\big(\min\{\alpha^2, 1\}\lambda_n\big).$$

Since $(x_0)_\#\lambda_n = \mu$, we have

$$0 \le \eta_n \le \mu, \qquad \eta_n(\mathbb{R}^d) \le \int \alpha^2 \, d\lambda_n \le \varepsilon_n^2 \longrightarrow 0.$$

As $q^2 \in L^1(\mu)$, Lemma C.1 yields $\int q^2 \, d\eta_n \to 0$. Consequently,

$$\int q_2^2 \min\{\alpha^2, 1\} \, d\lambda_n \le 2\delta_n^2 + 2 \int q^2 \, d\eta_n \longrightarrow 0.$$

Cauchy–Schwarz therefore gives

$$\int q_2 \alpha \min\{\alpha, 1\} \, d\lambda_n \le \varepsilon_n \left(\int q_2^2 \min\{\alpha^2, 1\} \, d\lambda_n \right)^{1/2} = o(\varepsilon_n).$$

Multiplying (104) by r_0^2 and integrating now yields

$$\begin{aligned} &\left| \int_{\{r_0 > 0\}} \big(\mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_1, \mathfrak{y}_2) - \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_2) \big) \, dA_n + 2 \int \langle \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_2), \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_1) \rangle_{\mathrm{WFR}} \, d\lambda_n \right| \\ &\qquad \le \varepsilon_n^2 + C \int q_2 \alpha \min\{\alpha, 1\} \, d\lambda_n = o(\varepsilon_n). \end{aligned}$$

Replacing $\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_2)$ by $\mathcal{L}(x_0)$ in this pairing changes the expression by at most

$$\begin{aligned} &2 \left| \int \langle \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_2) - \mathcal{L}(x_0), \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_1) \rangle_{\mathrm{WFR}} \, d\lambda_n \right| \\ &\qquad \le 2\delta_n \left(\int \| \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_1) \|_{\mathrm{WFR}}^2 \, d\lambda_n \right)^{1/2} \le 2\delta_n \varepsilon_n = o(\varepsilon_n). \end{aligned}$$

Finally, on $\{r_0 = 0\}$, the cost difference is

$$\mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_1, \mathfrak{y}_2) - \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_2) = r_1^2 - 2 r_1 r_2 K_{x_2}(x_1),$$

where the product term is interpreted as zero if $r_1 r_2 = 0$. Since $\mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_1) = r_1^2$ on this set,

$$\int_{\{r_0 = 0\}} r_1^2 \, dA_n \le \varepsilon_n^2.$$

Using $|K_{x_2}| \le 1$, Cauchy–Schwarz, and the definition of m_n, we conclude that

$$\begin{aligned} \int_{\{r_0 = 0\}} \big| \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_1, \mathfrak{y}_2) - \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_2) \big| \, dA_n &\le \int_{\{r_0 = 0\}} r_1^2 \, dA_n + 2 \left(\int_{\{r_0 = 0\}} r_1^2 \, dA_n \right)^{1/2} \sqrt{m_n} \\ &\le \varepsilon_n^2 + 2\varepsilon_n \sqrt{m_n} = o(\varepsilon_n). \end{aligned}$$

Combining the estimates on $\{r_0 > 0\}$ and $\{r_0 = 0\}$ proves (102). □

5.3 Proof of differentiability and characterization

Proof of Theorem 5.1. We first prove $(i) \Rightarrow (ii)$. Set $D := \mathrm{WFR}^2(\mu, \hat{\mu})$ and suppose that $F_{\hat{\mu}}$ is WFR differentiable at μ, with gradient $\mathcal{V}$.

The descending metric slope of $F_{\hat{\mu}}$ at μ equals $\sqrt{D}$. Indeed, the triangle inequality gives

$$\frac{\max\{F_{\hat{\mu}}(\mu) - F_{\hat{\mu}}(\nu), 0\}}{\mathrm{WFR}(\mu,\nu)} \le \frac{\sqrt{D} + \mathrm{WFR}(\nu,\hat{\mu})}{2}, \qquad \nu \neq \mu.$$

For $D > 0$, any constant-speed geodesic $(\boldsymbol{\mu}_t)_{t\in[0,1]}$ from μ to $\hat{\mu}$ satisfies

$$\frac{F_{\hat{\mu}}(\mu) - F_{\hat{\mu}}(\boldsymbol{\mu}_t)}{\mathrm{WFR}(\mu,\boldsymbol{\mu}_t)} = \frac{(2-t)}{2}\sqrt{D} \longrightarrow \sqrt{D}.$$

For $D = 0$, the slope is zero since $F_{\hat{\mu}}$ attains its minimum at μ. Proposition 4.5 therefore gives

$$\|\mathcal{V}\|_\mu = \sqrt{D}.$$

Let $\mathcal{L}$ be any regular barycentric logarithm from μ to $\hat{\mu}$, and let $(\boldsymbol{\mu}_t)_{t\in[0,1]}$ be an associated geodesic. Since $F_{\hat{\mu}}(\boldsymbol{\mu}_t) = (1-t)^2 D/2$, Proposition 4.7 gives

$$\frac{(1-t)^2 D - D}{2} = t\langle \mathcal{V}, \mathcal{L}\rangle_\mu + o(t).$$

Consequently, $\langle \mathcal{V}, \mathcal{L}\rangle_\mu = -D$. Using $\|\mathcal{L}\|_\mu \le \sqrt{D}$, we obtain

$$D = -\langle \mathcal{V}, \mathcal{L}\rangle_\mu \le \|\mathcal{V}\|_\mu \|\mathcal{L}\|_\mu \le D.$$

Equality in Cauchy–Schwarz implies

$$\mathcal{L} = -\mathcal{V}, \qquad \|\mathcal{L}\|_\mu = \sqrt{D}.$$

Since $\mathcal{L}$ was arbitrary, this proves (ii). If $D = 0$, the bound $\|\mathcal{L}\|_\mu \le \sqrt{D}$ directly gives the same conclusion.

We next prove *(ii)* $\Rightarrow$ *(iii)*. Fix $A \in \mathrm{Opt}\Gamma_{\mathfrak{C}}(\mu,\hat{\mu})$ and write $\mathfrak{y}_0 = [x_0, r_0]$, $\mathfrak{y}_2 = [x_2, r_2]$. If $A(\mathfrak{C}^2) = 0$ the claim is trivial; otherwise we may assume that A is a probability measure, by constant dilation, which preserves the homogeneous marginals, the cost, and all source-weighted logarithmic integrals below. Let $\mathbf{\Pi}$ be a probability measure on $\mathrm{Geo}(\mathfrak{C})$ that generates A, which exists by Theorem 2.7. Let $Q(d\mathfrak{y}_0, d\mathfrak{y}_2) = r_0^2 A(d\mathfrak{y}_0, d\mathfrak{y}_2)$. Its spatial source marginal is $(x_0)_\# Q = \mu$, so we may disintegrate it as $Q = \int_{\mathbb{R}^d} Q_x \, d\mu(x)$. Let

$$\mathcal{L}_A(x) \coloneqq \int_{\mathfrak{C}^2} \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_2)\, dQ_x(\mathfrak{y}_0,\mathfrak{y}_2) = \mathcal{V}^{\mathbf{\Pi}}(x)$$

be the corresponding barycenter, where the last equality follows from $(e_0,e_1)_\#\mathbf{\Pi} = A$.

Proposition 3.8 and Theorem 3.11 show that $\mathcal{L}_A$ is a regular barycentric logarithm from μ to $\hat{\mu}$. Thus, by (ii),

$$\mathcal{L}_A = \mathcal{L}, \qquad \|\mathcal{L}_A\|_\mu = \mathrm{WFR}(\mu,\hat{\mu}).$$

Therefore, there is no energy loss due to barycentric averaging. The cone energy identity gives

$$\int_{\{r_0>0\}} r_0^2 \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_2)\|_{\mathrm{WFR}}^2 \, dA = \mathrm{WFR}(\mu,\hat{\mu})^2 - \int_{\{r_0=0\}} r_2^2 \, dA.$$

Moreover, the definition of the barycenter and the homogeneous marginal constraint imply

$$\int_{\{r_0>0\}} r_0^2 \, \langle \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_2), \mathcal{L}_A(x_0)\rangle_{\mathrm{WFR}} \, dA = \|\mathcal{L}_A\|_\mu^2,$$

$$\int_{\{r_0>0\}} r_0^2 \|\mathcal{L}_A(x_0)\|_{\mathrm{WFR}}^2 \, dA = \|\mathcal{L}_A\|_\mu^2.$$

Expanding the square therefore yields the exact norm deficit

$$\mathrm{WFR}(\mu,\hat{\mu})^2 - \|\mathcal{L}_A\|_\mu^2 = \int_{\{r_0>0\}} r_0^2 \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_2) - \mathcal{L}_A(x_0)\|_{\mathrm{WFR}}^2 \, dA + \int_{\{r_0=0\}} r_2^2 \, dA. \tag{105}$$

Its left-hand side vanishes. Both terms on the right are nonnegative, and $r_0^2 > 0$ on $\{r_0 > 0\}$. Hence every optimal A satisfies

$$\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_2) = \mathcal{L}(x_0) \quad A\text{-a.e. on } \{r_0>0\}, \qquad \int_{\{r_0=0\}} r_2^2 \, dA = 0.$$

We recover common Borel maps T, q from $\mathcal{L} = (\xi,\zeta)$. Set

$$a(x) \coloneqq 1 + \frac{\zeta(x)}{2}, \qquad q(x) \coloneqq \sqrt{a(x)^2 + \|\xi(x)\|^2}.$$

Optimal cone pairs are reduced. Therefore, on $\{r_0 > 0, r_2 > 0\}$, writing $\|x_2 - x_0\| \le \pi/2$, we have

$$a(x_0) = \frac{r_2}{r_0}\cos(\|x_2 - x_0\|), \qquad \xi(x_0) = \frac{r_2}{r_0}\sin(x_2 - x_0)$$

On $\{r_0 > 0, r_2 = 0\}$, both $a(x_0)$ and $\xi(x_0)$ vanish. Consequently,

$$a \ge 0 \quad \mu\text{-a.e.}, \qquad q(x_0) = \frac{r_2}{r_0} \quad A\text{-a.e. on } \{r_0>0\}.$$

After modifying the representative on a μ-null set, assume $a \ge 0$ everywhere and define

$$T(x) \coloneqq \begin{cases} x + \dfrac{\arccos(a(x)/q(x))}{\|\xi(x)\|}\xi(x), & \xi(x) \neq 0, \\ x, & \xi(x) = 0. \end{cases}$$

This defines a Borel map: when $\xi(x) \neq 0$, the identity $q(x)^2 = a(x)^2 + \|\xi(x)\|^2$ ensures that $q(x) > 0$ and $a(x)/q(x) \in [0,1]$.

To verify that T recovers the target position, consider a pair with $r_0 > 0$ and $\xi(x_0) \neq 0$. Then $r_2 = q(x_0)r_0 > 0$, and the preceding identities give

$$\frac{a(x_0)}{q(x_0)} = \cos(\|x_2 - x_0\|), \qquad \frac{\xi(x_0)}{\|\xi(x_0)\|} = \frac{x_2 - x_0}{\|x_2 - x_0\|}.$$

Since reducedness gives $\|x_2 - x_0\| \in (0, \pi/2]$, we have $\arccos(a(x_0)/q(x_0)) = \|x_2 - x_0\|$, hence $T(x_0) = x_2$.

If $q(x_0) > 0$ and $\xi(x_0) = 0$, then $q(x_0)\sin(\|x_2 - x_0\|) = \|\xi(x_0)\| = 0$. Since $\|x_2 - x_0\| \in [0, \pi/2]$, this implies $\|x_2 - x_0\| = 0$, so again $T(x_0) = x_0 = x_2$.

If $q(x_0) = 0$, the target is the vertex and the value of $T(x_0)$ is irrelevant. Thus every optimal A satisfies

$$\mathfrak{y}_2 = [T(x_0), q(x_0)r_0] \quad A\text{-a.e. on } \{r_0>0\}.$$

The maps are common to all optimizers because they depend only on $\mathcal{L}$. Together with the vanishing creation term, this proves (93), hence (iii).

Finally, we prove $(iii) \Rightarrow (i)$ and the stated expansion. Assume (iii), and use the corresponding maps T, q, optimal cone plan $A^\star$, and field $\mathcal{L}$ defined above. In particular, $\mathcal{L} \in \mathrm{T}_\mu\mathcal{M}(\mathbb{R}^d)$ and $\|\mathcal{L}\|_\mu^2 = \mathrm{WFR}^2(\mu,\hat{\mu})$.

Fix $\mu_n \to \mu$ in WFR and arbitrary $\gamma^n \in \mathrm{OptSC}(\mu, \mu_n)$. Write

$$d_n \coloneqq \mathrm{WFR}(\mu, \mu_n), \qquad D \coloneqq \mathrm{WFR}^2(\mu, \hat{\mu}), \qquad D_n \coloneqq \mathrm{WFR}^2(\mu_n, \hat{\mu}).$$

Terms with $d_n = 0$ are immediate and can be omitted. By Proposition 3.9, with the default choice $u = 1$, choose a reduced optimal cone coupling B_n representing the prescribed barycentric logarithm $\overline{\mathrm{Log}}_{\gamma^n}$. Thus

$$\begin{aligned} \int_{\mathfrak{C}^2} \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_1)\, dB_n &= d_n^2, \\ \int_{\{r_0>0\}} r_0^2 \left\langle \mathcal{L}(x_0), \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_1) \right\rangle_{\mathrm{WFR}} dB_n &= \left\langle \mathcal{L}, \overline{\mathrm{Log}}_{\gamma^n} \right\rangle_{\mu}. \end{aligned} \tag{106}$$

These identities are preserved by the dilations and vertex–vertex modifications in Lemma B.2.

Upper bound. Glue B_n and $A^\star$ at their common source μ, using Lemma B.2 after relabelling the coordinates, and denote the resulting triple plan by A_n^+. Its $(0,2)$ marginal is a dilation of $A^\star$, up to vertex–vertex mass. Consequently,

$$\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_2) = \mathcal{L}(x_0) \quad A_n^+\text{-a.e. on } \{r_0 > 0\}, \qquad \int_{\{r_0=0\}} r_2^2\, dA_n^+ = 0.$$

Moreover,

$$\int_{\mathfrak{C}^3} \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_1)\, dA_n^+ = d_n^2, \qquad \int_{\mathfrak{C}^3} \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_2)\, dA_n^+ = D.$$

The $(1,2)$ marginal is admissible for $(\mu_n, \hat{\mu})$. Applying Lemma 5.6 and (106) gives

$$\frac{D_n - D}{2} \le \frac{1}{2} \int_{\mathfrak{C}^3} \left(\mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_1, \mathfrak{y}_2) - \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_2) \right) dA_n^+ = -\left\langle \mathcal{L}, \overline{\mathrm{Log}}_{\gamma^n} \right\rangle_{\mu} + o(d_n). \tag{107}$$

Lower bound. Choose $C_n \in \mathrm{Opt}\Gamma_{\mathfrak{C}}(\mu_n, \hat{\mu})$ and glue it to B_n at their common endpoint μ_n, again using Lemma B.2. Denote the resulting triple plan by A_n^- and set

$$E_n := (\pi_{02})_{\#} A_n^-.$$

Then $E_n \in \Gamma_{\mathfrak{C}}(\mu, \hat{\mu})$, and the prescribed pair costs satisfy

$$\int_{\mathfrak{C}^3} \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_1)\, dA_n^- = d_n^2, \qquad \int_{\mathfrak{C}^3} \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_1, \mathfrak{y}_2)\, dA_n^- = D_n.$$

The cone triangle inequality and Minkowski's inequality yield

$$\sqrt{D} \le \left(\int_{\mathfrak{C}^2} \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_2)\, dE_n \right)^{1/2} \le d_n + \sqrt{D_n}.$$

Since

$$\left| \sqrt{D_n} - \sqrt{D} \right| \le d_n \longrightarrow 0,$$

the couplings E_n are asymptotically optimal:

$$\int_{\mathfrak{C}^2} \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0, \mathfrak{y}_2)\, dE_n \longrightarrow D.$$

Lemma 5.5 therefore gives

$$\int_{\{r_0>0\}} r_0^2 \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_2) - \mathcal{L}(x_0)\|_{\mathrm{WFR}}^2 \, dA_n^- \longrightarrow 0, \qquad \int_{\{r_0=0\}} r_2^2 \, dA_n^- \longrightarrow 0.$$

Thus Lemma 5.6 applies to A_n^- as well. Using admissibility of E_n and (106), we obtain

$$\frac{D_n - D}{2} \geq \frac{1}{2}\int_{\mathfrak{C}^3} \big(\mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_1,\mathfrak{y}_2) - \mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_0,\mathfrak{y}_2)\big)\, dA_n^- = -\left\langle \mathcal{L}, \overline{\mathrm{Log}}_{\gamma^n} \right\rangle_{\mu} + o(d_n). \tag{108}$$

Combining (107) and (108) proves (96). Since $\mathcal{L} \in \mathrm{T}_\mu\mathcal{M}\big(\mathbb{R}^d\big)$, this is WFR differentiability at μ, with

$$\nabla_{\mathrm{WFR}} F_{\hat{\mu}}(\mu) = -\mathcal{L}.$$

□

6 Restriction to probability measures

In this last section, we consider the restriction of our calculus to the space of probability measures.

A time-dependent tangent vector $\mathcal{V}_t = (\xi_t, \zeta_t)$ acts on a narrowly continuous curve of measures through the continuity equation with reaction

$$\partial_t \mu_t + \mathrm{div}(\xi_t \mu_t) = \zeta_t \mu_t, \qquad t \in [0,1], \tag{109}$$

with the standard integrability conditions

$$\int_0^1 \int_{\mathbb{R}^d} \|\xi_t(x)\|^2 \, d\mu_t(x) dt < +\infty, \quad \int_0^1 \int_{\mathbb{R}^d} |\zeta_t(x)|^2 \, d\mu_t(x) dt < +\infty.$$

In particular, testing against compactly supported cutoffs converging to $\varphi \equiv 1$ yields

$$\frac{d}{dt}\mu_t(\mathbb{R}^d) = \int_{\mathbb{R}^d} \zeta_t(x)\, d\mu_t(x) \tag{110}$$

for almost every t, and hence a curve taking values in $\mathcal{P}(\mathbb{R}^d)$ must satisfy

$$\int_{\mathbb{R}^d} \zeta_t(x)\, d\mu_t(x) = 0 \tag{111}$$

for almost every t. This motivates defining, for $\mu \in \mathcal{P}(\mathbb{R}^d)$,

$$T_\mu\mathcal{P}(\mathbb{R}^d) \coloneqq \left\{ (\xi,\zeta) \in T_\mu\mathcal{M}\big(\mathbb{R}^d\big) \colon \int_{\mathbb{R}^d} \zeta(x)\, d\mu(x) = 0 \right\}. \tag{112}$$

Since the map $(\xi,\zeta) \longmapsto \int_{\mathbb{R}^d} \zeta \, d\mu$ is continuous on $T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$, the space $T_\mu\mathcal{P}(\mathbb{R}^d)$ is a closed subspace of $T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$. Moreover, one has $(0,c) \in T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$ for every constant $c \in \mathbb{R}$, as follows by approximating constant potentials by compactly supported smooth cutoffs. As a consequence, when $\mu \in \mathcal{P}(\mathbb{R}^d)$, the WFR-orthogonal projection of $T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$ onto $T_\mu\mathcal{P}(\mathbb{R}^d)$ is given by

$$\Pi_\mu(\xi,\zeta) \coloneqq \left(\xi, \zeta - \int_{\mathbb{R}^d} \zeta(x)\, d\mu(x)\right). \tag{113}$$

Thus, the projection leaves the transport component unchanged and only acts on the reaction component by removing its mean.

The proposed construction is analogous to the standard Riemannian-submanifold setting, where the gradient of the restriction of a smooth function to a submanifold is obtained by orthogonally projecting the ambient gradient onto the tangent space of the submanifold.

Definition 6.1. We say that $F\colon \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$ is WFR differentiable at $\mu \in \mathcal{P}(\mathbb{R}^d)$ if there exists

$$\mathcal{V} = (\xi, \zeta) \in T_\mu \mathcal{P}(\mathbb{R}^d)$$

such that, for every sequence $(\mu_n)_{n\in\mathbb{N}} \subset \mathcal{P}(\mathbb{R}^d)$ satisfying $\mathrm{WFR}(\mu_n, \mu) \to 0$, and every sequence of optimal semicouplings

$$\boldsymbol{\gamma}^n = (\gamma_0^n, \gamma_1^n) \in \mathrm{OptSC}(\mu, \mu_n),$$

one has

$$F(\mu_n) = F(\mu) + \langle \mathcal{V}, \Pi_\mu \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \rangle_\mu + o\big(\mathrm{WFR}(\mu, \mu_n)\big) \tag{114}$$

as $n \to +\infty$.

The previous definition offers an advantage over the classical Wasserstein differentiability notion in that it works in the full space of probabilities without requiring any finiteness conditions on the second moments. Moreover, differently from what happens for flat derivatives, the reaction component of the gradient is canonical, since adding a constant to it would make the gradient not tangent anymore.

We notice that, since the gradient is assumed to lie in $T_\mu \mathcal{P}(\mathbb{R}^d)$, by standard properties of the projection

$$\langle \mathcal{V}, \Pi_\mu \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \rangle_\mu = \langle \Pi_\mu \mathcal{V}, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \rangle_\mu = \langle \mathcal{V}, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \rangle_\mu$$

so that taking the projection also of the logarithm maps to make them tangent to $\mathcal{P}(\mathbb{R}^d)$ is irrelevant and one can rewrite (114) as

$$F(\mu_n) = F(\mu) + \langle \mathcal{V}, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \rangle_\mu + o\big(\mathrm{WFR}(\mu, \mu_n)\big)$$

Similarly to the case of the ambient gradients in $\mathcal{M}\big(\mathbb{R}^d\big)$ we have the following result.

Proposition 6.2 (Strength and uniqueness on probability measures)**.** *Let $F\colon \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$, let $\mu \in \mathcal{P}(\mathbb{R}^d)$, and let $\widetilde{\mathcal{V}} \in T_\mu \mathcal{P}(\mathbb{R}^d)$ satisfy* (114)*. Then, for every sequence $(\mu_n)_{n\in\mathbb{N}} \subset \mathcal{P}(\mathbb{R}^d)$ and every sequence of semicouplings $\boldsymbol{\gamma}^n \in \mathrm{SC}(\mu, \mu_n)$ such that $\mathcal{C}(\boldsymbol{\gamma}^n) \to 0$, one has*

$$F(\mu_n) = F(\mu) + \langle \widetilde{\mathcal{V}}, \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \rangle_\mu + o\left(\sqrt{\mathcal{C}(\boldsymbol{\gamma}^n)}\right). \tag{115}$$

In particular, if F is differentiable at μ according to Definition 6.1, its gradient in $T_\mu \mathcal{P}(\mathbb{R}^d)$ is unique, and we denote it by

$$\widetilde{\nabla}_{\mathrm{WFR}} F(\mu) = \big(\widetilde{\nabla}^T F(\mu), \widetilde{\nabla}^R F(\mu)\big). \tag{116}$$

Proof. The proof of strength in the ambient space, see Theorem 4.3, applies unchanged when the endpoints are restricted in $\mathcal{P}(\mathbb{R}^d)$.

We now prove uniqueness. Let $\widetilde{\mathcal{V}}_1$ and $\widetilde{\mathcal{V}}_2$ be two elements of $T_\mu \mathcal{P}(\mathbb{R}^d)$ satisfying (114). For $\varphi \in C_c^\infty(\mathbb{R}^d)$, set

$$T \coloneqq \nabla \varphi, \qquad \rho \coloneqq 4\left(\varphi - \int_{\mathbb{R}^d} \varphi \, d\mu\right).$$

For $\varepsilon > 0$, consider

$$\mu_\varepsilon \coloneqq (\mathrm{Id} + \varepsilon T)_\# (q_\varepsilon^2 \mu),$$

where

$$q_\varepsilon \coloneqq \frac{1 + \varepsilon\rho/2}{\sqrt{Z_\varepsilon}}, \qquad Z_\varepsilon \coloneqq 1 + \frac{\varepsilon^2}{4} \int_{\mathbb{R}^d} \rho^2 \, d\mu.$$

Since $\int \rho \, d\mu = 0$, one has $\int q_\varepsilon^2 \, d\mu = 1$, and hence $\mu_\varepsilon \in \mathcal{P}(\mathbb{R}^d)$. Define

$$G_\varepsilon \coloneqq (\mathrm{Id}, \mathrm{Id} + \varepsilon T), \qquad \boldsymbol{\gamma}^\varepsilon \coloneqq \big((G_\varepsilon)_\# \mu, (G_\varepsilon)_\# (q_\varepsilon^2 \mu)\big) \in \mathrm{SC}(\mu, \mu_\varepsilon).$$

The normalized square-root weight can be written as

$$q_\varepsilon = 1 + \frac{\varepsilon}{2} \rho_\varepsilon, \qquad \rho_\varepsilon \coloneqq \frac{\rho}{\sqrt{Z_\varepsilon}} + \frac{2}{\varepsilon} \left(\frac{1}{\sqrt{Z_\varepsilon}} - 1 \right).$$

Since $Z_\varepsilon = 1 + O(\varepsilon^2)$ and μ is a probability measure, we have $Z_\varepsilon^{-1/2} - 1 = O(\varepsilon^2)$ and

$$\|\rho_\varepsilon - \rho\|_{L^2(\mu)} \le |Z_\varepsilon^{-1/2} - 1| \left(\|\rho\|_{L^2(\mu)} + \frac{2}{\varepsilon} \right) = O(\varepsilon).$$

Thus $\rho_\varepsilon \to \rho$ in $L^2(\mu)$. Lemma A.1, with $T_\varepsilon = T$, therefore yields

$$\frac{\overline{\mathrm{Log}}_{\boldsymbol{\gamma}^\varepsilon}}{\varepsilon} \longrightarrow (T, \rho) \quad \text{in } H(\mu), \qquad \sqrt{\mathcal{C}(\boldsymbol{\gamma}^\varepsilon)} = O(\varepsilon).$$

Applying (115) to each $\widetilde{\mathcal{V}}_i$, we obtain

$$F(\mu_\varepsilon) - F(\mu) = \varepsilon \langle \widetilde{\mathcal{V}}_i, (T, \rho) \rangle_\mu + o(\varepsilon), \qquad i = 1, 2.$$

Subtracting these identities, dividing by ε, and passing to the limit gives

$$\left\langle \widetilde{\mathcal{V}}_1 - \widetilde{\mathcal{V}}_2, \left(\nabla \varphi, 4 \left(\varphi - \int_{\mathbb{R}^d} \varphi \, d\mu \right) \right) \right\rangle_\mu = 0.$$

The centered potential fields appearing here are the images of $(\nabla\varphi, 4\varphi)$ under Π_μ. They are therefore dense in $T_\mu \mathcal{P}(\mathbb{R}^d)$. Since $\widetilde{\mathcal{V}}_1 - \widetilde{\mathcal{V}}_2 \in T_\mu \mathcal{P}(\mathbb{R}^d)$, we conclude that $\widetilde{\mathcal{V}}_1 = \widetilde{\mathcal{V}}_2$. □

Remark 6.3 (Metric slope on probability measures). The identification of the metric slope extends to the restriction of WFR to $\mathcal{P}(\mathbb{R}^d)$. Indeed, if $F \colon \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$ is WFR differentiable at μ, then

$$|\partial^- F|_{\mathcal{P}}(\mu) = \|\widetilde{\nabla}_{\mathrm{WFR}} F(\mu)\|_\mu,$$

where

$$|\partial^- F|_{\mathcal{P}}(\mu) \coloneqq \limsup_{\substack{\nu \to \mu, \ \nu \neq \mu \\ \nu \in \mathcal{P}(\mathbb{R}^d)}} \frac{(F(\mu) - F(\nu))_+}{\mathrm{WFR}(\mu, \nu)}.$$

The proof follows as in the ambient case, using Proposition 6.2 and the mass-normalized graph perturbations constructed in its proof for the lower bound.

WFR differentiable functions are differentiable according to Definition 6.1 when restricted to $\mathcal{P}(\mathbb{R}^d)$.

Proposition 6.4. *Assume $F \colon \mathcal{M}(\mathbb{R}^d) \to \mathbb{R}$ is Wasserstein–Fisher–Rao differentiable at $\mu \in \mathcal{P}(\mathbb{R}^d)$. Then its restriction $F_{|\mathcal{P}} \colon \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$ is Wasserstein–Fisher–Rao differentiable at μ and*

$$\widetilde{\nabla}_{\mathrm{WFR}} F(\mu) = \Pi_\mu \nabla_{\mathrm{WFR}} F(\mu). \tag{117}$$

Proof. Write

$$\widetilde{\mathcal{V}} \coloneqq \Pi_\mu \nabla_{\mathrm{WFR}} F(\mu) \in T_\mu \mathcal{P}(\mathbb{R}^d).$$

Let $\mu_n \in \mathcal{P}(\mathbb{R}^d)$ satisfy $\mathrm{WFR}(\mu_n, \mu) \to 0$, and let $\boldsymbol{\gamma}^n = (\gamma_0^n, \gamma_1^n) \in \mathrm{OptSC}(\mu, \mu_n)$. By the WFR differentiability of F in $\mathcal{M}\big(\mathbb{R}^d\big)$,

$$F(\mu_n) - F(\mu) = \big\langle \nabla_{\mathrm{WFR}} F(\mu), \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \big\rangle_\mu + o\big(\mathrm{WFR}(\mu, \mu_n)\big).$$

The definition of Π_μ gives

$$\big\langle \nabla_{\mathrm{WFR}} F(\mu), \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \big\rangle_\mu - \big\langle \Pi_\mu \nabla_{\mathrm{WFR}} F(\mu), \Pi_\mu \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \big\rangle_\mu = \frac{1}{4} \left(\int_{\mathbb{R}^d} \nabla^R F(\mu)\, d\mu \right) \left(\int_{\mathbb{R}^d} \overline{\mathrm{Log}}^R_{\boldsymbol{\gamma}^n}\, d\mu \right).$$

Choose a common dominating measure λ^n and write $a_n = d\gamma_0^n / d\lambda^n$ and $b_n = d\gamma_1^n / d\lambda^n$. Since μ and μ_n are probability measures, (2) yields

$$\begin{aligned} \mathrm{WFR}(\mu, \mu_n)^2 &= 2 - 2 \int_{\mathbb{R}^d \times \mathbb{R}^d} \sqrt{a_n(x,y) b_n(x,y)} \cos\left(\|y - x\| \wedge \frac{\pi}{2} \right) d\lambda^n(x,y) \\ &= - \int_{\mathbb{R}^d \times \mathbb{R}^d} \mathrm{Log}^R_{\boldsymbol{\gamma}^n}(x,y)\, d\gamma_0^n(x,y) \\ &= - \int_{\mathbb{R}^d} \overline{\mathrm{Log}}^R_{\boldsymbol{\gamma}^n}\, d\mu. \end{aligned}$$

Consequently, the difference of the two pairings is $O(\mathrm{WFR}(\mu, \mu_n)^2) = o(\mathrm{WFR}(\mu, \mu_n))$, and hence

$$F(\mu_n) - F(\mu) = \Big\langle \widetilde{\mathcal{V}}, \Pi_\mu \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^n} \Big\rangle_\mu + o\big(\mathrm{WFR}(\mu, \mu_n)\big),$$

which is precisely the WFR differentiability of $F_{|\mathcal{P}}$ at μ. □

Remark 6.5 (Relation with spherical Hellinger–Kantorovich geometry). Although WFR geodesics between probability measures need not preserve mass, the intrinsic geometry on $\mathcal{P}(\mathbb{R}^d)$ is described by the spherical Hellinger–Kantorovich distance. With our normalization, this distance satisfies

$$\mathrm{SHK}(\mu, \nu) = \arccos\left(1 - \frac{1}{2} \mathrm{WFR}^2(\mu, \nu) \right), \qquad \mu, \nu \in \mathcal{P}(\mathbb{R}^d);$$

see [21]. In particular,

$$\mathrm{SHK}(\mu, \nu) = \mathrm{WFR}(\mu, \nu) + O\big(\mathrm{WFR}^3(\mu, \nu)\big) \qquad \text{as } \mathrm{WFR}(\mu, \nu) \to 0.$$

Thus replacing WFR by SHK in the little-o remainder of our differentiability definition on $\mathcal{P}(\mathbb{R}^d)$ leaves the definition unchanged. The zero-mean condition on the reaction component and the projection formula (117) reflect the unit-mass constraint at the tangent level.

The main result of this section is the following one.

Theorem 6.6 (Wasserstein–flat decomposition of the WFR gradient)**.** *Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ and let $F \colon \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$ be WFR differentiable at μ, with*

$$\widetilde{\nabla}_{\mathrm{WFR}} F(\mu) = \big(\widetilde{\nabla}^T F(\mu), \widetilde{\nabla}^R F(\mu) \big).$$

Then $F|_{\mathcal{P}_2(\mathbb{R}^d)}$ is Wasserstein differentiable at μ and

$$\widetilde{\nabla}^T F(\mu) = \nabla_{W_2} F(\mu).$$

If, in addition, F is flat differentiable at μ (cf. Definition 6.9), then

$$\widetilde{\nabla}_{\mathrm{WFR}} F(\mu) = \left(\nabla_{W_2} F(\mu), 4 \left[\frac{\delta F}{\delta \mu}(\mu, \cdot) - \int_{\mathbb{R}^d} \frac{\delta F}{\delta \mu}(\mu, y)\, d\mu(y) \right] \right).$$

The proof is split into the next two subsections. We first identify the transport component with the Wasserstein gradient. We then study reaction perturbations and show that, whenever a flat derivative exists, the reaction component is its canonical zero-mean representative.

6.1 The Wasserstein component

If we restrict to $\mathcal{P}_2\big(\mathbb{R}^d\big)$, we recover Wasserstein differentiability. In this sense, we can think of Wasserstein–Fisher–Rao calculus as an extension of Wasserstein calculus, and of the Wasserstein gradient as the partial derivative corresponding to spatial transport.

For completeness we recall the definition of a Wasserstein differentiable function.

Definition 6.7 (Wasserstein differentiability)**.** Let $F\colon \mathcal{P}_2\big(\mathbb{R}^d\big) \to \mathbb{R}$. We say that F is Wasserstein differentiable at μ if there exists

$$\xi \in T_\mu \mathcal{P}_2\big(\mathbb{R}^d\big) \coloneqq \overline{\{\nabla\psi\colon \psi \in C_c^\infty(\mathbb{R}^d)\}}^{L^2(\mu)}$$

such that, for every sequence $(\mu_n)_{n\in\mathbb{N}} \subset \mathcal{P}_2\big(\mathbb{R}^d\big)$ satisfying $W_2(\mu_n,\mu) \to 0$, and every sequence of optimal couplings γ^n one has

$$F(\mu_n) = F(\mu) + \int_{\mathbb{R}^d\times\mathbb{R}^d} \langle \xi(x), y-x\rangle \, d\gamma_n(x,y) + o\big(W_2(\mu,\mu_n)\big) \tag{118}$$

as $n \to +\infty$. If such a ξ exists, it is unique, we denote it by $\nabla_{W_2} F(\mu)$ and call it *Wasserstein gradient* of F at μ.

The main result is the following, which follows from Proposition 6.2 by using specific semicouplings built from optimal couplings.

Theorem 6.8. *Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ and let $F\colon \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$ be WFR differentiable at μ. Then the restriction of F to $\mathcal{P}_2(\mathbb{R}^d)$ is Wasserstein differentiable at μ and*

$$\nabla_{W_2} F(\mu) = \widetilde{\nabla}^T F(\mu).$$

Proof. Let $(\mu_n)_{n\in\mathbb{N}} \subset \mathcal{P}_2(\mathbb{R}^d)$ satisfy

$$d_n \coloneqq W_2(\mu,\mu_n) \longrightarrow 0,$$

and let γ_n be a sequence of optimal couplings between μ and μ_n. That is

$$d_n^2 = \int_{\mathbb{R}^d\times\mathbb{R}^d} \|x-y\|^2 \, d\gamma_n(x,y).$$

Consider the pure-transport semicoupling $\boldsymbol{\gamma}^n = (\gamma_n,\gamma_n)$. Taking $\lambda_n = \gamma_n$, the corresponding densities satisfy $a_n = b_n = 1$ and its semicoupling cost satisfies

$$\mathcal{C}(\boldsymbol{\gamma}^n) = \int_{\mathbb{R}^d\times\mathbb{R}^d} 2\left(1-\cos\Big(\|y-x\| \wedge \frac{\pi}{2}\Big)\right) d\gamma_n(x,y) \le \int_{\mathbb{R}^d\times\mathbb{R}^d} |x-y|^2 \, d\gamma_n(x,y) = d_n^2.$$

In particular, the cost of $\boldsymbol{\gamma}^n$ converges to zero. Therefore, by Proposition 6.2,

$$\begin{aligned} F(\mu_n) - F(\mu) = &\int_{\mathbb{R}^d\times\mathbb{R}^d} \widetilde{\nabla}^T F(\mu)(x) \cdot \mathrm{Log}^T_{\boldsymbol{\gamma}^n}(x,y) \, d\gamma_n(x,y) \\ &+ \frac{1}{4}\int_{\mathbb{R}^d\times\mathbb{R}^d} \widetilde{\nabla}^R F(\mu)(x) \, \mathrm{Log}^R_{\boldsymbol{\gamma}^n}(x,y) \, d\gamma_n(x,y) + o(d_n). \end{aligned} \tag{119}$$

We first show that the reaction contribution is negligible. Set $\delta = \|x - y\|$ and define

$$h_R(\delta) \coloneqq \begin{cases} 0, & \delta = 0, \\ \dfrac{2(\cos(\delta) - 1)}{\delta}, & 0 < \delta < \frac{\pi}{2}, \\ -\dfrac{2}{\delta}, & \delta \geq \frac{\pi}{2}. \end{cases}$$

Then h_R is bounded, $h_R(\delta) \to 0$ as $\delta \to 0$, and

$$\mathrm{Log}^R_{\boldsymbol{\gamma}^n}(x, y) = |x - y| h_R(|x - y|).$$

Consequently, by the Cauchy–Schwarz inequality, by setting

$$H^F_R(x, y) \coloneqq \left\| \widetilde{\nabla}^R F(\mu)(x) \right\|^2 |h_R(\|x - y\|)|^2$$

we have

$$\left| \int_{\mathbb{R}^d \times \mathbb{R}^d} \widetilde{\nabla}^R F(\mu)(x) \, \mathrm{Log}^R_{\boldsymbol{\gamma}^n}(x, y) \, d\gamma^n n(x, y) \right| \leq d_n \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} H^F_R(x, y) \, d\gamma^n(x, y) \right)^{1/2}.$$

The second factor converges to zero by Corollary C.1.2. Therefore

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} \widetilde{\nabla}^R F(\mu)(x) \, \mathrm{Log}^R_{\boldsymbol{\gamma}^n}(x, y) \, d\gamma_n(x, y) = o(d_n). \tag{120}$$

We next consider the transport contribution. We have

$$\mathrm{Log}^T_{\boldsymbol{\gamma}^n}(x, y) = \kappa(\|x - y\|)(y - x),$$

where

$$\kappa(\delta) \coloneqq \begin{cases} 1 & \delta = 0 \\ \dfrac{\sin \delta}{\delta}, & 0 < \delta \leq \frac{\pi}{2} \\ 0, & \delta > \frac{\pi}{2}. \end{cases}$$

Hence

$$\mathrm{Log}^T_{\boldsymbol{\gamma}^n}(x, y) - (y - x) = h_T(\|x - y\|)(y - x),$$

where $h_T(\delta) \coloneqq \kappa(\delta) - 1$. The function h_T is bounded and satisfies

$$h_T(\delta) \longrightarrow 0 \qquad \text{as } \delta \to 0.$$

Therefore, for

$$H^F_T(x, y) \coloneqq \left\| \widetilde{\nabla}^T F(\mu)(x) \right\|^2 |h_T(\|x - y\|)|^2$$

it holds

$$\left| \int_{\mathbb{R}^d \times \mathbb{R}^d} \widetilde{\nabla}^T F(\mu)(x) \cdot \left(\mathrm{Log}^T_{\boldsymbol{\gamma}^n}(x, y) - (y - x) \right) \, d\gamma_n(x, y) \right| \leq d_n \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} H^F_T(x, y) \, d\gamma_n(x, y) \right)^{1/2}.$$

Again, the second factor converges to zero by Corollary C.1.2. It follows that

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} \widetilde{\nabla}^T F(\mu)(x) \cdot \mathrm{Log}^T_{\boldsymbol{\gamma}^n}(x, y) \, d\gamma_n(x, y) = \int_{\mathbb{R}^d \times \mathbb{R}^d} \widetilde{\nabla}^T F(\mu)(x) \cdot (y - x) \, d\gamma_n(x, y) + o(d_n). \tag{121}$$

Combining (119), (120), and (121), we obtain

$$F(\mu_n) - F(\mu) = \int_{\mathbb{R}^d \times \mathbb{R}^d} \widetilde{\nabla}^T F(\mu)(x) \cdot (y - x)\, d\gamma_n(x, y) + o\big(W_2(\mu, \mu_n)\big).$$

Finally, since

$$\big(\widetilde{\nabla}^T F(\mu), \widetilde{\nabla}^R F(\mu)\big) \in T_\mu \mathcal{M}\big(\mathbb{R}^d\big),$$

there exists a sequence $(\varphi_k)_{k\in\mathbb{N}} \subset C_c^\infty(\mathbb{R}^d)$ such that

$$(\nabla\varphi_k, 4\varphi_k) \longrightarrow \big(\widetilde{\nabla}^T F(\mu), \widetilde{\nabla}^R F(\mu)\big)$$

in $H(\mu)$. In particular,

$$\nabla\varphi_k \longrightarrow \widetilde{\nabla}^T F(\mu) \qquad \text{in } L^2(\mu; \mathbb{R}^d).$$

Thus $\widetilde{\nabla}^T F(\mu) \in T_\mu \mathcal{P}_2(\mathbb{R}^d)$ and the proof is completed. □

We have the following immediate corollary.

Corollary 6.8.1. *Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ and $\hat{\mu} \in \mathcal{M}\big(\mathbb{R}^d\big)$ satisfy the equivalent conditions of Theorem 5.1. Then the restriction of $F_{\hat{\mu}}$ to $\mathcal{P}_2(\mathbb{R}^d)$ is Wasserstein differentiable at μ and*

$$\nabla_{W_2}\big(F_{\hat{\mu}}|_{\mathcal{P}_2(\mathbb{R}^d)}\big)(\mu) = -\mathcal{L}^T,$$

where $\mathcal{L}$ is the unique regular barycentric logarithm from μ to $\hat{\mu}$. Equivalently, with the maps T, q of Theorem 5.1,

$$\nabla_{W_2}\big(F_{\hat{\mu}}|_{\mathcal{P}_2(\mathbb{R}^d)}\big)(\mu)(x) = q(x)\frac{\sin\big(\|T(x) - x\|\big)}{\|T(x) - x\|}(x - T(x)) \qquad \mu\text{-a.e.},$$

where $\sin\theta/\theta = 1$ at $\theta = 0$.

6.2 The reaction component and flat directions

We now compare WFR differentiability with flat differentiability. We first recall the local notion of flat differentiability that we shall use.

Definition 6.9 (Flat differentiability)**.** Let $F\colon \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$ and $\mu \in \mathcal{P}(\mathbb{R}^d)$. We say that F is flat differentiable at μ if there exists a function

$$\frac{\delta F}{\delta \mu}(\mu, \cdot) \in C_b(\mathbb{R}^d)$$

such that, for every $\nu \in \mathcal{P}(\mathbb{R}^d)$,

$$\lim_{\varepsilon \downarrow 0} \frac{F\big((1-\varepsilon)\mu + \varepsilon\nu\big) - F(\mu)}{\varepsilon} = \int_{\mathbb{R}^d} \frac{\delta F}{\delta \mu}(\mu, x)\, d(\nu - \mu)(x). \tag{122}$$

Remark 6.10. The flat derivative is determined only up to an additive constant.

Every first-order Wasserstein perturbation is also a first-order WFR perturbation, which allowed us to derive Wasserstein differentiability from WFR differentiability. Flat perturbations, on the other hand, form a larger class: first-order flat perturbations may create mass on sets that are negligible for the reference measure, and such creation need not occur at order ε in the WFR distance.

The following result identifies the flat directions that are directly controlled by WFR differentiability.

Proposition 6.11 (Absolutely continuous flat directions)**.** *Let $F\colon \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$ be WFR differentiable at $\mu \in \mathcal{P}(\mathbb{R}^d)$. Let $\nu \in \mathcal{P}(\mathbb{R}^d)$, and consider its Lebesgue decomposition with respect to μ to be $\nu = h\mu + \nu^\perp$. For $\varepsilon \in [0,1]$, set*

$$\mu^\varepsilon := (1-\varepsilon)\mu + \varepsilon\nu$$

and define the diagonal semicoupling $\boldsymbol{\gamma}^\varepsilon$ as

$$\gamma_0^\varepsilon := (\mathrm{Id}, \mathrm{Id})_\# \mu, \qquad \gamma_1^\varepsilon := (\mathrm{Id}, \mathrm{Id})_\# \mu^\varepsilon,$$

then

$$\mathcal{C}(\boldsymbol{\gamma}^\varepsilon) = \int_{\mathbb{R}^d} \left(\sqrt{1+\varepsilon(h-1)} - 1\right)^2 d\mu + \varepsilon \nu^\perp(\mathbb{R}^d). \tag{123}$$

In particular, if

$$\nu = h\mu, \qquad h - 1 \in L^2(\mu),$$

then

$$\lim_{\varepsilon \downarrow 0} \frac{F(\mu^\varepsilon) - F(\mu)}{\varepsilon} = \frac{1}{4} \int_{\mathbb{R}^d} \widetilde{\nabla}^R F(\mu)(x)\, d(\nu - \mu)(x). \tag{124}$$

Thus F admits a flat directional derivative along every absolutely continuous direction whose density belongs to $1 + L^2(\mu)$.

Proof. The measure

$$\lambda := (\mathrm{Id}, \mathrm{Id})_\# (\mu + \nu^\perp)$$

dominates both components of $\boldsymbol{\gamma}^\varepsilon$. On the part corresponding to μ, the densities are $a = 1$ and $b = 1 + \varepsilon(h-1)$, whereas on a μ-negligible set carrying $\nu^\perp$ they are $a = 0$ and $b = \varepsilon$. Since all pairs lie on the diagonal, the first part contributes

$$\int_{\mathbb{R}^d} \left(\sqrt{1+\varepsilon(h(x)-1)} - 1\right)^2 d\mu(x)$$

to the semicoupling cost, while the singular part is pure target creation and contributes $\varepsilon\, \nu^\perp(\mathbb{R}^d)$. This proves (123).

Suppose that $\nu = h\mu$ and $h - 1 \in L^2(\mu)$. Set $T_\varepsilon = 0$ and

$$\rho_\varepsilon := \frac{2(h-1)}{\sqrt{1+\varepsilon(h-1)} + 1}.$$

Then

$$\sqrt{1+\varepsilon(h-1)} = 1 + \frac{\varepsilon}{2}\rho_\varepsilon.$$

Moreover, $\rho_\varepsilon \to h - 1$ pointwise and $|\rho_\varepsilon| \le 2|h-1|$, so the convergence also holds in $L^2(\mu)$. Lemma A.1 gives

$$\frac{\overline{\mathrm{Log}}_{\boldsymbol{\gamma}^\varepsilon}}{\varepsilon} \longrightarrow (0, h-1) \quad \text{in } H(\mu), \qquad \frac{\mathcal{C}(\boldsymbol{\gamma}^\varepsilon)}{\varepsilon^2} \longrightarrow \frac{1}{4}\int_{\mathbb{R}^d} (h-1)^2\, d\mu.$$

Since $\sqrt{\mathcal{C}(\boldsymbol{\gamma}^\varepsilon)} = O(\varepsilon)$, Proposition 6.2 gives

$$F(\mu^\varepsilon) - F(\mu) = \left\langle \widetilde{\nabla}_{\mathrm{WFR}} F(\mu), \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^\varepsilon} \right\rangle_\mu + o(\varepsilon).$$

Dividing by ε and passing to the limit yields

$$\begin{aligned}\lim_{\varepsilon\downarrow 0}\frac{F(\mu^\varepsilon)-F(\mu)}{\varepsilon} &= \Big\langle \widetilde{\nabla}_{\mathrm{WFR}}F(\mu),(0,h-1)\Big\rangle_\mu \\ &= \frac{1}{4}\int_{\mathbb{R}^d}\widetilde{\nabla}^R F(\mu)(x)(h(x)-1)\,d\mu(x) \\ &= \frac{1}{4}\int_{\mathbb{R}^d}\widetilde{\nabla}^R F(\mu)(x)\,d(\nu-\mu)(x). \qquad \square\end{aligned}$$

The preceding result also clarifies the relation between the reaction component of the WFR gradient and the flat derivative. Suppose that F is additionally flat differentiable at μ. Comparing (124) with (122), we obtain

$$\int_{\mathbb{R}^d}\left(\frac{\delta F}{\delta\mu}(\mu,x)-\frac{1}{4}\widetilde{\nabla}^R F(\mu)(x)\right)(h(x)-1)\,d\mu(x)=0$$

for every $h\geq 0$ with $\int h\,d\mu=1$ and $h-1\in L^2(\mu)$. In particular, taking $h=1+\varepsilon\psi$ for any bounded zero-mean function ψ and sufficiently small $\varepsilon>0$, we deduce that the expression in parentheses is orthogonal to every bounded zero-mean function. By density in the zero-mean subspace of $L^2(\mu)$, it must be constant μ-a.e. Thus

$$\frac{\delta F}{\delta\mu}(\mu,x)=\frac{1}{4}\widetilde{\nabla}^R F(\mu)(x)+c_\mu,\qquad \mu\text{-a.e.},\tag{125}$$

for some constant c_μ. It is important to note that the ambiguity in (125) belongs exclusively to the flat derivative. Indeed, flat variations inside $\mathcal{P}(\mathbb{R}^d)$ have zero total mass, and therefore

$$\int_{\mathbb{R}^d} c\,d(\nu-\mu)=0$$

for every constant c. Consequently, the flat derivative cannot distinguish between two representatives differing by an additive constant.

By contrast, the reaction component $\widetilde{\nabla}^R F(\mu)$ of the WFR gradient is uniquely determined by the fact that $\widetilde{\nabla}_{\mathrm{WFR}}F(\mu)\in T_\mu\mathcal{P}(\mathbb{R}^d)$. In this sense, $\frac{1}{4}\widetilde{\nabla}^R F(\mu)$ provides a canonical representative, in $L^2(\mu)$, of the flat derivative along absolutely continuous directions.

The preceding proposition concerns flat perturbations that are also first-order WFR perturbations. Not every flat perturbation has this property. Although singular mass can be generated by transport at order ε, this becomes impossible when new mass is placed beyond the transport threshold. The following example shows that, in this case, the intrinsic WFR scale is $\sqrt{\varepsilon}$, and consequently WFR differentiability does not imply flat differentiability.

Example 6.12. Fix $1<p<2$ and $\hat{\mu}\in\mathcal{P}(\mathbb{R}^d)$, and consider $F\colon\mathcal{P}(\mathbb{R}^d)\to\mathbb{R}$ defined by

$$F(\mu)\coloneqq \mathrm{WFR}(\mu,\hat{\mu})^p.$$

Then F is WFR differentiable at $\hat{\mu}$, with

$$\widetilde{\nabla}_{\mathrm{WFR}}F(\hat{\mu})=(0,0)$$

given that

$$F(\mu)-F(\hat{\mu})=\mathrm{WFR}(\mu,\hat{\mu})^p=o\big(\mathrm{WFR}(\mu,\hat{\mu})\big).$$

We now fix $d = 1$, set

$$\widehat{\mu} = \delta_0, \qquad \nu = \delta_L, \qquad |L| > \frac{\pi}{2},$$

and consider the segment

$$\mu^\varepsilon = (1-\varepsilon)\delta_0 + \varepsilon\delta_L.$$

Since 0 and L lie beyond the transport threshold, mass at 0 cannot be effectively transported to L. It follows that

$$\mathrm{WFR}^2(\delta_0, \mu^\varepsilon) = \left(1 - \sqrt{1-\varepsilon}\right)^2 + \varepsilon = 2\left(1 - \sqrt{1-\varepsilon}\right). \tag{126}$$

Consequently,

$$\mathrm{WFR}^2(\delta_0, \mu^\varepsilon) = \varepsilon + O(\varepsilon^2), \qquad \mathrm{WFR}(\delta_0, \mu^\varepsilon) \sim \sqrt{\varepsilon}.$$

Therefore

$$\frac{F(\mu^\varepsilon) - F(\hat{\mu})}{\varepsilon} = \frac{\left[2\left(1 - \sqrt{1-\varepsilon}\right)\right]^{p/2}}{\varepsilon} \sim \varepsilon^{p/2-1} \longrightarrow +\infty.$$

Thus F does not possess a finite flat directional derivative at $\widehat{\mu}$ in the direction $\delta_L - \delta_0$, and in particular it is not flat differentiable at $\hat{\mu}$.

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A Expansions of costs and logarithms

A.1 Calculations along cone geodesics

We prove Lemmas 2.8 and 2.9, using the notation of Section 2.3.

Proof of Lemma 2.8. In the first case $r, s > 0$, direct differentiation of $\boldsymbol{m}_t$ yields the formula for $\dot{\boldsymbol{m}}_t$, and the chain rule proves the formula for $\dot{\boldsymbol{r}}_t$ whenever $\boldsymbol{r}_t > 0$. If $\|y - x\| = 0$, then $\boldsymbol{x}_t \equiv x$, and the formula for $\dot{\boldsymbol{x}}_t$ is immediate. If $0 < \|y - x\| < \pi$, then $v \neq 0$ and

$$\boldsymbol{x}_t = x + \operatorname{Arg}\left(1 + tu + \mathrm{i}t\|v\|\right) \frac{v}{\|v\|},$$

where the complex number inside the argument never vanishes.

Consequently, direct differentiation gives

$$\dot{\boldsymbol{x}}_t = \frac{v}{(1+tu)^2 + t^2\|v\|^2}.$$

If $\|y - x\| \geq \pi$, then $v = 0$ and $\boldsymbol{x}_t$ is constant on each side of the vertex time $\tau = r/(r+s)$. Thus the same formula holds whenever $\boldsymbol{r}_t > 0$. Since $s > 0$, the velocity formulas also extend continuously to $t = 1$ and give the corresponding left derivatives there.

When $s > 0$, substituting the velocity formulas and the expression for $\boldsymbol{r}_t$ gives

$$\boldsymbol{r}_t^2\|\dot{\boldsymbol{x}}_t\|^2 + |\dot{\boldsymbol{r}}_t|^2 = r^2\big(u^2 + \|v\|^2\big) = r^2 + s^2 - 2rsK_y(x) = \mathrm{d}^2_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_1).$$

If $\mathfrak{y}_1 = \mathfrak{o}$, the radial formula gives $\dot{\boldsymbol{x}}_t = 0$ and $\dot{\boldsymbol{r}}_t = -r$, so the same identity follows from $\mathrm{d}^2_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{o}) = r^2$. □

Proof of Lemma 2.9. Suppose first that $r, s > 0$ and $\|y - x\| < \pi$. From the formula for $\boldsymbol{m}_t$ we have $\boldsymbol{m}_t > 0$ in $[0, 1]$. Hence the geodesic never reaches the vertex. Since $\Phi_\varphi(\mathfrak{y}_t) = \boldsymbol{m}_t\varphi(\boldsymbol{x}_t)$ we obtain from the chain rule

$$\frac{d}{dt}\Phi_\varphi(\mathfrak{y}_t) = \dot{\boldsymbol{m}}_t\varphi(\boldsymbol{x}_t) + rs\,\nabla\varphi(\boldsymbol{x}_t) \cdot \mathbf{sin}(y - x).$$

At $t = 0$, using the expression for $\operatorname{Log}_{\mathfrak{C}}(\mathfrak{y}_0, \mathfrak{y}_1)$ in Section 2.3, this is precisely the claimed initial derivative.

Moreover, $\dot{\boldsymbol{m}}_t$ is affine in t and $|rs\cos(\|y - x\|)| \leq rs$, so

$$|\dot{\boldsymbol{m}}_t| \leq \max\{|\dot{\boldsymbol{m}}_0|, |\dot{\boldsymbol{m}}_1|\} \leq 2\big(\max\{r^2, s^2\} + rs\big) \leq 3(r^2 + s^2), \qquad rs\|\mathbf{sin}(y - x)\| \leq rs \leq \frac{1}{2}(r^2 + s^2).$$

Thus the formula for the derivative yields the claimed derivative bound. Integrating over $[0,t]$ gives the difference-quotient bound.

For the quadratic estimate, assume first that $\varphi \in C_b^2(\mathbb{R}^d)$. Differentiating the velocity formulas of Lemma 2.8 gives, for $t \in (0,1)$,

$$\ddot{\boldsymbol{r}}_t = \boldsymbol{r}_t \|\dot{\boldsymbol{x}}_t\|^2, \qquad \ddot{\boldsymbol{x}}_t = -2\frac{\dot{\boldsymbol{r}}_t}{\boldsymbol{r}_t}\dot{\boldsymbol{x}}_t.$$

Differentiating $\Phi_\varphi(\mathfrak{y}_t) = \boldsymbol{r}_t^2 \varphi(\boldsymbol{x}_t)$ twice and using (20) yields

$$\frac{d^2}{dt^2}\Phi_\varphi(\mathfrak{y}_t) = \boldsymbol{r}_t^2 \dot{\boldsymbol{x}}_t^\top \big(D^2\varphi(\boldsymbol{x}_t) + 2\varphi(\boldsymbol{x}_t) I\big)\dot{\boldsymbol{x}}_t + 2\boldsymbol{r}_t \dot{\boldsymbol{r}}_t \nabla\varphi(\boldsymbol{x}_t)\cdot\dot{\boldsymbol{x}}_t + 2\varphi(\boldsymbol{x}_t)|\dot{\boldsymbol{r}}_t|^2.$$

Since

$$2\boldsymbol{r}_t\|\dot{\boldsymbol{x}}_t\||\dot{\boldsymbol{r}}_t| \le \boldsymbol{r}_t^2\|\dot{\boldsymbol{x}}_t\|^2 + |\dot{\boldsymbol{r}}_t|^2,$$

the constant-speed identity gives

$$\left|\frac{d^2}{dt^2}\Phi_\varphi(\mathfrak{y}_t)\right| \le \Big(\big\|D^2\varphi\big\|_{\infty,\mathrm{op}} + 2\|\varphi\|_\infty + \|\nabla\varphi\|_\infty\Big)\,\mathrm{d}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)^2.$$

Taylor's formula with integral remainder therefore yields the claimed quadratic estimate.

If $r,s>0$ and $\|y-x\| \ge \pi$, set $\tau := r/(r+s)$. Then

$$\Phi_\varphi(\mathfrak{y}_t) = \begin{cases} (r+s)^2(t-\tau)^2\varphi(x), & 0 \le t \le \tau, \\ (r+s)^2(t-\tau)^2\varphi(y), & \tau < t \le 1. \end{cases}$$

Its initial right derivative is $-2r(r+s)\varphi(x)$, which agrees with the claimed formula since $\mathrm{Log}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1) = (0,-r-s)$. The map is continuously differentiable across $t=\tau$, with derivative zero there, and

$$\left|\frac{d}{dt}\Phi_\varphi(\mathfrak{y}_t)\right| \le 2(r+s)\max\{r,s\}\|\varphi\|_\infty \le 3(r^2+s^2)\|\varphi\|_\infty.$$

This proves the derivative and difference-quotient bounds. Moreover, its first derivative is absolutely continuous and

$$\left|\frac{d^2}{dt^2}\Phi_\varphi(\mathfrak{y}_t)\right| \le 2\|\varphi\|_\infty(r+s)^2 = 2\|\varphi\|_\infty \mathrm{d}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{y}_1)^2$$

for almost every $t \in (0,1)$. Taylor's formula with integral remainder again yields the claimed quadratic estimate.

If one endpoint is the vertex, the geodesic is radial and

$$\Phi_\varphi(\mathfrak{y}_t([x,r],\mathfrak{o})) = (1-t)^2 r^2\varphi(x), \qquad \Phi_\varphi(\mathfrak{y}_t(\mathfrak{o},[y,s])) = t^2 s^2 \varphi(y).$$

In particular, the initial right derivatives are respectively $-2r^2\varphi(x)$ and 0. The former agrees with the claimed initial derivative since $\mathrm{Log}_{\mathfrak{C}}(\mathfrak{y}_0,\mathfrak{o}) = (0,-r)$, while the derivative, difference-quotient, and quadratic bounds follow directly. If both endpoints are the vertex, all claims are immediate.

Finally, standard mollification extends the quadratic estimate to $\varphi \in C_b^{1,1}(\mathbb{R}^d)$ with the stated constant; see [14, Appendix C.5 and Section 5.3.1]. □

A.2 Graph semicouplings

Lemma A.1 (Linearization of graph semicouplings)**.** *Let $\mu \in \mathcal{M}(\mathbb{R}^d)$ and suppose that*

$$(T_\varepsilon, \rho_\varepsilon) \longrightarrow (T, \rho) \qquad \text{in } H(\mu)$$

as $\varepsilon \to 0$. Set

$$X_\varepsilon \coloneqq \mathrm{Id} + \varepsilon T_\varepsilon, \qquad q_\varepsilon \coloneqq 1 + \frac{\varepsilon}{2}\rho_\varepsilon, \qquad G_\varepsilon \coloneqq (\mathrm{Id}, X_\varepsilon),$$

and define

$$\mu^\varepsilon \coloneqq (X_\varepsilon)_\#(q_\varepsilon^2 \mu), \qquad \boldsymbol{\gamma}^\varepsilon \coloneqq \big((G_\varepsilon)_\#\mu,\, (G_\varepsilon)_\#(q_\varepsilon^2\mu)\big).$$

Then $\boldsymbol{\gamma}^\varepsilon \in \mathrm{SC}(\mu, \mu^\varepsilon)$ and

$$\frac{\overline{\mathrm{Log}}_{\boldsymbol{\gamma}^\varepsilon}}{\varepsilon} \longrightarrow (T, \rho) \qquad \text{in } H(\mu). \tag{127}$$

Moreover,

$$\frac{\mathcal{C}(\boldsymbol{\gamma}^\varepsilon)}{\varepsilon^2} \longrightarrow \|(T, \rho)\|_\mu^2. \tag{128}$$

The same conclusions hold for one-sided limits.

Proof. It is straightforward to check that $\boldsymbol{\gamma}^\varepsilon \in \mathrm{SC}(\mu, \mu^\varepsilon)$. Let $\lambda^\varepsilon \coloneqq (G_\varepsilon)_\#\mu$. Then λ^ε dominates both γ_0^ε and γ_1^ε. Writing $a^\varepsilon = d\gamma_0^\varepsilon/d\lambda^\varepsilon$ and $b^\varepsilon = d\gamma_1^\varepsilon/d\lambda^\varepsilon$, the graph structure gives the pulled-back densities

$$\tilde{a}^\varepsilon(x) \coloneqq a^\varepsilon(G_\varepsilon(x)) = 1, \qquad \tilde{b}^\varepsilon(x) \coloneqq b^\varepsilon(G_\varepsilon(x)) = q_\varepsilon^2(x), \qquad \mu\text{-a.e.}$$

Thus

$$\mathcal{C}(\boldsymbol{\gamma}^\varepsilon) = \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y, a^\varepsilon(x,y), b^\varepsilon(x,y))\, d\lambda^\varepsilon(x,y) = \int_{\mathbb{R}^d} c_\varepsilon(x)\, d\mu(x),$$

where

$$c_\varepsilon(x) \coloneqq \tilde{a}^\varepsilon(x) + \tilde{b}^\varepsilon(x) - 2\sqrt{\tilde{a}^\varepsilon(x)\tilde{b}^\varepsilon(x)}\cos\left(\|\varepsilon T_\varepsilon(x)\| \wedge \frac{\pi}{2}\right).$$

Similarly, the graph structure gives

$$\mathcal{L}_\varepsilon(x) \coloneqq \mathrm{Log}_{\boldsymbol{\gamma}^\varepsilon}(x, X_\varepsilon(x)) = \overline{\mathrm{Log}}_{\boldsymbol{\gamma}^\varepsilon}(x).$$

We can rewrite the pointwise cost as

$$c_\varepsilon(x) = (|q_\varepsilon(x)| - 1)^2 + 2|q_\varepsilon(x)|\left(1 - \cos\left(\|\varepsilon T_\varepsilon(x)\| \wedge \frac{\pi}{2}\right)\right).$$

Using

$$\big||q_\varepsilon(x)| - 1\big| \le \frac{|\varepsilon|}{2}|\rho_\varepsilon(x)|, \qquad 0 \le 2\left(1 - \cos\left(\|\varepsilon T_\varepsilon(x)\| \wedge \frac{\pi}{2}\right)\right) \le \min\{\varepsilon^2\|T_\varepsilon(x)\|^2, 2\},$$

together with Young's inequality and the pointwise estimate in the proof of Lemma 3.5, we obtain

$$\frac{\|\mathcal{L}_\varepsilon\|_{\mathrm{WFR}}^2}{\varepsilon^2} \le \frac{c_\varepsilon}{\varepsilon^2} \le \frac{3}{2}\|T_\varepsilon\|^2 + \frac{1}{2}|\rho_\varepsilon|^2. \tag{129}$$

Fix any sequence $\varepsilon_n \to 0$ with $\varepsilon_n \neq 0$. By strong convergence in $H(\mu)$ and [5, Theorem 4.9], applied componentwise, we may extract a subsequence, not relabeled, such that

$$(T_{\varepsilon_n}, \rho_{\varepsilon_n}) \longrightarrow (T, \rho) \qquad \mu\text{-a.e.},$$

and there exists $\Psi \in L^2(\mu)$ satisfying

$$\|T_{\varepsilon_n}\| + |\rho_{\varepsilon_n}| \le \Psi \qquad \mu\text{-a.e. for every } n.$$

At μ-almost every point, eventually $q_{\varepsilon_n} > 0$ and $\|\varepsilon_n T_{\varepsilon_n}\| < \pi/2$. Therefore,

$$\mathcal{L}_{\varepsilon_n}(x) = \left(q_{\varepsilon_n}(x) \frac{\sin\left(\|\varepsilon_n T_{\varepsilon_n}(x)\|\right)}{\|\varepsilon_n T_{\varepsilon_n}(x)\|} \varepsilon_n T_{\varepsilon_n}(x),\ 2\left(q_{\varepsilon_n}(x)\cos\left(\|\varepsilon_n T_{\varepsilon_n}(x)\|\right) - 1\right) \right),$$

with the continuous convention $\sin\theta/\theta = 1$ at $\theta = 0$, and

$$\frac{c_{\varepsilon_n}}{\varepsilon_n^2} = \frac{1}{4}|\rho_{\varepsilon_n}|^2 + 2q_{\varepsilon_n}\frac{1-\cos\left(\|\varepsilon_n T_{\varepsilon_n}\|\right)}{\varepsilon_n^2}.$$

Consequently,

$$\frac{\mathcal{L}_{\varepsilon_n}}{\varepsilon_n} \longrightarrow (T, \rho), \qquad \frac{c_{\varepsilon_n}}{\varepsilon_n^2} \longrightarrow \|T\|^2 + \frac{1}{4}|\rho|^2 \qquad \mu\text{-a.e.}$$

Together with (129) and the common majorant Ψ, dominated convergence proves both conclusions along this subsequence. Since every sequence tending to zero admits such a subsequence, the full limits follow. □

B Radial dilations and gluing

B.1 Radial dilations

We collect the dilation identities used in the cone constructions. For a finite measure on $\mathrm{Geo}(\mathfrak{C})$ with finite endpoint radial second moments, homogeneous time marginals are defined by the same formula as for probability measures. They are finite, since $r_t \le (1-t)r_0 + tr_1$ along every cone geodesic.

Lemma B.1 (Dilation and probability normalization)**.** *Let $A \in \mathcal{M}_2(\mathfrak{C}^N)$ and let $\vartheta\colon \mathfrak{C}^N \to (0,\infty)$ be Borel measurable with $\int \vartheta^2\, dA < \infty$. Define*

$$D_\vartheta\big((\mathfrak{y}_i)_{i=1}^N\big) \coloneqq \big([x_i, r_i/\vartheta]\big)_{i=1}^N, \qquad \mathrm{dil}_{\vartheta,2}\, A \coloneqq (D_\vartheta)_\#(\vartheta^2 A), \tag{130}$$

where $\mathfrak{y}_i = [x_i, r_i]$ and ϑ is evaluated at the whole tuple. Then $\mathrm{dil}_{\vartheta,2}\, A \in \mathcal{M}_2(\mathfrak{C}^N)$ and

$$\int \Phi\, d(\mathrm{dil}_{\vartheta,2}\, A) = \int \vartheta^2 \Phi(D_\vartheta(\mathfrak{y}))\, dA(\mathfrak{y}) = \int \Phi\, dA \tag{131}$$

for every integrable Borel function Φ that is jointly 2-homogeneous in the radial variables. In particular, dilation preserves homogeneous marginals and integrated squared cone distances.

The normalized cone logarithm is jointly 0-homogeneous:

$$\mathcal{L}_{\mathfrak{C}}([x,\theta r],[y,\theta s]) = \mathcal{L}_{\mathfrak{C}}([x,r],[y,s]), \qquad \theta > 0. \tag{132}$$

Consequently, dilation also preserves integrable source-weighted logarithmic pairings, squared logarithmic errors, and creation masses, that is, integrals of the forms

$$r_i^2\, \langle W(x_i), \mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_i, \mathfrak{y}_j)\rangle_{\mathrm{WFR}}, \qquad r_i^2\, \|\mathcal{L}_{\mathfrak{C}}(\mathfrak{y}_i, \mathfrak{y}_j) - W(x_i)\|_{\mathrm{WFR}}^2,$$
$$r_j^2 \mathbf{1}_{\{r_i = 0\}},$$

where $W\colon \mathbb{R}^d \to \mathbb{R}^d \times \mathbb{R}$ is Borel measurable. The first two expressions are set to zero when $r_i = 0$. These statements apply on reduced pairs as in Remark 3.3, and also on arbitrary cone pairs when the same normalization of the cone logarithm is used.

Finally, every finite measure $\widetilde{\mathbf{\Pi}}$ on $\mathrm{Geo}(\mathfrak{C})$ with finite endpoint radial second moments admits a probability normalization $\mathbf{\Pi}$ preserving all homogeneous time marginals, endpoint cost, and integrable source-weighted logarithmic pairings. The normalization preserves concentration on $\mathrm{Geo}^*(\mathfrak{C})$.

Proof. The dilation map is Borel measurable and $\mathrm{dil}_{\vartheta,2}\, A(\mathfrak{C}^N) = \int \vartheta^2\, dA < \infty$. Joint homogeneity gives

$$\int \Phi\, d(\mathrm{dil}_{\vartheta,2}\, A) = \int \vartheta^2 \Phi\big(D_\vartheta(\mathfrak{y})\big)\, dA(\mathfrak{y}) = \int \Phi\, dA.$$

Applying this to $\sum_i r_i^2$ proves the finite second-moment assertion. Applying it to $r_i^2\varphi(x_i)$ and $\mathrm{d}_{\mathfrak{C}}^2(\mathfrak{y}_i, \mathfrak{y}_j)$ proves the marginal and cost identities.

For $r > 0$, the cone-geodesic formulas give

$$\mathrm{Log}_{\mathfrak{C}}([x,\theta r],[y,\theta s]) = \big(\mathrm{Log}^T_{\mathfrak{C}}([x,r],[y,s]), \theta\, \mathrm{Log}^R_{\mathfrak{C}}([x,r],[y,s])\big)\,.$$

Since $N_{\theta r}(\xi,\theta\zeta) = N_r(\xi,\zeta)$, this proves (132); the convention at the source vertex gives the same identity there. The displayed source-weighted expressions are therefore jointly 2-homogeneous, so (131) applies.

For path measures, use the same convention

$$(D_\vartheta \mathfrak{y})_t \coloneqq [\boldsymbol{x}_t, \boldsymbol{r}_t/\vartheta(\mathfrak{y})].$$

The factor is constant along each path, so this map preserves cone geodesics and reducedness. Set $M \coloneqq \widetilde{\mathbf{\Pi}}(\mathrm{Geo}(\mathfrak{C}))$. If $M > 0$, define

$$\mathbf{\Pi} \coloneqq \frac{1}{M}(D_{M^{-1/2}})_{\#}\widetilde{\mathbf{\Pi}}. \tag{133}$$

This is a probability measure: its radii are multiplied by $\sqrt{M}$ and its weight is divided by M. Every jointly 2-homogeneous path observable therefore has the same integral before and after normalization. Apply this to $r_t^2\varphi(x_t)$, the endpoint cost, and the source-weighted logarithmic pairings. If $M = 0$, use the probability measure concentrated on the constant vertex path; all these integrals vanish for both measures. □

B.2 Gluing prescribed cone couplings

The next result follows by normalizing the common coordinate and then applying the usual gluing construction; see [24, Lemma 7.10 and the proof of Lemma 7.11]. We give the proof to record explicitly the preservation of the prescribed couplings up to dilation and an atom at the vertex pair.

Lemma B.2 (Gluing prescribed cone couplings)**.** *Let $\mu_0, \mu_1, \mu_2 \in \mathcal{M}\big(\mathbb{R}^d\big)$ and*

$$A_{01} \in \Gamma_{\mathfrak{C}}(\mu_0,\mu_1), \qquad A_{12} \in \Gamma_{\mathfrak{C}}(\mu_1,\mu_2).$$

There exist positive Borel functions $\vartheta_{ij}\colon \mathfrak{C}^2 \to (0,\infty)$, constants $c_{ij} \geq 0$, and $A \in \mathcal{M}_2(\mathfrak{C}^3)$ such that, for $(i,j) \in \{(0,1),(1,2)\}$,

$$(\pi_{ij})_{\#}A = \mathrm{dil}_{\vartheta_{ij},2}\big(A_{ij} + c_{ij}\delta_{(\mathfrak{o},\mathfrak{o})}\big), \qquad (\pi_1)_{\#}A = \iota_{\#}\mu_1 + \delta_{\mathfrak{o}},$$

where $\iota(x) := [x, 1]$ and π_i, π_{ij} are the coordinate projections.

Moreover, for every Borel function Φ jointly 2-homogeneous in the radial variables and integrable with respect to A_{ij},

$$\int_{\mathfrak{C}^3} \Phi(\mathfrak{y}_i, \mathfrak{y}_j)\, dA = \int_{\mathfrak{C}^2} \Phi\, dA_{ij}.$$

In particular, the prescribed pair costs and source-weighted logarithmic pairings are preserved.

Proof. For each $(i, j) \in \{(0, 1), (1, 2)\}$, let ρ_{ij} denote the radius of the coordinate corresponding to μ_1. Set

$$z_{ij} := A_{ij}(\{\rho_{ij} = 0\}), \qquad c_{ij} := \mathbf{1}_{\{z_{ij}=0\}}, \qquad M_{ij} := z_{ij} + c_{ij} > 0,$$

and define

$$\vartheta_{ij} := \begin{cases} \rho_{ij}, & \rho_{ij} > 0, \\ M_{ij}^{-1/2}, & \rho_{ij} = 0. \end{cases}$$

These are positive Borel functions satisfying

$$\int \vartheta_{ij}^2\, d\big(A_{ij} + c_{ij}\delta_{(\mathfrak{o},\mathfrak{o})}\big) = \mu_1(\mathbb{R}^d) + 1 < \infty.$$

Thus the dilated plans

$$B_{ij} := \mathrm{dil}_{\vartheta_{ij},2}\big(A_{ij} + c_{ij}\delta_{(\mathfrak{o},\mathfrak{o})}\big)$$

belong to $\mathcal{M}_2(\mathfrak{C}^2)$. On $\{\rho_{ij} > 0\}$, the common coordinate has radius one and the weight is multiplied by ρ_{ij}^2. On $\{\rho_{ij} = 0\}$, its ordinary mass becomes $M_{ij}^{-1}(z_{ij} + c_{ij}) = 1$. Hence both plans have the same ordinary marginal in their common coordinate:

$$\sigma := \iota_{\#}\mu_1 + \delta_{\mathfrak{o}}.$$

Disintegrate

$$B_{01}(d\mathfrak{y}_0, d\mathfrak{y}_1) = \kappa^0_{\mathfrak{y}_1}(d\mathfrak{y}_0)\, \sigma(d\mathfrak{y}_1), \qquad B_{12}(d\mathfrak{y}_1, d\mathfrak{y}_2) = \sigma(d\mathfrak{y}_1)\, \kappa^2_{\mathfrak{y}_1}(d\mathfrak{y}_2),$$

where the conditional measures are probabilities, and set

$$A(d\mathfrak{y}_0, d\mathfrak{y}_1, d\mathfrak{y}_2) := \kappa^0_{\mathfrak{y}_1}(d\mathfrak{y}_0)\, \sigma(d\mathfrak{y}_1)\, \kappa^2_{\mathfrak{y}_1}(d\mathfrak{y}_2).$$

Its prescribed pair marginals are B_{01} and B_{12}, and its middle marginal is σ. Moreover,

$$\int_{\mathfrak{C}^3} (r_0^2 + r_1^2 + r_2^2)\, dA = \mu_0(\mathbb{R}^d) + \mu_1(\mathbb{R}^d) + \mu_2(\mathbb{R}^d) < \infty,$$

so $A \in \mathcal{M}_2(\mathfrak{C}^3)$. Finally, a jointly 2-homogeneous real-valued function satisfies $\Phi(\mathfrak{o}, \mathfrak{o}) = 0$. Applying (131) to the augmented pair plans therefore gives

$$\int \Phi(\mathfrak{y}_i, \mathfrak{y}_j)\, dA = \int \Phi\, dB_{ij} = \int \Phi\, dA_{ij},$$

as required. □

C Technical results

C.1 Vanishing-measure estimates

Lemma C.1 (Vanishing dominated measures)**.** *Let $(X, \mathcal{B})$ be a measurable space, let $\mu \in \mathcal{M}^+(X)$ be a finite measure, and let $(\eta_n)_{n\in\mathbb{N}} \subset \mathcal{M}^+(X)$ satisfy*

$$0 \le \eta_n \le \mu \qquad \text{for every } n \in \mathbb{N} \tag{134}$$

and

$$\eta_n(X) \longrightarrow 0. \tag{135}$$

Then, for every $f \in L^1(\mu)$,

$$\int_X |f|\, d\eta_n \longrightarrow 0. \tag{136}$$

Proof. Let $f \in L^1(\mu)$. For every $M > 0$, we have

$$\int_X |f|\, d\eta_n = \int_{|f|\le M} |f|\, d\eta_n + \int_{|f|>M} |f|\, d\eta_n \le M\eta_n(X) + \int_{|f|>M} |f|\, d\mu,$$

where we used $\eta_n \le \mu$ in the second term. Therefore,

$$\limsup_{n\to+\infty} \int_X |f|\, d\eta_n \le \int_{|f|>M} |f|\, d\mu.$$

Since $f \in L^1(\mu)$, the right-hand side converges to zero as $M \to +\infty$. This proves the claim. □

Corollary C.1.1. *Under the assumptions of Lemma C.1, for every $g \in L^2(\mu; \mathbb{R}^m)$ one has*

$$\int_X |g|^2\, d\eta_n \longrightarrow 0. \tag{137}$$

Corollary C.1.2. *Let $\mu \in \mathcal{M}(\mathbb{R}^d)$ and let $(\gamma^n)_{n\in\mathbb{N}} \subset \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^d)$ satisfy*

$$(\pi^1)_\# \gamma^n = \mu, \qquad \int_{\mathbb{R}^d\times\mathbb{R}^d} \|x - y\|^2\, d\gamma^n(x, y) \longrightarrow 0.$$

Let $h : [0, +\infty) \to \mathbb{R}$ be bounded and Borel measurable, with

$$h(0) = 0, \qquad \lim_{r\downarrow 0} h(r) = 0.$$

Then, for every $g \in L^2(\mu; \mathbb{R}^m)$,

$$\int_{\mathbb{R}^d\times\mathbb{R}^d} \|g(x)\|^2 |h(\|x - y\|)|^2\, d\gamma^n(x, y) \longrightarrow 0.$$

Proof. We may assume $\|h\|_\infty > 0$. Define

$$\eta_n \coloneqq (\pi^1)_\# \left(\frac{|h(\|x - y\|)|^2}{\|h\|_\infty^2} \gamma^n \right).$$

Clearly, $0 \le \eta_n \le \mu$. Moreover, for every $\delta > 0$,

$$\eta_n(\mathbb{R}^d) \le \frac{\mu(\mathbb{R}^d)}{\|h\|_\infty^2} \sup_{0\le r\le\delta} |h(r)|^2 + \frac{1}{\delta^2} \int_{\mathbb{R}^d\times\mathbb{R}^d} \|x - y\|^2\, d\gamma^n(x, y).$$

Taking first $n \to +\infty$ and then $\delta \downarrow 0$, we obtain $\eta_n(\mathbb{R}^d) \to 0$. The conclusion now follows from Corollary C.1.1, since

$$\int_{\mathbb{R}^d\times\mathbb{R}^d} \|g(x)\|^2 |h(\|x - y\|)|^2\, d\gamma^n(x, y) = \|h\|_\infty^2 \int_{\mathbb{R}^d} \|g(x)\|^2\, d\eta_n(x).$$

□

C.2 Approximation and tangency

Lemma C.2. *Let $(\xi,\zeta)\in H(\mu)$ and consider for $R>0$ smooth cutoff functions $\chi_R\in C_c^\infty(\mathbb{R}^d)$ satisfying $0\le\chi_R\le 1$ and*

$$\chi_R\equiv 1 \text{ on } B_{R/2},\qquad \operatorname{supp}\chi_R \Subset B_R,\qquad \|\nabla\chi_R\|_\infty\le\frac{C}{R}, \tag{138}$$

for some $C>0$ independent of R. Then $(\chi_R\xi,\chi_R\zeta)\to(\xi,\zeta)$ in $H(\mu)$.

Proof. Since $\chi_R\to 1$ pointwise and $0\le\chi_R\le 1$, dominated convergence yields

$$\|(\chi_R-1)\xi\|_{L^2(\mu)}\to 0,\qquad \|(\chi_R-1)\zeta\|_{L^2(\mu)}\to 0.$$

Moreover,

$$\|\nabla\chi_R\cdot\xi\|_{L^2(\mu)}\le\frac{C}{R}\|\xi\|_{L^2(\mu)}\longrightarrow 0.$$

Consequently, $\xi\chi_R\to\xi$ and $\zeta\chi_R\to\zeta$ in $L^2(\mu)$ as desired. □

Lemma C.3. *Let $\varphi\in C_b^{1,1}(\mathbb{R}^d)$. For every $\mu\in\mathcal{M}\big(\mathbb{R}^d\big)$*

$$(\nabla\varphi,4\varphi)\in T_\mu\mathcal{M}\big(\mathbb{R}^d\big).$$

Proof. Let $\varphi_R\coloneqq\chi_R\varphi$, with χ_R as in Lemma C.2. Mollification approximates φ_R and its gradient uniformly by smooth compactly supported functions and their gradients. Since μ is finite, this gives $(\nabla\varphi_R,4\varphi_R)\in T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$.

Moreover,

$$(\nabla\varphi_R,4\varphi_R)=(\chi_R\nabla\varphi,\chi_R 4\varphi)+(\varphi\nabla\chi_R,0)\longrightarrow(\nabla\varphi,4\varphi)\quad\text{in } H(\mu),$$

by Lemma C.2 and $\|\varphi\nabla\chi_R\|_{L^2(\mu)}\le CR^{-1}\|\varphi\|_{L^2(\mu)}$. The conclusion follows from the closedness of $T_\mu\mathcal{M}\big(\mathbb{R}^d\big)$. □

Lemma C.4. *Let $\mathcal{W}=(w,\eta)\in T_\mu\mathcal{M}\big(\mathbb{R}^d\big)^\perp$. Then there exists a family of normal vectors $\mathcal{W}_R=(w_R,\eta_R)\in T_\mu\mathcal{M}\big(\mathbb{R}^d\big)^\perp$, $R>0$, with compact support contained in*

$$B_R\coloneqq\big\{x\in\mathbb{R}^d\colon\|x\|<R\big\},$$

such that $\mathcal{W}_R\to\mathcal{W}$ in $H(\mu)$ as $R\to+\infty$.

Proof. For χ_R as in Lemma C.2, define

$$w_R\coloneqq\chi_R w,\qquad \eta_R\coloneqq\chi_R\eta+\nabla\chi_R\cdot w.$$

Both components belong to $L^2(\mu)$ and vanish μ-a.e. outside the compact set $\operatorname{supp}\chi_R$.

We first verify that $\mathcal{W}_R\in T_\mu\mathcal{M}\big(\mathbb{R}^d\big)^\perp$. Indeed for every $\varphi\in C_c^\infty(\mathbb{R}^d)$, orthogonality of $\mathcal{W}$, tested against $\chi_R\varphi$, gives

$$\begin{aligned}0&=\int_{\mathbb{R}^d}\left(\nabla(\chi_R\varphi)\cdot w+\chi_R\varphi\eta\right)d\mu\\&=\int_{\mathbb{R}^d}\left(\nabla\varphi\cdot(\chi_R w)+\varphi(\chi_R\eta+\nabla\chi_R\cdot w)\right)d\mu\\&=\int_{\mathbb{R}^d}\left(\nabla\varphi\cdot w_R+\varphi\eta_R\right)d\mu.\end{aligned}$$

Convergence follows from Lemma C.2. □